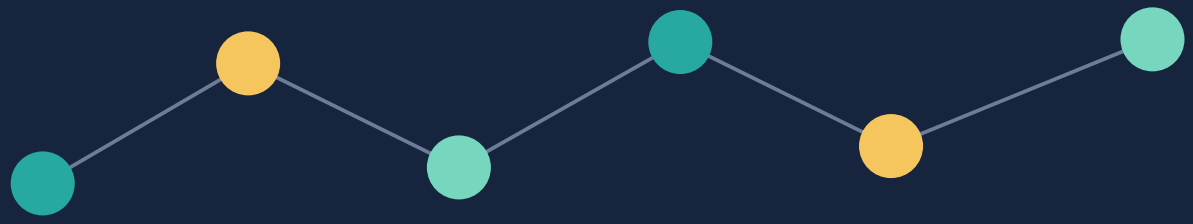
GATE 2027 / COMPLETE SYLLABUS EDITION

Engineering Mathematics

Start from the basics. Build every syllabus idea. Then practise.

Section 1: Engineering Mathematics



LEARN → SEE → WORK THROUGH → RETRIEVE → PRACTISE

1 Understand

Each chapter starts in plain language.

2 Visualize

Follow the flow and redraw it.

3 Work

Trace the worked example with pencil.

4 Recall

Answer before checking the solution.

A beginner learning guide + illustrated recall/practice + your supplied reference material

2027 syllabus coverage map

Use this as a route map: every item below has a beginner lesson in this PDF. The supplied handbook/chapter PDFs are gathered in the reference section after the learning and practice pages.

- [ ] **Logic and sets**

  Propositions • Connectives and quantifiers • Set operations

- [ ] **Relations, functions, orders and lattices**

  Relations • Equivalence and classes • Functions • Partial order and lattice

- [ ] **Monoids and groups**

  Closure and associativity • Monoid • Group • Abelian group

- [ ] **Graphs: connectivity, matching and colouring**

  Basic graph language • Connectivity • Matching • Colouring

- [ ] **Counting, recurrences and generating functions**

  Counting rules • Recurrence relations • Generating functions

- [ ] **Matrices, determinants and linear systems**

  Matrix operations • Determinant • Linear systems • Inverse

- [ ] **Eigenvalues, eigenvectors and LU**

  Eigenpair • Meaning • LU decomposition • Why LU helps

- [ ] **Limits, continuity and differentiability**

  Limits • Continuity • Differentiability • Useful link

- [ ] **Maxima, minima and the mean value theorem**

  Critical points • Classifying extrema • Mean value theorem • Interpretation

2027 syllabus coverage map (continued)

- [ ] **Integration**

  Antiderivatives • Definite integral • Area and sign • Substitution

- [ ] **Random variables, distributions and Bayes**

  Random variable • Named distributions • Summary statistics • Conditional probability and Bayes

- [ ] **Matrix powers, eigenvalue facts and nullity**

  Power shortcut • Cayley-Hamilton • Determinant and eigenvalues • Rank-nullity

- [ ] **Integration by parts and probability traps**

  Integration by parts • Definite limits • Conditional probability • Independence and Bayes

Your route through this subject

Follow the lessons in order. Tick a topic only after you can explain its picture without looking; a tick means “I can teach it,” not “I have seen it.”

1. Logic and sets
2. Relations, functions, orders and lattices
3. Monoids and groups
4. Graphs: connectivity, matching and colouring
5. Counting, recurrences and generating functions
6. Matrices, determinants and linear systems
7. Eigenvalues, eigenvectors and LU
8. Limits, continuity and differentiability
9. Maxima, minima and the mean value theorem
10. Integration
11. Random variables, distributions and Bayes
12. Matrix powers, eigenvalue facts and nullity
13. Integration by parts and probability traps

THE 20-MINUTE LEARNING LOOP

5 min	5 min	7 min	3 min
Read	Redraw	Solve	Recall

IF THIS SUBJECT FEELS COMPLETELY NEW

That is okay. Start with the first lesson, keep a pencil beside you, and redraw its picture once. Attempt the recall prompt before opening the answer guide.

Logic and sets

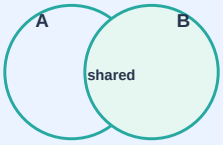
START WITH THE IDEA

Logic gives exact rules for statements; set language gives a precise way to collect and compare objects. These are the building blocks for proofs, databases, circuits and probability.

A CONCRETE FIRST EXAMPLE

Let A={1,2,3} and B={3,4}. Then A intersection B={3}; A union B={1,2,3,4}. “Every item is in A” is false if even one counterexample exists.

PICTURE IT



IMAGINE IT THIS WAY

Treat a set like a labeled basket and a proposition like a yes/no test. Draw two overlapping circles before simplifying a set expression.

THE IDEAS, ONE BY ONE

Propositions

A proposition is a statement that is true or false. A predicate becomes a statement after a value or quantifier is supplied.

Connectives and quantifiers

AND, OR, NOT, implication and equivalence combine claims. “For every” is stronger than “there exists”; negating them swaps the quantifier.

Set operations

Union collects items in either set; intersection keeps shared items; difference removes items; complement means everything in the chosen universe except the set.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT When you negate “every,” use “there exists a counterexample.”

TRY FROM MEMORY

Cover this page. Explain Logic and sets using the picture, then give one example before looking again.

Relations, functions, orders and lattices

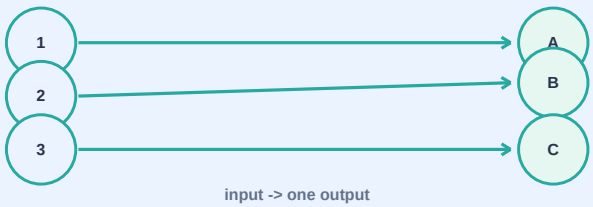
START WITH THE IDEA

A relation records which pairs are connected. A function is a stricter relation: every input has exactly one output. Order and lattice ideas describe structured comparisons.

A CONCRETE FIRST EXAMPLE

On integers, “a divides b” is a partial order. For 2 and 3, the meet is gcd(2,3)=1 and the join is lcm(2,3)=6.

PICTURE IT



IMAGINE IT THIS WAY

Think of a relation as a web of arrows; a function allows one outgoing arrow per input. An order is a ranking that may leave some pairs incomparable.

THE IDEAS, ONE BY ONE

Relations

Reflexive means every item relates to itself; symmetric reverses pairs; antisymmetric forbids two distinct items relating both ways; transitive chains relations.

Equivalence and classes

An equivalence relation is reflexive, symmetric and transitive. It partitions objects into non-overlapping classes of equivalent objects.

Functions

Injective means no two inputs share an output; surjective means every target is reached; bijective means both and therefore invertible.

Partial order and lattice

A partial order is reflexive, antisymmetric and transitive. A lattice provides each pair a greatest lower bound (meet) and least upper bound (join).

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT Do not call an order total unless every pair can be compared.

TRY FROM MEMORY

Cover this page. Explain Relations, functions, orders and lattices using the picture, then give one example before looking again.

Monoids and groups

START WITH THE IDEA

Algebraic structures describe a set plus an operation and its rules. The same checklist helps distinguish a monoid from a group.

A CONCRETE FIRST EXAMPLE

Whole numbers under addition form a monoid: 0 is the identity. Integers under addition form a group because every n has inverse -n.

PICTURE IT



IMAGINE IT THIS WAY

Use the checklist: closure -> associativity -> identity -> inverse. Each extra property moves you to a more structured family.

THE IDEAS, ONE BY ONE

Closure and associativity

Closure keeps results inside the set. Associativity changes grouping, not order: (a*b)*c=a*(b*c).

Monoid

A monoid has an associative operation and an identity e with e*a=a*e=a.

Group

A group is a monoid in which each element a has an inverse a^-1 that combines with it to produce the identity.

Abelian group

If the operation is also commutative, a*b=b*a, the group is Abelian.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT Natural numbers under addition have no additive inverse for 5, so they are not a group.

TRY FROM MEMORY

Cover this page. Explain Monoids and groups using the picture, then give one example before looking again.

Graphs: connectivity, matching and colouring

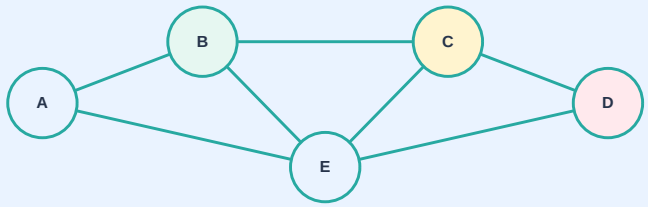
START WITH THE IDEA

Graphs model objects and their links. The GATE syllabus asks you to reason about whether vertices can reach one another and how edges or vertices can be selected or labelled.

A CONCRETE FIRST EXAMPLE

A triangle needs three colours because each vertex touches both others. In a path of four vertices, choosing the first and third edges gives a matching of size two.

PICTURE IT



IMAGINE IT THIS WAY

Picture a social network: connectivity asks whether there is a chain of introductions; matching pairs people without reusing anyone.

THE IDEAS, ONE BY ONE

Basic graph language

Vertices are objects; edges are links. Degree counts incident edges. A path is a walk without repeated vertices; a cycle returns to its start.

Connectivity

A graph is connected when every pair of vertices has a path. A bridge is an edge whose deletion disconnects the graph.

Matching

A matching is a set of edges with no shared endpoints. A perfect matching covers every vertex.

Colouring

A proper vertex colouring assigns different colours to adjacent vertices. The chromatic number is the smallest number of colours needed.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT A matching is a set of edges, while a vertex cover is a set of vertices touching every edge.

TRY FROM MEMORY

Cover this page. Explain Graphs: connectivity, matching and colouring using the picture, then give one example before looking again.

Counting, recurrences and generating functions

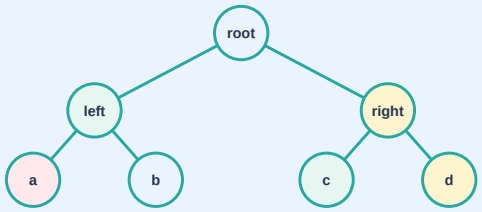
START WITH THE IDEA

Counting avoids listing every possibility. A recurrence defines a sequence using earlier values; a generating function packages the sequence so algebra can reveal its pattern.

A CONCRETE FIRST EXAMPLE

Choose 2 of 5 people: C(5,2)=5·4/(2·1)=10. Fibonacci has F_0=0,F_1=1,F■=F■■_1+F■■_2; its first values are 0,1,1,2,3.

PICTURE IT



IMAGINE IT THIS WAY

Order matters for a password, not for a committee. A generating function is a shelf of coefficients indexed by x powers.

THE IDEAS, ONE BY ONE

Counting rules

Use the sum rule for alternatives and the product rule for sequential choices. Permutations care about order; combinations do not.

Recurrence relations

A recurrence plus base cases defines a sequence. Solve by expansion, characteristic roots for linear recurrences, or a suitable generating function.

Generating functions

For sequence a_0,a_1,... use A(x)=sum a■x^n. Coefficients store the sequence; multiplication combines choices and convolves counts.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT Write base cases before solving a recurrence; they select the intended solution.

TRY FROM MEMORY

Cover this page. Explain Counting, recurrences and generating functions using the picture, then give one example before looking again.

Matrices, determinants and linear systems

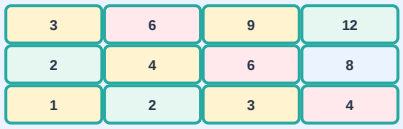
START WITH THE IDEA

A matrix is a rectangular table of coefficients. A linear system asks for values that satisfy all equations simultaneously; matrix operations provide a compact way to solve and analyze it.

A CONCRETE FIRST EXAMPLE

For 2x+y=5 and x-y=1, add equations to get 3x=6, so x=2 and y=1. The coefficient determinant is -3, not zero, so the solution is unique.

PICTURE IT



IMAGINE IT THIS WAY

Elimination is like removing one unknown at a time while preserving the solution set.

THE IDEAS, ONE BY ONE

Matrix operations

Addition is entrywise; multiplication combines rows with columns and is generally not commutative.

Determinant

For a square matrix, determinant signals invertibility: det(A)!=0 means a unique solution for every right-hand side.

Linear systems

Gaussian elimination uses row operations to reach echelon form. Pivots reveal rank, consistency and free variables.

Inverse

If A is nonsingular, Ax=b has solution x=A^-1b, though elimination is usually the practical calculation method.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT A zero determinant means no unique inverse; the system may have no solution or infinitely many.

TRY FROM MEMORY

Cover this page. Explain Matrices, determinants and linear systems using the picture, then give one example before looking again.

Eigenvalues, eigenvectors and LU

START WITH THE IDEA

Some vectors keep their direction when a matrix transforms them. Eigenvalues quantify the stretch. LU decomposition reuses a matrix factorization to solve systems efficiently.

A CONCRETE FIRST EXAMPLE

For A=[[2,1],[1,2]], v=[1,1] gives Av=[3,3]=3v, so lambda=3. In LU solving, do not skip the forward-substitution intermediate y.

PICTURE IT

1 Eigenpair → 2 Meaning → 3 Why LU helps

IMAGINE IT THIS WAY

Eigenvectors are the directions a transformation does not turn; LU is a two-step funnel that turns one hard solve into two easy ones.

THE IDEAS, ONE BY ONE

Eigenpair

Av=lambdav: nonzero v is an eigenvector and lambda its eigenvalue. Find candidates from det(A-lambdaI)=0, then solve for v.

Meaning

Eigenvectors are invariant directions; eigenvalues may describe scaling, stability or repeated behavior.

LU decomposition

Factor A=LU, where L is lower triangular and U is upper triangular. Solve Ly=b first, then Ux=y.

Why LU helps

When many right-hand sides use the same A, factor once and perform inexpensive substitutions for each b.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT The zero vector is never an eigenvector.

TRY FROM MEMORY

Cover this page. Explain Eigenvalues, eigenvectors and LU using the picture, then give one example before looking again.

Limits, continuity and differentiability

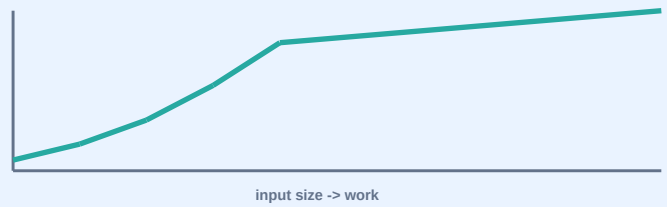
START WITH THE IDEA

Limits describe what a function approaches. Continuity says the approached value agrees with the function value. Differentiability measures local rate of change and implies continuity at that point.

A CONCRETE FIRST EXAMPLE

For f(x)=x^2, (f(2+h)-f(2))/h = ((2+h)^2-4)/h = 4+h, whose limit is 4. So f'(2)=4.

PICTURE IT



IMAGINE IT THIS WAY

Zoom in: a smooth curve looks like a straight line. At a corner, zooming never produces one unique tangent slope.

THE IDEAS, ONE BY ONE

Limits

Check left and right behavior. Algebraic simplification, standard limits and squeeze arguments handle common indeterminate forms.

Continuity

At x=a require f(a) to exist, the limit to exist, and lim f(x)=f(a).

Differentiability

The derivative is the limit of a difference quotient and represents tangent slope or instantaneous rate.

Useful link

Differentiability implies continuity, but continuity does not guarantee differentiability; a sharp corner is continuous but not smooth.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT Check one-sided limits at piecewise boundaries; a two-sided limit can fail even when both sides are finite.

TRY FROM MEMORY

Cover this page. Explain Limits, continuity and differentiability using the picture, then give one example before looking again.

Maxima, minima and the mean value theorem

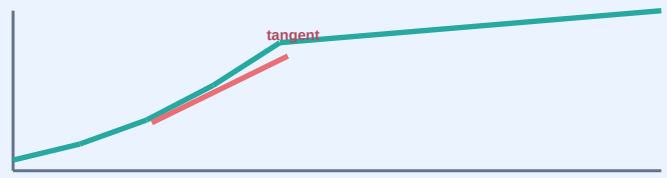
START WITH THE IDEA

Derivatives reveal where a function rises or falls. The mean value theorem guarantees a point where instantaneous slope matches the average slope over an interval.

A CONCRETE FIRST EXAMPLE

For f(x)=x^2 on [1,3], average slope=(9-1)/2=4. Since f'(x)=2x, c=2 satisfies f'(2)=4.

PICTURE IT



IMAGINE IT THIS WAY

Average slope is a secant; the theorem finds a tangent parallel to that secant.

THE IDEAS, ONE BY ONE

Critical points

Candidates occur where f'=0 or f' does not exist, plus endpoints on a closed interval.

Classifying extrema

Use sign changes of f' or the second derivative test, then compare endpoints when seeking absolute extrema.

Mean value theorem

If f is continuous on [a,b] and differentiable on (a,b), some c satisfies f'(c)=(f(b)-f(a))/(b-a).

Interpretation

A car's average speed over a trip equals its speed at least once during the trip, under the theorem's assumptions.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT Verify the theorem's continuity and differentiability assumptions before applying it.

TRY FROM MEMORY

Cover this page. Explain Maxima, minima and the mean value theorem using the picture, then give one example before looking again.

Integration

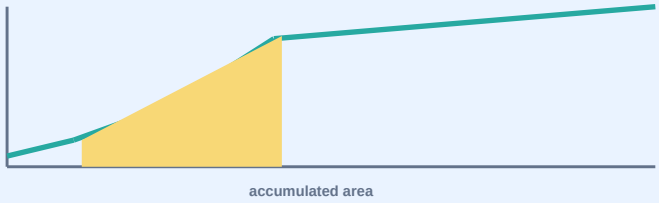
START WITH THE IDEA

Integration adds tiny contributions into a total. A definite integral is signed area/accumulated change; an antiderivative reverses differentiation.

A CONCRETE FIRST EXAMPLE

∫_0^2 x dx=[x^2/2]_0^2=2. This is the triangular area under y=x from 0 to 2.

PICTURE IT



IMAGINE IT THIS WAY

Derivative turns accumulated change into a local rate; integration rebuilds total change from rates.

THE IDEAS, ONE BY ONE

Antiderivatives

If F'=f, then ∫f(x)dx=F(x)+C. The constant disappears on differentiation.

Definite integral

∫_a■ f(x)dx=F(b)-F(a), by the fundamental theorem of calculus.

Area and sign

Areas below the x-axis count negatively; split intervals or use absolute value when the question asks geometric area.

Substitution

Use u=g(x) when the integrand contains a function and its derivative, turning a complicated expression into a simpler one.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT Do not confuse signed area with total geometric area.

TRY FROM MEMORY

Cover this page. Explain Integration using the picture, then give one example before looking again.

Random variables, distributions and Bayes

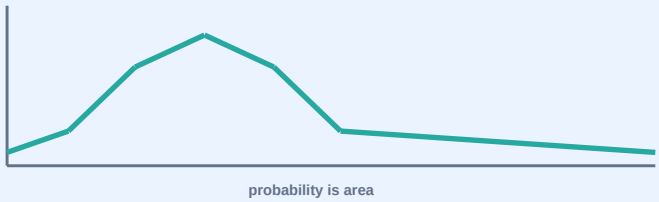
START WITH THE IDEA

Probability assigns weights to outcomes. A random variable turns outcomes into numbers; its distribution tells how likely each number or interval is.

A CONCRETE FIRST EXAMPLE

A fair coin has X=1 for heads and 0 for tails, so E[X]=0.5. If a test is rare-disease screening, Bayes combines sensitivity with the low base rate.

PICTURE IT



IMAGINE IT THIS WAY

Distribution is the full shape; mean and spread are only its summary. Bayes is a base-rate-aware evidence update.

THE IDEAS, ONE BY ONE

Random variable

Discrete variables have countable values with a PMF; continuous variables use a density, whose area over an interval is probability.

Named distributions

Bernoulli/binomial model success counts; Poisson models counts over a fixed interval; uniform gives equal density/probability over a range; normal is bell-shaped; exponential models waiting times.

Summary statistics

Mean is the balance point, median the middle, mode the most frequent value, and standard deviation the typical spread around the mean.

Conditional probability and Bayes

P(A|B)=P(A intersection B)/P(B). Bayes reverses evidence: P(A|B)=P(B|A)P(A)/P(B).

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT For continuous variables, P(X=a)=0; probabilities come from intervals under the density curve.

TRY FROM MEMORY

Cover this page. Explain Random variables, distributions and Bayes using the picture, then give one example before looking again.

Matrix powers, eigenvalue facts and nullity

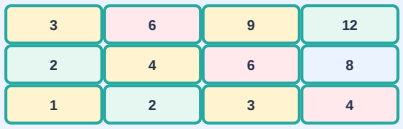
START WITH THE IDEA

Matrix questions become manageable when you track what the matrix does to vectors and use the characteristic equation to reduce powers. For a rectangular matrix, nullity counts the free directions that survive A x = 0.

A CONCRETE FIRST EXAMPLE

For A=[[1,2],[2,-1]], A squared=[[5,0],[0,5]]=5I. Then A to the eighth=(5I) to the fourth=625I.

PICTURE IT



IMAGINE IT THIS WAY

Treat A as a machine acting on vectors. The characteristic equation is its built-in shortcut card for repeated use.

THE IDEAS, ONE BY ONE

Power shortcut

If A squared equals cI, then A to an even power is a scalar multiple of I; compute small powers before multiplying repeatedly.

Cayley-Hamilton

Every square matrix satisfies its own characteristic polynomial. Rearrange that equation to reduce A squared or higher powers to lower powers.

Determinant and eigenvalues

The product of eigenvalues, counted with algebraic multiplicity, equals det(A); their sum equals trace(A). det(A)=0 means at least one zero eigenvalue.

Rank-nullity

For an m by n matrix, rank(A)+nullity(A)=n. Therefore a homogeneous system with more variables than rows has nonzero solutions.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT Use the dimension n of the domain in rank-nullity, not the number of rows.

TRY FROM MEMORY

Cover this page. Explain Matrix powers, eigenvalue facts and nullity using the picture, then give one example before looking again.

Integration by parts and probability traps

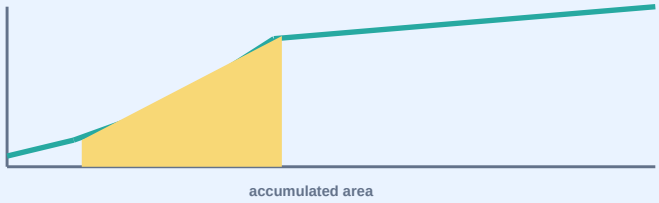
START WITH THE IDEA

Integration by parts reverses the product rule. Probability questions become safer when you separate conditional probability, independence, and the full sample space.

A CONCRETE FIRST EXAMPLE

For integral from 1 to x of t ln(t) dt, choose u=ln(t), dv=t dt. The antiderivative is (t squared/2)ln(t) - t squared/4; evaluate at both bounds.

PICTURE IT



IMAGINE IT THIS WAY

Integration by parts is product rule run backward. Conditional probability is a camera zoomed in on the given event.

THE IDEAS, ONE BY ONE

Integration by parts

Integral u dv = uv - integral v du. Choose u to become simpler when differentiated; choose dv that is easy to integrate.

Definite limits

Find an antiderivative, substitute the upper endpoint, subtract the lower endpoint, and keep the sign of each term.

Conditional probability

P(A given B)=P(A and B)/P(B). Restrict the sample space to B before counting.

Independence and Bayes

Independence means P(A and B)=P(A)P(B). Bayes reverses conditioning by combining likelihood with the prior probability.

HOW TO USE IT ON A PROBLEM

Name the objects and assumptions. Write the definition or equation first, solve one step at a time, then check units, boundary cases and whether the result is plausible.

WATCH OUT Do not infer independence from a pair of probabilities alone; test the product identity.

TRY FROM MEMORY

Cover this page. Explain Integration by parts and probability traps using the picture, then give one example before looking again.

Past-paper trail and coverage check

WHAT THE PAPERS MATCH

Verified anchors across years: 2018 Q17/Q26 (eigenvalues/matrices), Q28 (logic), Q44 (probability); 2022 Q34/Q36/Q45 (limits, generating functions, LU); 2023 Q17/Q27/Q30 (determinants, calculus, integration); 2024 CS1 Q11/Q12/Q49/Q63; 2025 CS2 Q11/Q12/Q14/Q42/Q44; 2026 CS1 Q13/Q20/Q32/Q58.

WHAT THIS EDITION ADDS

Added full worked lessons for matrix powers/Cayley-Hamilton, rank-nullity, integration by parts, conditional probability and independence.

SYLLABUS BOUNDARY

Not a separate 2027 topic: advanced numerical-analysis methods beyond the listed calculus outcomes. Prioritize the listed limits, derivatives, extrema, MVT and integration.

HOW TO USE THIS CHECK

A question is a match when its tested skill is taught in the lesson pages, even if the story or numbers are new. Anchors are selected verified examples, not an exhaustive list; the separate Coverage and Gap Audit PDF identifies years whose scans could not be checked question by question.

Close the notes: try before you peek

For each card, explain the idea from memory and recreate the lesson example on paper.

1. **Logic and sets**

   Without notes: define Set operations and explain why this example works. Then recreate the steps on the lesson page.

2. **Relations, functions, orders and lattices**

   Without notes: define Partial order and lattice and explain why this example works. Then recreate the steps on the lesson page.

3. **Monoids and groups**

   Without notes: define Abelian group and explain why this example works. Then recreate the steps on the lesson page.

4. **Graphs: connectivity, matching and colouring**

   Without notes: define Colouring and explain why this example works. Then recreate the steps on the lesson page.

5. **Counting, recurrences and generating functions**

   Without notes: define Generating functions and explain why this example works. Then recreate the steps on the lesson page.

6. **Matrices, determinants and linear systems**

   Without notes: define Inverse and explain why this example works. Then recreate the steps on the lesson page.

Close the notes: try before you peek

For each card, explain the idea from memory and recreate the lesson example on paper.

7 **Eigenvalues, eigenvectors and LU**

Without notes: define Why LU helps and explain why this example works. Then recreate the steps on the lesson page.

8 **Limits, continuity and differentiability**

Without notes: define Useful link and explain why this example works. Then recreate the steps on the lesson page.

9 **Maxima, minima and the mean value theorem**

Without notes: define Interpretation and explain why this example works. Then recreate the steps on the lesson page.

10 **Integration**

Without notes: define Substitution and explain why this example works. Then recreate the steps on the lesson page.

11 **Random variables, distributions and Bayes**

Without notes: define Conditional probability and Bayes and explain why this example works. Then recreate the steps on the lesson page.

12 **Matrix powers, eigenvalue facts and nullity**

Without notes: define Rank-nullity and explain why this example works. Then recreate the steps on the lesson page.

Close the notes: try before you peek

For each card, explain the idea from memory and recreate the lesson example on paper.

13

Integration by parts and probability traps

Without notes: define Independence and Bayes and explain why this example works. Then recreate the steps on the lesson page.

Compare with the worked examples

Use this only after attempting the recall prompts. The full reasoning is on the numbered lesson page.

1. LESSON 1: Logic and sets

Let A={1,2,3} and B={3,4}. Then A intersection B={3}; A union B={1,2,3,4}. “Every item is in A” is false if even one counterexample exists.

2. LESSON 2: Relations, functions, orders and lattices

On integers, “a divides b” is a partial order. For 2 and 3, the meet is gcd(2,3)=1 and the join is lcm(2,3)=6.

3. LESSON 3: Monoids and groups

Whole numbers under addition form a monoid: 0 is the identity. Integers under addition form a group because every n has inverse -n.

4. LESSON 4: Graphs: connectivity, matching and colouring

A triangle needs three colours because each vertex touches both others. In a path of four vertices, choosing the first and third edges gives a matching of size two.

5. LESSON 5: Counting, recurrences and generating functions

Choose 2 of 5 people: C(5,2)=5·4/(2·1)=10. Fibonacci has F_0=0,F_1=1,F■=F■■_1+F■■_2; its first values are 0,1,1,2,3.

6. LESSON 6: Matrices, determinants and linear systems

For 2x+y=5 and x-y=1, add equations to get 3x=6, so x=2 and y=1. The coefficient determinant is -3, not zero, so the solution is unique.

Compare with the worked examples

Use this only after attempting the recall prompts. The full reasoning is on the numbered lesson page.

7. LESSON 7: Eigenvalues, eigenvectors and LU

For A=[[2,1],[1,2]], v=[1,1] gives Av=[3,3]=3v, so lambda=3. In LU solving, do not skip the forward-substitution intermediate y.

8. LESSON 8: Limits, continuity and differentiability

For f(x)=x^2, (f(2+h)-f(2))/h = ((2+h)^2-4)/h = 4+h, whose limit is 4. So f'(2)=4.

9. LESSON 9: Maxima, minima and the mean value theorem

For f(x)=x^2 on [1,3], average slope=(9-1)/2=4. Since f'(x)=2x, c=2 satisfies f'(2)=4.

10. LESSON 10: Integration

∫_0^2 x dx=[x^2/2]_0^2=2. This is the triangular area under y=x from 0 to 2.

11. LESSON 11: Random variables, distributions and Bayes

A fair coin has X=1 for heads and 0 for tails, so E[X]=0.5. If a test is rare-disease screening, Bayes combines sensitivity with the low base rate.

12. LESSON 12: Matrix powers, eigenvalue facts and nullity

For A=[[1,2],[2,-1]], A squared=[[5,0],[0,5]]=5I. Then A to the eighth=(5I) to the fourth=625I.

Compare with the worked examples

Use this only after attempting the recall prompts. The full reasoning is on the numbered lesson page.

13. LESSON 13: Integration by parts and probability traps

For integral from 1 to x of t ln(t) dt, choose u=ln(t), dv=t dt. The antiderivative is (t squared/2)ln(t) - t squared/4; evaluate at both bounds.

Your practice and reference library

01 **Beginner lessons**

Plain-language explanation, visual sequence, example and watch-outs for every listed 2027 topic cluster.

02 **Recall and GATE-style practice**

The illustrated recap section includes questions and a separate answer explanation.

03 **Detailed reference**

Your supplied subject handbook and chapter PDFs follow for extra depth, reading and worked material.

Included source documents

Illustrated recall and GATE-style practice	MATH - Engineering Mathematics - GATE 2027 Illustrated Notes.pdf	13 pages
Detailed supplied reference	DM.pdf	43 pages
Detailed supplied reference	Engineering Mathematics.pdf	43 pages

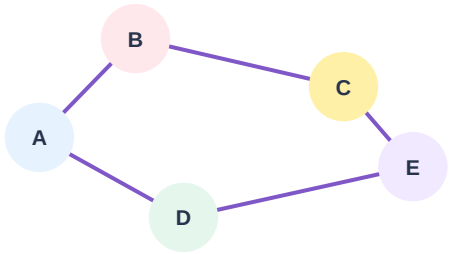
The official 2027 syllabus map precedes the lessons. Original reference pages are retained so you can use them for slower reading and extra practice.

Engineering Mathematics

GATE 2027 | VISUAL LEARNING NOTES | STARTING FROM THE BASICS

Turn definitions into structures, then structures into answers.

PICTURE IT



premises -> conclusion

HOW TO USE THIS NOTEBOOK

1. **Learn the idea**
   Read the plain-language explanation before memorizing a formula.
2. **See it**
   Study the labeled picture and redraw it from memory.
3. **Follow a solve**
   Work through the examples step by step.
4. **Test yourself**
   Attempt the GATE-style questions before opening the key.

THE PATH YOU WILL FOLLOW

- Discrete mathematics
- Linear algebra and calculus
- Probability and statistics

IF YOU FEEL COMPLETELY NEW

That is okay. Start with the first topic page, keep a pencil beside you, and copy each diagram once. Do not jump straight to the answer key. The questions are meant to come after the concepts.

Your route through this subject

Follow the topics in order. Each page teaches one idea, shows a picture, gives a recall cue and points out a common trap.

DISCRETE MATHEMATICS

- [ ] Propositional and first-order logic.
- [ ] Sets, relations, functions, partial orders and lattices.
- [ ] Monoids and groups.
- [ ] Graphs: connectivity, matching and colouring.
- [ ] Combinatorics: counting, recurrence relations and generating functions.

LINEAR ALGEBRA AND CALCULUS

- [ ] Matrices, determinants and systems of linear equations.
- [ ] Eigenvalues/eigenvectors and LU decomposition.
- [ ] Limits, continuity, differentiability, maxima/minima, mean value theorem and integration.

PROBABILITY AND STATISTICS

- [ ] Random variables; uniform, normal, exponential, Poisson and binomial distributions.
- [ ] Mean, median, mode, standard deviation, conditional probability and Bayes theorem.

THE 20-MINUTE LEARNING LOOP

5 min	5 min	7 min	3 min
Read	Redraw	Solve	Recall

Tick a topic only after you can explain its picture without looking. A tick means "I can teach it," not "I have seen it."

BEFORE EACH STUDY SESSION

Pick one topic. Read its idea, redraw its diagram, follow the worked examples, then answer questions with the notes closed. Circle only the step where you got stuck - that is your next revision target.

CONCEPT 01 / MATH

Logic, predicates and quantifiers

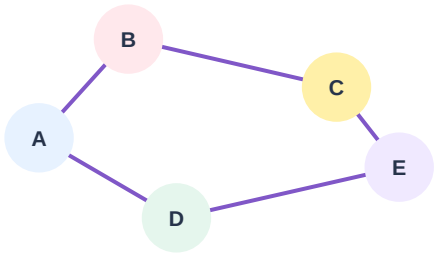
START WITH THE IDEA

A proposition is true or false. Implication p -> q is false only when p is true and q false; it is equivalent to not-p OR q. Quantifiers need their domain: forall x P(x) differs from exists x P(x).

A CONCRETE FIRST EXAMPLE

A proposition is a statement that is either true or false. A predicate becomes a proposition once its variable is assigned or quantified. "For all" and "there exists" have different meanings and cannot be swapped casually. For domain {1,2,3}, "there exists x: x>2" is true (x=3); "for every x: x>2" is false.

PICTURE IT



premises -> conclusion

IMAGINE IT THIS WAY

A quantifier is a database filter: forall checks every row; exists needs one witness.

HOW TO USE IT ON A PROBLEM

Negate forall by making it exists-not; negate exists by making it forall-not.

WATCH OUT

Changing quantifier order can change the statement.

TRY FROM MEMORY

Cover the page and explain Logic, predicates and quantifiers using the diagram. Give one example before looking again.

CONCEPT 02 / MATH

Sets, relations and functions

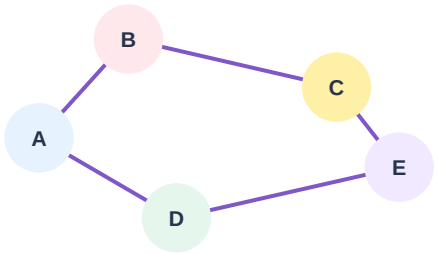
START WITH THE IDEA

A relation is a subset of A x B. Check reflexive, symmetric and transitive properties separately. An equivalence relation partitions a set into classes. A function assigns each input exactly one output; it can be many-to-one.

A CONCRETE FIRST EXAMPLE

A set collects objects; a relation tells which pairs are connected; a function gives each input exactly one output. Equivalence relations group objects into classes, while partial orders describe comparable structure. For f(x)=x^2 on {-2,-1,1,2}, inputs -2 and 2 share output 4; this is a function but not one-to-one.

PICTURE IT



objects connected by pairs

IMAGINE IT THIS WAY

A function is a one-ticket-per-person assignment; two people may share a seat label, but one person cannot get two outputs.

HOW TO USE IT ON A PROBLEM

For injective, compare inputs; for surjective, cover every target element.

WATCH OUT

Do not confuse a relation with a function or an inverse relation with an inverse function.

TRY FROM MEMORY

Cover the page and explain Sets, relations and functions using the diagram. Give one example before looking again.

CONCEPT 03 / MATH

Partial orders, lattices, monoids and groups

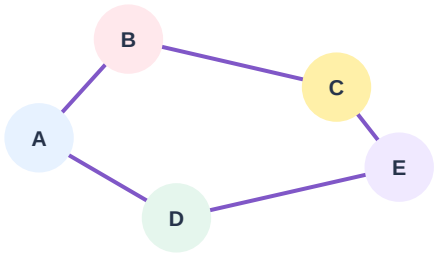
START WITH THE IDEA

A partial order is reflexive, antisymmetric and transitive; incomparable elements are allowed. A lattice has a meet (greatest lower bound) and join (least upper bound) for every pair. A group has identity and inverses; a monoid requires identity but not inverses.

A CONCRETE FIRST EXAMPLE

A partial order is reflexive, antisymmetric and transitive. A lattice is a poset where every pair has a greatest lower bound and least upper bound. A group adds an identity and inverses to an associative operation. Under divisibility, 2 and 3 have greatest common divisor 1 and least common multiple 6; these are their meet and join.

PICTURE IT



ordered nodes with bounds

IMAGINE IT THIS WAY

A hierarchy is a partial order: two teams can be incomparable. A group is a reversible move system.

HOW TO USE IT ON A PROBLEM

Test algebraic properties from the definition, one at a time.

WATCH OUT

A total order compares every pair; a partial order need not.

TRY FROM MEMORY

Cover the page and explain Partial orders, lattices, monoids and groups using the diagram. Give one example before looking again.

CONCEPT 04 / MATH

Graph structure

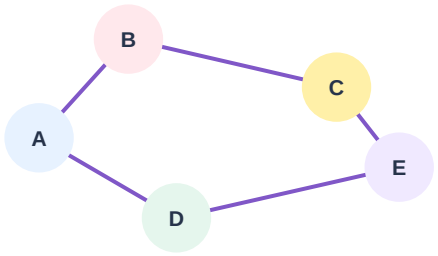
START WITH THE IDEA

Degree sum equals 2|E| in an undirected graph. Connectivity asks for paths; a matching uses no shared endpoints; proper colouring gives adjacent vertices different colours. A tree with n vertices has n-1 edges.

A CONCRETE FIRST EXAMPLE

A graph is made of vertices and edges. Degree counts incident edges; a tree is connected and has no cycle. Matching chooses edges that share no endpoints, and coloring assigns labels so adjacent vertices differ. A tree with 6 vertices has 5 edges. If one more edge is added between existing vertices, a cycle appears.

PICTURE IT



vertices + edges

IMAGINE IT THIS WAY

A graph is a route map: vertices are places and edges are usable roads.

HOW TO USE IT ON A PROBLEM

Use the question's graph type (directed/undirected, simple/multigraph) before applying a theorem.

WATCH OUT

A matching is not the same as a perfect matching; the latter covers every vertex.

TRY FROM MEMORY

Cover the page and explain Graph structure using the diagram. Give one example before looking again.

CONCEPT 05 / MATH

Counting, recurrences and generating functions

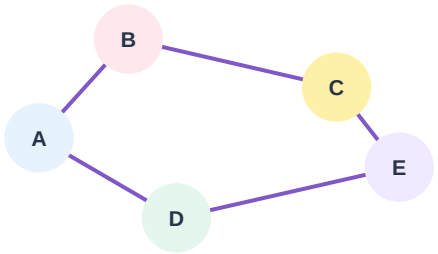
START WITH THE IDEA

Use sum rule for alternatives and product rule for sequential choices. Permutations care about order; combinations do not. A recurrence defines terms from earlier terms; generating functions encode the sequence as coefficients of a power series.

A CONCRETE FIRST EXAMPLE

Counting begins by asking whether order matters and whether repetition is allowed. Recurrences define a sequence from earlier terms; generating functions encode the whole sequence as coefficients of a power series. Choose 2 people from 5: order does not matter, so use C(5,2)=10. Ordered pairs would instead give 5 x 4=20.

PICTURE IT



count cases without overlap

IMAGINE IT THIS WAY

Choosing a committee is a combination; assigning roles to its members is a permutation.

HOW TO USE IT ON A PROBLEM

For a recurrence, identify base cases before expanding or solving.

WATCH OUT

Do not count the same arrangement twice when objects repeat or positions are indistinguishable.

TRY FROM MEMORY

Cover the page and explain Counting, recurrences and generating functions using the diagram. Give one example before looking again.

CONCEPT 06 / MATH

Linear algebra: systems, eigenpairs and LU

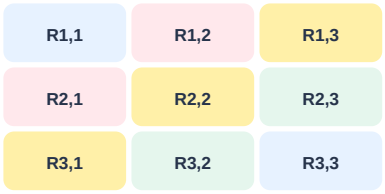
START WITH THE IDEA

Rank compares independent rows/columns and determines consistency. Determinant zero means a square matrix is singular. Eigenpairs satisfy Av=lambda v. LU factors A=LU and reuses triangular solves for multiple right-hand sides.

A CONCRETE FIRST EXAMPLE

A linear system Ax=b asks for a vector x that satisfies every equation at once. LU factors A into triangular pieces so the same system can be solved by forward and back substitution. Eigenvectors keep their direction under A; eigenvalues give the scale. For A=[[2,1],[1,2]], vector [1,1] is scaled to [3,3], so it is an eigenvector with eigenvalue 3.

PICTURE IT



row x column

IMAGINE IT THIS WAY

LU is a reusable prep station: factor once, then solve many orders cheaply.

HOW TO USE IT ON A PROBLEM

For Ax=b, solve Ly=b then Ux=y after factoring A=LU.

WATCH OUT

Do not divide by a zero pivot; use a valid pivot strategy or state the limitation.

TRY FROM MEMORY

Cover the page and explain Linear algebra: systems, eigenpairs and LU using the diagram. Give one example before looking again.

CONCEPT 07 / MATH

Calculus

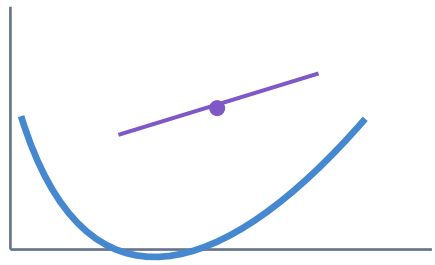
START WITH THE IDEA

Limits describe approach; continuity means no jump, and differentiability means a well-defined local slope. Use the mean value theorem to connect average and instantaneous rates. For maxima/minima, test critical points and interval endpoints; integration accumulates area.

A CONCRETE FIRST EXAMPLE

A derivative is the local slope; an integral accumulates area or total change. Critical points occur where the derivative is zero or undefined. The mean value theorem connects an average change over an interval to a tangent slope somewhere inside it. For f(x)=x^2, f' (x)=2x. The slope at x=3 is 6; the area under y=x from 0 to 2 is 2.

PICTURE IT



tangent slope = derivative

IMAGINE IT THIS WAY

A derivative is a speedometer; an integral is distance accumulated from the speed trace.

HOW TO USE IT ON A PROBLEM

Check continuity/differentiability conditions before applying a theorem.

WATCH OUT

An interior critical point is not automatically the global maximum or minimum.

TRY FROM MEMORY

Cover the page and explain Calculus using the diagram. Give one example before looking again.

CONCEPT 08 / MATH

Probability and statistics

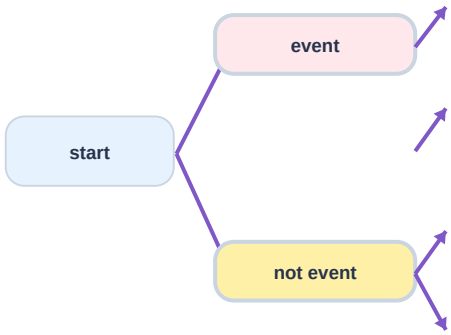
START WITH THE IDEA

Name the random variable and support. Binomial counts successes in fixed independent trials; Poisson models event counts; exponential models waiting time; normal is bell-shaped; uniform spreads mass evenly over an interval. Mean locates; standard deviation measures spread. Bayes reverses conditioning.

A CONCRETE FIRST EXAMPLE

A random variable assigns a number to each outcome. Its distribution describes probabilities; the mean is its long-run average and variance measures spread. Conditional probability updates a probability after evidence is known. For a fair coin, X=1 on heads and 0 on tails: E[X]=0.5. Bayes' rule reverses a conditional using the base rates.

PICTURE IT



multiply along a path; add paths

IMAGINE IT THIS WAY

A clinic updates its belief after a test; a call center counts arrivals and waiting time.

HOW TO USE IT ON A PROBLEM

For conditional probability, update the sample space before computing.

WATCH OUT

Independent does not mean mutually exclusive; uncorrelated does not generally imply independent.

TRY FROM MEMORY

Cover the page and explain Probability and statistics using the diagram. Give one example before looking again.

Let's solve it together

Cover the answer with your hand. Try each step first, then compare your reasoning.

EXAMPLE 1: LU solve

QUESTION

Solve 2x + y = 5 and 4x + 3y = 11.

WORK THROUGH THE STEPS

A=[[2,1],[4,3]]. Eliminate with multiplier 2: U=[[2,1],[0,1]], L=[[1,0],[2,1]]. Solve Ly=[5,11] gives y=[5,1]. Solve Ux=y: x2=1, 2x1+1=5, so

ANSWER (x,y)=(2,1).

Check: write the key rule beside your answer. If it does not fit the rule, redo the model before recalculating.

EXAMPLE 2: Bayes update

QUESTION

A test has sensitivity 0.9, false-positive rate 0.1, and prevalence 0.02. Find P(disease | positive).

WORK THROUGH THE STEPS

P(+)=0.9(0.02)+0.1(0.98)=0.116. Bayes:

ANSWER 9/58, about 0.155. A positive result is not a 90% disease probability.

Check: write the key rule beside your answer. If it does not fit the rule, redo the model before recalculating.

Now you try

Attempt these without the notes. The question types are marked MCQ, MSQ or NAT. Answers and explanations are on the next page.

MCQ Which is equivalent to p -> q?

(A) p AND q (B) not-p OR q (C) p OR not-q (D) not-p AND q

NAT A tree has 12 vertices. How many edges does it have?

Enter an integer.

NAT For A=[[2,1],[4,3]], use no row swaps. What is the first elimination multiplier in LU?

Enter a number.

MCQ If X is Binomial(n=4,p=1/2), what is P(X=2)?

(A) 1/4 (B) 3/8 (C) 1/2 (D) 5/8

MSQ For any undirected graph, which statements are always true?

(A) Sum of degrees is 2|E| (B) Odd-degree count is even (C) Every connected graph is a tree

Exam habit: decide what is being asked before doing arithmetic. Show units, assumptions and one line of reasoning.

Answers with the why

Read the explanation even when your answer is correct. The goal is to learn the reasoning pattern.

1. B

WHY

Implication fails only at p=true,q=false; not-p OR q has the same truth table.

2. 11

WHY

Every tree on n vertices has n-1 edges.

3. 2

WHY

The multiplier is A21/A11=4/2=2.

4. B

WHY

C(4,2)(1/2)^4=6/16=3/8.

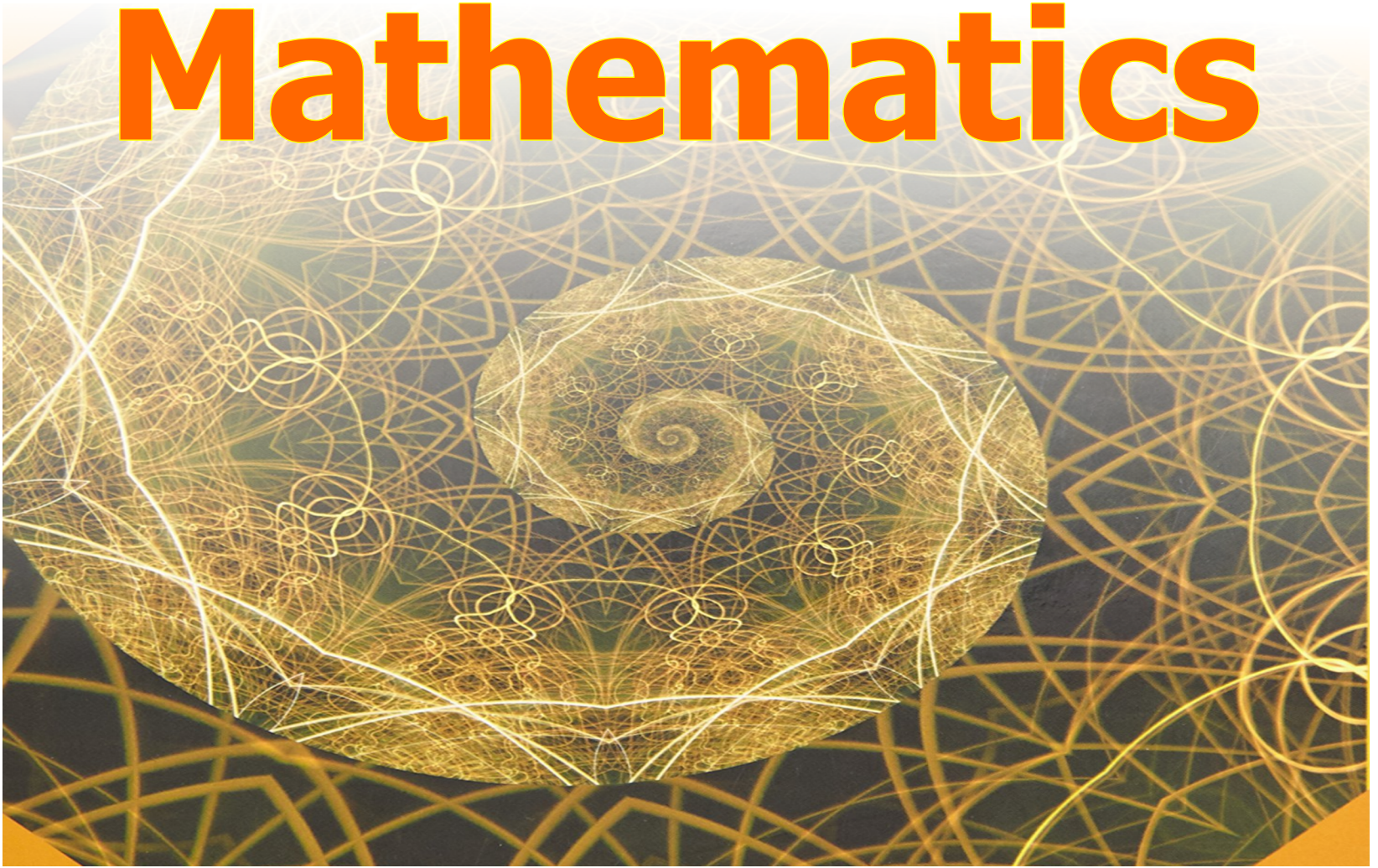
5. A, B

WHY

The handshaking lemma gives A; parity gives B. A cycle can make a connected graph not a tree.

Review loop: mark each missed concept page, redraw its picture, and retry a similar problem tomorrow.

Discrete Mathematics



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Email: support@pw.live

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Discrete Mathematics

INDEX

1. Graph Theory ..... 2.1 – 2.10
2. Logic Handbook ..... 2.11 – 2.17
3. Set Theory ..... 2.18 – 2.35
4. Combinatorics ..... 2.36 – 2.40

1 GRAPH THEORY

1.1 Basics of Graphs:

The number of vertices of odd degree in a graph is always even.

Every graph has an even number of odd vertices.

- Based on Parallel edges & self-loops graphs are classified into 3 types

	Parallel graph	Self-loops
Simple graph	×	×
Multi graph	✓	×
Pseudograph	✓	✓

(We mainly discuss simple graph in our syllabus)

1.1.1 Theorem1:

Maximum degree of a vertex, in a simple graph with n vertices, is ≤ n – 1.

Note:

Hand Shaking Lemma: Sum of all degrees = 2 × sum of all edges.

1.1.2 Theorem2 :

Maximum no. of edges, in a simple graph with n vertices, is $\leq n_{c_2} = \frac{n(n-1)}{2}$

Note: No of different graphs possible with n distinct vertices is $= 2^{\frac{n(n-1)}{2}}$

No. of different graphs possible with n distinct vertices and 'e' edges is $\left[\frac{n(n-1)}{2}\right]_{C_e}$

1.1.3 Degree sequence:

If the degrees of a graph are written in increasing order or decreasing order, we call it a degree sequence

- Not all degree sequence forms simple graph.
- The degree sequence which forms simple graph is called graphical

1.1.4 Theorem 3:

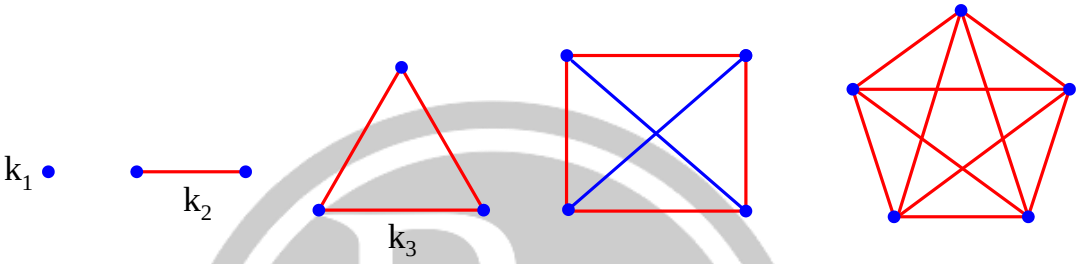
In a simple graph at least two vertices with have same degree ($n \geq 2$)
Example: {5, 4, 3, 2, 1} is not graphical

1.1.5 Theorem 4:

Max degrees in a given graph G is denoted as $\Delta(G)$ & Min degree is denoted as $\delta(G)$

$$\delta(G) \leq \frac{2e}{n} \leq \Delta(G) \leq n-1$$

1.1.6 Complete Graph (k_n) ($n \geq 1$):



Degree of every vertex is n – 1

(or)

There direct edge b/w every pair of vertices

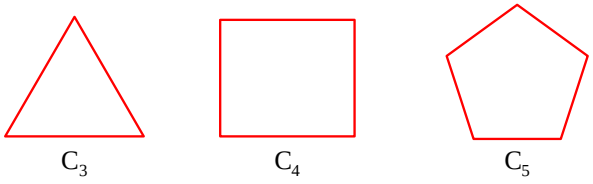
- no. of edges is,

$$e = \frac{n(n-1)}{2}$$

1.1.7 Regular Graph:

A graph in which degree of all vertices is same is called a regular graph.
$n.\delta(G) = 2e = n.\Delta(G)$

1.1.8 Cycle Graph (C_n) ($n \geq 3$):



If given degree sequence is all 2's, then it's not guaranteed that it's a cycle graph
G: [2, 2, 2, 2, 2, 2]

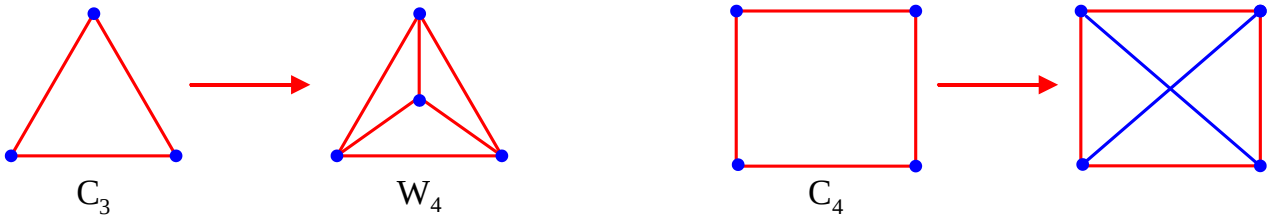


Degree of all vertices is 2 in a cycle graph

- Every Cn is a regular graph
- Number of Edges = n = Number of Vertex

1.1.9 Wheel Graph (w_n) ($n \geq 4$):

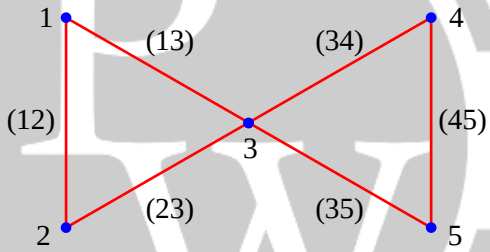
A wheel graph w_n is obtained by adding a vertex (hub) to c_{n-1}(Cycle Graph) such that this vertex is adjacent to all the other vertices.



Number of edges = 2 (n - 1)

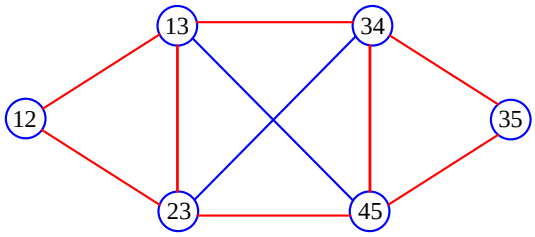
1.1.10 Line Graph (L(G)):

Consider below graph, G



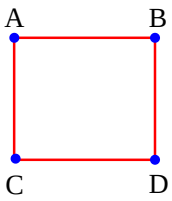
Step to construct L(G)

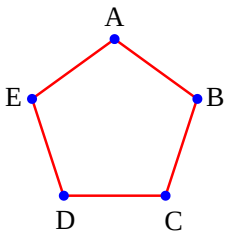
(i) define edges in G
(ii) label these edges as vertices in L(G)
(iii) Connect vertices with common number.

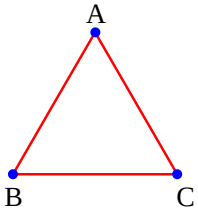


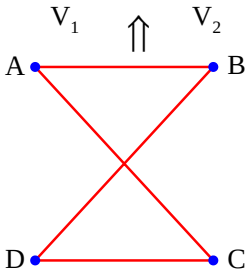
Line graph of every cycle graph is also a cycle.

1.1.11 Bipartite (L(G)):

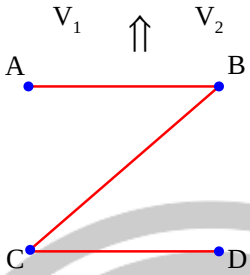








$\therefore C_4$ is a bipartite graph



Now adding E in either of the sets violates the property of bipartite graph

$\therefore C_5$ is not a bipartite graph



adding C is not possible

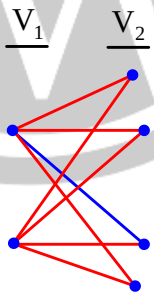
$\therefore C_3$ is not a bipartite graph

Note: Bi – partite graphs do not contain odd length cycle.

1.1.12 Complete bipartite graph ($k_{m,n}$) :

Each vertex in set v_1 is adjacent to all vertices in set v_2.

Example:



$K_2, 4$ $\quad |v_1| = m$

$| v_2 | = n$

In $k_{m, n}$ number of vertices = m + n

number of edges = mn

$\Delta(k_{m, n}) = \max(m, n)$, $\quad \delta(k_{m, n}) = \min(m, n)$

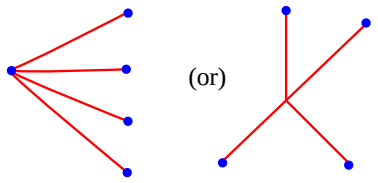
1.1.13 Theorem 5 :

- Maximum no. of edges passible is bipartite graph of n vertices is $\leq \left\lfloor \frac{n^2}{4} \right\rfloor$

1.1.14 Star graph ($k_{1,n-1}$) :

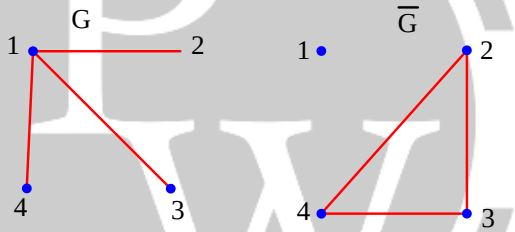
It is complete bipartite graph with one vertex in one set and rest of the vertices in other set.
(or)
Star graph (k_1, k_{n-1}) is complete bipartite graph possible with n vertices and minimum no. of edges.

Eg: Star graph of 5 vertices, k_1, 4 is
total no of edges in $k_{1, n-1} = n - 1$
$\Delta (k_{1, n-1}) = n - 1$, $\delta (k_{1, n-1}) = 1$



1.1.15 Complement graph $(\bar{G})$:

For a graph G, complement of G $(\bar{G})$ is the graph which contains all the vertices present in G and does not contain the edges present in G.



$$\therefore \quad G + \bar{G} = k_n \Rightarrow e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$$

If the degree of vertex v is x in graph G, then the degree of vertex v in $\bar{G}$ is [(n-1)-x]
If $d_1, d_2 \ldots\ldots\ldots d_n$ is degree sequence for G then
$(n-1-d_1), (n1-d_2), \ldots.. (n-1-d_n)$ is degree sequence of $\bar{G}$

Example:
consider a graph of degree sequence {5, 2, 2, 2, 2, 1}.
What is the degree sequence of complement graph?
Total vertices, n = 6

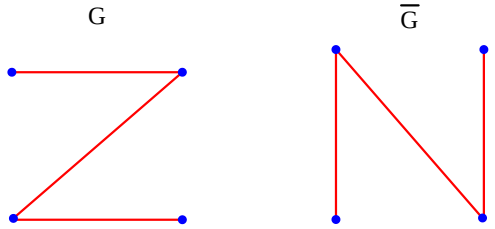
$$k_6 \rightarrow 5, 5, 5, 5, 5, 5$$
$$G \rightarrow 5, 2, 2, 2, 2, 1$$
$$\bar{G} \rightarrow \{0, 3, 3, 3, 3, 4\}$$

Isomorphic Graphs:
Let G1 = (V1, E1) and G2 = (V2, E2) be two undirected graphs. A function f: V1 $\rightarrow$ V2 is called a graph isomorphism if (a) f is one-to-one and onto, and (b) for all a, b $\in$ V1. {a, b} $\in$ E1 if and only if {f(a), f(b)} $\in$ E2. When such a function exists, G1 and G2 are called isomorphic graphs.

(ii) Self-complement: $(G \equiv \bar{G})$

It is a graph which is isomorphic to-its own complement.

i.e., $G = \bar{G}$



It is clear that above two graphs are self-complement to each other.

w.r.t

$$e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$$

Let e be no. of edges in G

$$e\ te = \frac{n(n-1)}{2}$$

- $e = \frac{n(n-1)}{4}$ i.e., no. of edges in a self-complement graph

Complement of star graph $k_{1, n-1}$ gives one isolated vertex and a complete graph k_{n-1}

n must be congruent to 0 or 1 mod 4.

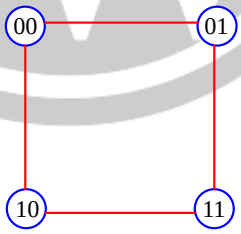
Hypercube (Q_n):

Q_1

2' vertices – 0, 1

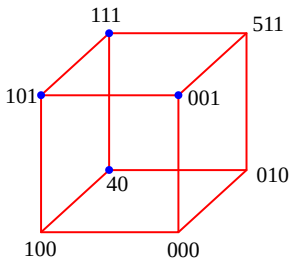
Q_2

4' vertices – 00, 01, 10, 11



Every cycle in hypercube B of even length (think why). So, every hypercube is bipartite graph

Q_3: 2^3 vertices



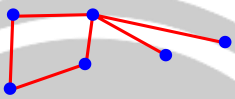
Number of edges = $\frac{n \times 2^n}{2} = n \times 2^{n-1}$

1.2 Connectivity

Name	Repeated Vertex (Vertices)	Repeated Edge(s)	Open	Closed
Walk (open)	Yes	Yes	Yes	
Walk (closed)	Yes	Yes		Yes
Trail	Yes	No	Yes	
Circuit	Yes	No		Yes
Path	No	No	Yes	
Cycle	No	No		Yes

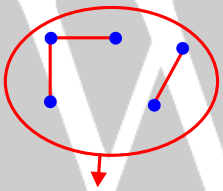
1.2.1 Connected graph:

For every two pair of vertices, there must exist a path between them.



1.2.2 Disconnected graph:

If we can find at least one pair of vertices, such that there is no path available b/w these 2 verities, then the graph is said to be disconnected graph.



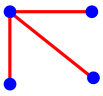
Disconnected graph

- If G is connected then $\bar{G}$ may be connected or disconnected

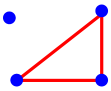


G Connected

$\overline{G}$ Connected



G Connected

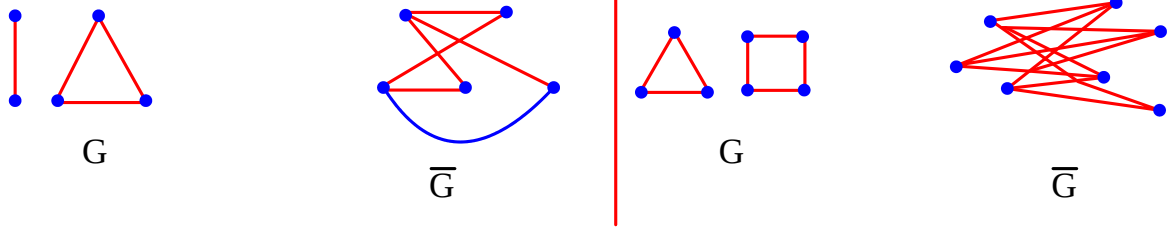


$\overline{G}$ disconnected

1.2.3 Theorem 6:

If G is disconnected, then $\bar{G}$ is connected.

Example:



1.2.4 Range of edges for a connected graph (k=1):

(k is connected components)

Tree

- Minimum no of edges required to get a possibility to make graph connected with in vertices is n-1.

$$(n-1)_{Tree} \le e \le \frac{n(n-1)}{2}$$

- The connected graph with n-1 edges is doesn't have a cycle.
- This graph is known as minimally connected graph.
- In this kind of graph, these will be a unique path b/w any two pair of vertices.
- This kind of graph is called a tree (a connected graph with no cycles)

1.2.5 Range of edges for a disconnected graph:

- Edges range b/w

$$n-k \le e \le \frac{(n-k)(n-k+1)}{2}$$

Proof:

Here lets say we have k components with $n_1, n_2 \ldots\ldots n_k$ components

$\therefore n_1 + n_2 + \ldots\ldots + n_2 : k$

For min no. of edge, each compared must be minimally connected.

$\therefore$ min no of edges is

$N_1 - 1 + n_2 - 1 + \ldots\ldots + n_k - 1$

$= (n_1 + n_2 + \ldots.. n_k) - k = n - k$

Note:

1. Let G be a graph of order n. If

deg u + deg v ≥ n – 1

nonadjacent vertices u and v of G, then G is connected and diam(G) ≤ 2.

2. If G is a graph of order n with $\delta(G) \ge (n-1)/2$, then G is connected.

3. A directed graph is strong connected if there is a path from a to b and from b to a whenever a and b are vertices in the graph.

4.A directed graph is weakly connected if there is a path between every two vertices in the underlying undirected graph.

5. $k(G) \le \lambda(G) \le \min_{v \in v} \deg(v)$.

6. Simple graph G with n vertices is connected if it has more than (n - 1) (n - 2)/2 edges.
7. Let G = (V, E) be a loop-free graph with n (≥2) vertices. If deg(v) ≥ (n - 1)/2 for all v ∈ V, then G has a Hamilton path.
8. If G = (V, E) is a loop-free undirected graph with $|V| = n \geq 3$, and if $|E| \geq \binom{n-1}{2} + 2$, then G has a Hamilton cycle.
9. If G = (V, E) is a loop-free undirected graph with $|V| = n \geq 3$, and $\deg(v) \geq n/2$ for all v ∈ V, then G has a Hamilton cycle.
(10) If G_1, G_2 are (loop-free) undirected graphs, G_1, G_2 are isomorphic if and only if $\bar{G}_1, \bar{G}_2$ are isomorphic.
(11) If G is an undirected graph or multigraph with no isolated vertices, then we can construct an Euler trail in G if and only if G is connected and has exactly two vertices of odd degree.
(12) $k(G) \leq \lambda(G) \leq \delta(G) \leq \Delta(G) \leq n-1$

1.3 Planarity

A graph (or multigraph) G is called planar if G can be drawn in the plane with its edges intersecting only at vertices of G. Such a drawing of G is called an embedding of G in the plane.

Kuratowski's Theorem. A graph is nonplanar if and only if it contains s subgraph that is homomorphic to either K_5 or $K_{3.3}$.

Let G = (V, E) be a connected planar graph or multigraph with |V| = ν and |E| = e. Let r be the number of regions in the plane determined by a planar embedding (or, depiction) of G; one of these regions has infinite area and is called the infinite region. Then ν – e + r = 2.
Let G = (V, E) be a loop-free connected planar graph with |V| = ν, |E| = e > 2, and r regions. Then 3r ≤ 2e and e ≤ 3ν – 6.

1.3.1 Coloring:

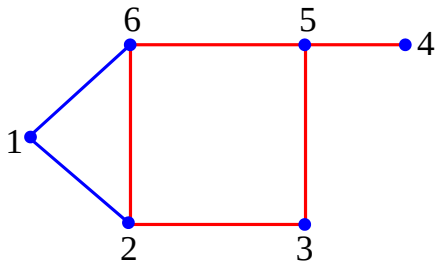
Note: Every MIS will always be MDS. But reverse need not to be true
Domination number ≤ Independence number.
α(G) = ≤ β(G

1.3.2 Theorem 7 :

Sum of size of minimum vertex cover and size of maximum independent set is equal to number of vertices

1.4 Covering:

It is set of edges such that all vertices should incident on at least one edge.



{16, 12, 65, 53, 54}
{16, 54, 23}, {16, 12, 53, 54}
{12, 16, 65, 23, 53, 62, 54}

Set of all edge is also a covering set

- It is also known as edge covering set.

1.4.1 Minimal Covering set:

It is a covering set from which we can't remove new elements(edge).
{16, 12, 53, 54} . {16, 54, 23} are MCS

1.4.2 Covering Number (C(G)):

It is no of edges present smallest covering set.
For above graph C(G) = 3

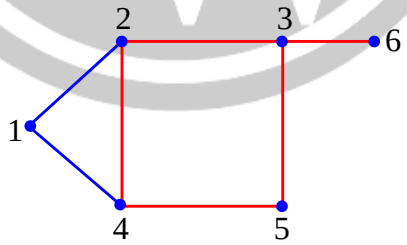
1.5 Perfect Matching:

A matching is said to be prefect matching if every vertex in the graph is matched
(or)
Induced degree of all the vertices is 1.

1.5.1 Induced degree:

The degree of a vertex in a matching is called indeed degree

Example:



{12, 45, 36} is perfect matching

Note: Every perfect matching is maximal, but reverse need not to be true.
If prefect matching exists, then no. of vertices will always be even but reverse need not to be true.
A graph may contain more than one perfect matching.
Total no. of prefect matching possible for a complete graph with 2n vertices is $\frac{(2n)!}{2^n.n!}$

❑❑❑

2 LOGIC HANDBOOK

2.1 Introduction

p	$\neg p$
T	F
F	T

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition "if p, then q."

The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or *antecedent* or *premise*) and q is called the conclusion (*or consequence*).

"if p, then q"	"p implies q"
"if p, q"	"p only if q"
"p is sufficient for q"	"a sufficient condition for q is p"
"q if p"	"q whenever p"
"q when p"	"q is necessary for p"
"a necessary condition for p is q"	"q follows from p"
"q unless $\neg p$"	"q provided that p"

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition "p if and only if q". The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications.

Operator	*Precedence*
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a tautology. A compound proposition that is always false is called a contradiction. A compound proposition that is neither a tautology nor a contradiction is called contingency.

$\rightarrow$ Every contingency is satisfiable, but reverse need not to be true 1.

$\rightarrow$ Every tautology is satisfiable, but revers need not to be true.

For any primitive statements p, q, r, any tautology 'T' and any contradiction 'F'.

(1) $\neg \neg p \Leftrightarrow p$ — Law of Double Negation

(2) $\neg (p \vee q) \Leftrightarrow \neg p \wedge \neg q$ — DeMorgan's Laws

$\neg (p \wedge q) \Leftrightarrow \neg p \vee \neg q$

(3) $p \vee q \Leftrightarrow q \vee p$ — Commutative Laws

$p \wedge q \Leftrightarrow q \wedge p$

(4) $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$ — Associative Laws

$p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$

(5) $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$ — Distributive Laws

$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

(6) $p \vee p \Leftrightarrow p$ — Idempotent Laws

$p \wedge p \Leftrightarrow p$

(7) $p \vee F \Leftrightarrow p$ — Identity Laws

$p \vee T \Leftrightarrow p$

(8) $p \vee \neg p \Leftrightarrow T$ — Inverse Laws

$p \wedge \neg p \Leftrightarrow F$

(9) $p \vee T \Leftrightarrow T$ — Domination Laws

$p \wedge F \Leftrightarrow F$

(10) $p \vee (p \wedge q) \Leftrightarrow p$ — Absorption Laws

$p \vee (p \wedge q) \Leftrightarrow p$

$p \rightarrow q \equiv \neg p \vee q$

$p \rightarrow q \equiv \neg q \rightarrow \neg p$

$p \vee q \equiv \neg p \rightarrow \neg q$

$p \wedge q \equiv \neg\ (p \rightarrow \neg q)$

$\neg (p \rightarrow q) \equiv p \wedge \neg q$

$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$

$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$

$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$

$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$

$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

$\neg (p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Rule of Inference	Related Logical Implications	Name of Rule
1) $\dfrac{p \quad p \rightarrow q}{\therefore q}$	$[p \wedge (p \rightarrow q)] \rightarrow q$	Rule of Detachment (Modus Ponens)
2) $\dfrac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	Law of the syllogism
3) $\dfrac{p \rightarrow q \quad \neg q}{\therefore \neg p}$	$[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$	Modus Tollens

4) $\dfrac{p}{q}$ $\therefore p \wedge q$		Rules of Conjunction
5) $\dfrac{p \vee q}{\neg p}$ $\therefore q$	$[(p \vee q) \wedge \neg p] \rightarrow q$	Rule of Disjunctive Syllogism
6) $\dfrac{\neg p \rightarrow F}{\therefore p}$	$(\neg p \rightarrow F_0) \rightarrow p$	Rule of Contradiction
7) $\dfrac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Rule of Conjunctive simplification
8) $\dfrac{p}{\therefore p \vee q}$	$p \rightarrow p \vee q$	Rule of Disjunctive Amplification
9) $\dfrac{p \wedge q}{p \rightarrow (q \rightarrow r)}$ $\therefore r$	$[(p \wedge q) \wedge [p \rightarrow (q \rightarrow r)]] \rightarrow r$	Rule of Conditional Proof
10) $\dfrac{p \rightarrow q}{q \rightarrow r}$ $\therefore (p \vee q) \rightarrow r$	$[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$	Rule for Proof by Cases
11) $\dfrac{p \rightarrow q}{r \rightarrow s}$ $\dfrac{p \vee r}{\therefore q \vee s}$	$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (q \vee s)$	Rule of the Constructive Dilemma
12) $\dfrac{p \rightarrow q}{r \rightarrow s}$ $\dfrac{\neg q \vee \neg s}{\therefore \neg p \vee \neg r}$	$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s)] \rightarrow (\neg p \vee \neg r)$	Rule of the Destructive Dilemma

$\exists x\, p(x)$	For some (at least one) a in the universe, $p(a)$ is true.	For every a in the universe, $p(a)$ is false.
$\forall x\, p(x)$	For every replacement a from The universe, $p(a)$ is true.	There is at least one replacement a from the universe for which $p(a)$ is false.
$\exists x \neg p(x)$	For at least one choice a in The universe, $p(a)$ is false, so Its negation $\neg p(a)$ is true.	For every replacement a in the universe, $p(a)$ is true.

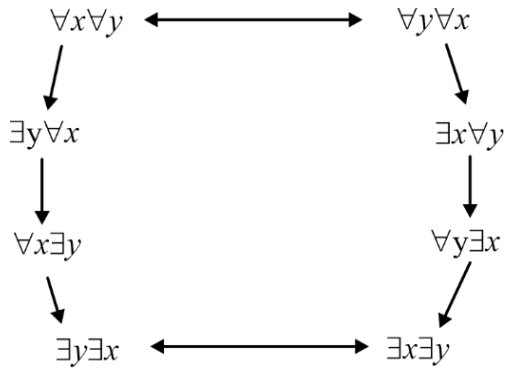
$\forall x \neg p(x)$	For every replacement a from The universe, $p(a)$ is false and its negation $\neg p(a)$ is true	There is at least one replacement a from the universe for which $\neg p(a)$ is false and $p(a)$ is true.

$$\neg [\forall x\, p(x)] \Leftrightarrow \exists x \neg p(x)$$
$$\neg [\exists x\, p(x)] \Leftrightarrow \forall x \neg p(x)$$
$$\neg [\forall x \neg p(x)] \Leftrightarrow \exists x \neg\neg p(x) \Leftrightarrow \exists x\, p(x)$$
$$\neg [\exists x \neg p(x)] \Leftrightarrow \forall x \neg\neg p(x) \Leftrightarrow \forall x\, p(x)$$

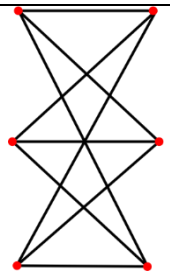
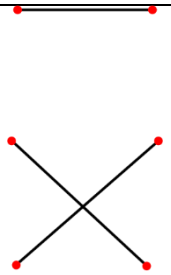
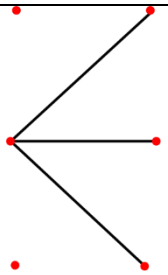
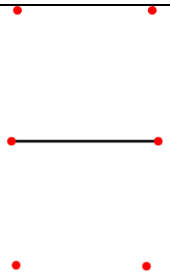
$\exists x\,[P(x) \vee Q(x)] \equiv \exists x\, P(x) \vee \exists x\, Q(x)$
$\forall x\,[P(x) \vee Q(x)] \equiv \forall x\, P(x) \vee \forall x\, Q(x)$
$\exists x\,[P(x) \wedge Q(x)] \rightarrow \exists x\, P(x) \wedge \exists x\, Q(x)$
$\forall x\,[P(x) \wedge \forall\, Q(x)] \rightarrow \forall x\,[P(x) \vee Q(x)]$
$\forall x\,[P(x) \rightarrow\ Q(x)] \rightarrow \forall x\, P(x) \rightarrow \forall x\, Q(x)$
$\forall x\,[P(x) \leftrightarrow\ Q(x)] \rightarrow \forall\, P(x) \leftrightarrow \forall x\, Q(x)$
$(\forall x)\, P(x) \wedge (\forall x)\, Q(x) \Leftrightarrow (\forall x)\,[P(x) \wedge Q(x)]$
$(\forall x)\, P(x) \wedge (\forall x)\, Q(x) \Leftrightarrow (\forall x)\,(\forall y)\,[P(x) \vee Q(y)]$
$(\exists x)\, P(x) \vee (\exists x)\, Q(x) \Leftrightarrow (\exists x)\,(\exists y)\,[P(x) \wedge Q(y)]$
$(\exists x)\, P(x) \vee (\exists x)\, Q(x) \Leftrightarrow (\exists x)\,[P(x) \vee Q(y)]$
$(\forall x)\, P(x) \wedge (\exists x)\, Q(x) \Leftrightarrow (\forall x)\,(\exists y)\,[\, P(x) \wedge Q(y)]$
$(\forall x)\, P(x) \vee (\exists x)\, Q(x) \Leftrightarrow (\forall x)\,(\exists y)\,[\, P(x) \vee Q(y)]$
$A \vee (\forall x)\, P(x) \Leftrightarrow (\forall x)\,[A \vee P(x)]$
$A \vee (\exists x)\, P(x) \Leftrightarrow (\exists x)\,[A \vee P(x)]$
$A \wedge (\forall x)\, P(x) \Leftrightarrow (\forall x)\,[A \wedge P(x)]$
$A \wedge (\exists x)\, P(x) \Leftrightarrow (\exists x)\,[A \wedge P(x)]$

Statement	When True?	When False?
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true	There is an x such that $P(x, y)$ is false for every y.
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y.	For every x there is a y for which $P(x, y)$ is false.

$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	The is a pair x, y for which $P(x, y)$ is true	$P(x, y)$ is false for every pair x, y



Rule of Inference	***Name***
$\dfrac{\forall x P(x)}{\therefore P(c)}$	Universal instantiation
$\dfrac{P(c) \text{for an arbitary } c}{\therefore \forall x P(x)}$	Universal generalization
$\dfrac{\exists x P(x)}{\therefore \text{for some element } c}$	Existential instantiation
$\dfrac{P(c) \text{for some elemet } c}{\therefore \exists x P(x)}$	Existential generalization

Number of non-equivalent propositional function possible with 'n' propositional variable			
2^{2^n}			
Nested Quantifier			
$\forall x \forall y$	$\forall x \exists y$	$\exists y \forall x$	$\exists x \exists y$
			
English statements		**Logical Expressions**	

All graphs are connected	$\forall x[G(x) \rightarrow C(x)]$
Not all graphs are connected	$\neg\forall x[G(x) \rightarrow C(x)]$
All graphs are not connected ≡ No graphs are connected	$\forall x[G(x) \rightarrow \neg C(x)] \equiv \forall x[G(x) \rightarrow \neg C(x)]$

❑❑❑

3 SET THEORY

3.1 Introduction

Collection of unordered, district and well defined objects.

- $\{1, 2\} = \{2, 1\} = \{1, 2, 1\}$ (order & repetition doesn't matter)

  Set representation: A {1, 2, 3} - Roaster form

  $A = \{z|z \in N \quad x \leq 3\}$ – Set builder form
- Set with one element is known as singleton set.
- Set with zero element is known as empty set or null set.
- No of elements in a set is known as cardinality of set.
- If a set A contains an element a, it is denoted as:

  $a \in A$.

3.1.1 Equal sets (Axiom of extension):

Two sets A and B are said to be equal iff they have same set of elements.

$A = B \Leftrightarrow \forall_x (x \in A \leftrightarrow x \in B) \Leftrightarrow A \subseteq B \wedge B \supseteq A$

3.1.2 Subset:

- A is said to be subset of B iff every element in A is also then in B. Denoted as $A \subseteq B$

  $\forall x (x \in A \Rightarrow x \in B)$

3.1.3 Proper subset:

- 'A' is said to be a proper subset of B if every element in A. exist in B and $A \neq B$. Denoted as $A \subset B$

  i.e., $A \subset B \Leftrightarrow \forall x (x \in A \rightarrow x \in B) \wedge \exists x. (x \in B \wedge x \notin A)$

  or

  $A \subset B \Leftrightarrow \forall x (x \in A \rightarrow x \in B) \wedge (|A| \neq |B|)$

3.1.4 Powerset:

Powerset of a set A is set of all subsets of A. Denoted as P(A)

- if |A| = n

  $|P(A)| = 2^n$

- For any set A

  $\phi \subseteq A$

  If A = ϕ, then

  $\phi \subset A$

3.1.5 Operations on set:

(i) **Union:**

$A \cup B = \{x | x \in A \vee x \in B\}$

(ii) **Intersection:**

$A \cap B = \{x | x \in A \wedge x \in B\}$

(iii) **Complement:**

If A is a set,

Complement of A, $\bar{A} = A^c = \{x | x \in \cup \wedge x \notin A\}$

i.e., $\bar{A} = U - A = U \cap \bar{A}$

(iv) **Difference:**

$A/B = A - B = \{x | x \in A \wedge x \notin B\} = A \cap \bar{B}$

$A - \bar{B}$ is also called complement of B w.r.t. to A.

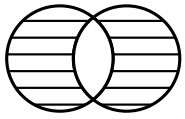
"complement of B in A"

(v) **Symmetric Difference:**

$A \Delta B = \{x | x \in A \vee x \in B$, but not both$\}$

Since it corresponds to XOR operator, it is also denoted as $A \oplus B$

$A \Delta B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$



- Two sets are said to be disjoint if their intersection ϕ.

3.1.6 Loss involving set operations:

- $\left.\begin{aligned} A\cup A&=A\\ A\cap A&=A\end{aligned}\right\}$ Idempotent law
- $\left.\begin{aligned} A\cap\phi&=\phi\\ A\cup U&=U\end{aligned}\right\}$ Domination law
- $\left.\begin{aligned} A\cap U&=A\\ A\cup\phi&=A\end{aligned}\right\}$ Indentity law
- $\left.\begin{aligned} A\cup(B\cup C)&=(A\cup B)\cup C\\ A\cup(B\cap C)&=(A\cap B)\cap C\\ A\oplus(B\oplus C)&=(A\oplus B)\oplus C\end{aligned}\right\}$ Associative law

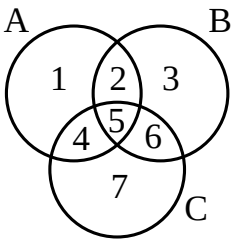
  $\because$ XOR operation is associative
- $\left.\begin{aligned} A\cup(B\cup C)&=(A\cup B)\cup(A\cup c)\\ A\cap(B\cap C)&=(A\cap B)\cup(A\cap C)\end{aligned}\right\}$ Distributive law
- $\left.\begin{aligned} A\cup(A\cap B)&=A\\ A\cap(A\cup B)&=A\end{aligned}\right\}$ absorption law
- $\left.\begin{aligned} \overline{A\cup B}&=\bar{A}\cup\bar{B}\\ \overline{A\cup B}&=\bar{A}\cap\bar{B}\end{aligned}\right\}$ Demorgan's law

Note:

$\left.\begin{aligned} A\oplus B=A\oplus C&\Rightarrow B=C\\ A\oplus C=B\oplus C&\Rightarrow A=C\end{aligned}\right\}$ *The proot can be obtained by relating this to XOR operation in* log*ic.*

3.1.7 Representation of sets using Venn diagrams:

Rather than direct calculations we sometimes represent sets with Venn diagram and see them as regions.
Let A, B, C be 3 sets



The region 1 is represented by $A \cap \bar{B} \cap \bar{c}$

The region 2 is represented by $A \cap B \cap \bar{c}$

Similarly, we have 8 regions including region outside of A, B, C

The principle of duality:

Let s denote a theorem dealing with the equality at two expressions which involve only operation $\cup$ and $\cap$. The dual of s(sd) (i.e., expression obtained by interchanging $\cup$ and $\cap$) is also a theorem.

Finding dual of $A \subseteq B$

$A \subseteq B$ can be written as

$A \cup B = B$

Dual is $A \cap B = B$

i.e., $B \subseteq A$

$\therefore$ Dual of $A \subseteq B$ is $B \subseteq A$

Duality principle should be applied only for general can but no for particular cases

3.1.8 Cartesian Product:

$A \times B = \{(a, b) | a \in A \wedge \text{be } B\}$

- Cartesian product is associative

  However

$$\underset{\text{it has ordered pairs t type } ((a,b),c)}{(A \times B) \times C} \neq \underset{(a,(b,c))}{A \times (B \times C)}$$

  $A \times B \neq B \times A$

  If $A \times B = B \times A$ then $(A = B \vee (A = \phi) \vee (B = \phi)$

- $A \times (B \cap C) = (A \times B) \cap (A \times C)$

  $A \times (B \cup C) = (A \times B) \cup (A \times C)$

  $\bar{A} \times \bar{B} \neq \overline{A \times B}$

$\bar{A} \times \bar{B} \leq \overline{A \times B}$

3.1.9 Multisets:

Multiset is an unordered collection of elements where on element can occur more than once.

Eg: $\{m_1.a_1, m_2.a_2 \text{ --- } m_n a_n\}$

Where m_1 is called multiplicity of set

3.1.10 Operations on multisets:

- Union: Maximum of multiplicities is considered.
- Intersection: Minimum of multiplicities is considered
- Difference: Difference of multiplicities is considered.

  If the difference is -ve, it is considered 0.
- Sum: Sum of multiplicities is considered.

  It is denoted as P + Q

3.2 Relations:

A binary relation from set A to set B is any subset of A × B.

If |A| = M and |B| = n, then |A × B| = mn

∴ number of relations possible from A to B = 2^{mn}

If relation B from A to A we say it relation on A.

∴ Number of relations possible on a set A = 2^{n^2}, |A| = n

Domain of relation : $\{ x \mid (x, y) \in R\}$

Range of relation : $\{y \mid (x, y) \in R\}$

- $(x, y) \in R$ is written as 'xRy' on 'k' relates y

3.2.1 Inverse of relations:

Inverse of a relation, $R^{-1} = \{(y, x) | (x, y) \in R\}$

3.2.2 Diagonal relation:

Diagonal relation on set A is

$\Delta_A = \{(a,a) \mid \forall a \in A\}$

3.2.3 Reflexive relations:

Relation R defined on set A is reflexive

$\Leftrightarrow aRa \; \forall a \in A$

i.e., if R is a reflexive relation then $\Delta_A \leq R$

3.2.4 Important type of relations:

Type of relation	Condition	No. of relatives possible	Union	Intersection
Reflective	aRa, $\forall a \in A$	2^{n^2-n}	✓	✓
Irreflective	$\forall a \in A$ (a $\not R$ a)	2^{n^2-n}	✓	✓
Symmetric	$\forall x, y \in A(xR_y \Rightarrow yR_x)$	$2^n.2^{\frac{n^2-n}{2}}$	✓	✓
Antisymmetric	$\forall x, \forall y \; (xR_y \wedge yR_x \Rightarrow x = y)$	$2^n 3^{\frac{n^2-n}{2}}$	×	✓
Asymmetric	$\forall x, \forall y \; (xR_y \Rightarrow y \not R_x)$	$3^{\frac{n^2-n}{2}}$	×	✓
Transitive	$\forall x, \forall y \; (\forall z(xR_y \wedge yR_z \Rightarrow xR_z)$	—	×	✓

- Note that all the above relations are defined on a single set.
- Also to prove any relation is not of certain type we need to P.T the logical formula for that relation is false.
- Every Asymmetric relation is antisymmetric relation.

3.2.5 Composition of relations:

If R is a relation from A to B, S is a relation from B to C, the composition of R & S is given by

SoR from A to C.

S.R = {(a, c)|(a, b)∈R and (b, c)∈S}

If R is relation on A, then composition is denoted as R^2, R^3.

Note:

If R is any relation on set A, then

$R_o\Delta_A = R$ and Δ_A oR = R

3.2.6 Closure:

Clause of a relation R under given property is smallest possible relation that contains R and Satisfy the prpperty.

(i) **Reflexive Closure (R*):**

Reflexive clouse of a relation R,

$R^{\#} = RU\Delta_A$

(ii) **Symmetric closure:**

$R^+ = RUR^{-1}$

(iii) **Transitive closure:**

- Finding transitive closure has no formula, but we have a procedure as shown below.
  Steps: Represent relation R with a directed graph such that whenever aRb draw a directed edge from a to b.

  Steps: Now from each vertex, find all reachable vertices and for each reachable vertex b from a add the ordered pair (a, b) to the closure.

- The standard procedure for finding transitive closes is
  $R\# = R \cup R^2 . \cup R^3 \cup \ldots\ldots . R^{n-1} \cup R^n$
  'n' is a positive integer such that $R^{n-1} = R^n$
- when a relation is represent as a graph,
  If (a, b) $\in R^n$, we can say that there exists an n-length path from a to b.
- when asked to find more than one closure, the order to be followed is reflexive, symmetric, transitive.
  Following any other order may give redundancy.
- The closure of a relation, if exists, under a property is intersection of the relations with that property containing R.
- If R is a transitive relation,
- $Rn \leq R \ \forall n \geq 1$
- also R^n is transitive relation.

3.2.7 Partion:

A partition of set s is dividing s in disjoint subsets such that

$A_1 \cup A_2 \cup \ldots \cup A_n = s$

$A_i \cap A_j = \phi \ \forall i, j \leq n$

3.2.8 Refinement:

Partition P_2 is called refinement of partition p_1.

if every partitioned subset of P_2 is subset to some partitioned subset of P_1.

3.2.9 Equivalence Relation:

A relation which is reflexive

Symmetric

Transitive

is called an equivalence relation.

- Equivalence relation creates partition.
- Each set of the partition is called an equivalence class.
- Every element of same equivalence class are related to each other.
- No two element of different classes of related to each other

- Equivalence class is represented by
  [a] R

Where a is any element of the equivalence classes

- It R is an equivalence relation, the below two sentences means the same
  (i) aRb
  (ii) $[a]_R = [b]_R$

  aRb & $[a]R \phi [b]$ also mean same.

  If R is an equivalence relation.

  aRb is read as "a is equivalent to b".

- If R_1, R_2 are two equivalence relations
  $R_1 \cup R_2$ need not to be an equivalence relation

  $R_1 \cup R_2$ is an equivalence relation.

3.2.10 Finding number of equivalence relations:

- No of equivalence relations on a set with n elements is given by

  Bell number $B_n = \sum_{k=1}^{n} s(n,k)$

  Where sterling 2nd kind number

  $$s(m,n) = \frac{1}{n!}\sum_{i=0}^{n}(-1)^i \, nc_i \, (n-i)^m$$

  The below is a shortcut to find the Bell number

  $B_0 \rightarrow (1)$

  $B_1 \rightarrow (1)$ 2

  $B_2 \rightarrow (2)$ 3 5

  $B_3 \rightarrow (5)$ 7 10 15

  $B_4 \rightarrow (15)$ 20 27 37 52

  Find no. of equivalence relations is nothing but find no. of way we can partition the set.

3.3 Partial Ordering Relations:

A relation R on a set A is called a partial order if R is reflexive, antisymmetric and transitive.

3.3.1 Poset:

A set 'A' together with a partial order R is called a poset

It is denoted as [A:R]. In a poset aRb is denoted as $a \leq b$. $a < b$ means $a \leq b$ and $a \neq b$.

3.3.2 To set:

A post [A;R] is called toset it every pair of element of set A are comparable.

Toset is also known as linearly ordered set (or) chain.

If [A, R] is a poset, then

$[A ; R^{-1}]$ is called dual of the poset.

i.e., these diagram of $[A : R^{-1}]$ can be obtained by turning the hasse diagram of [A, R] upside down.

3.3.3 Hasse Diagram (Poset Diagram):

- It is graphical representation of a poset.
  It is constructed as below:

  (i) Create vertex corresponding to every element of set A.
  (ii) All loops & Edges implied from transitivity are not shown.

- Let $[A, \leq]$ be a poset
  We say yes, covers $x \in s$, if $x < y$ such that there doesn't exist any $z \in s$, $x < z < y$
- Thus hasse diagram shows edges b/w two elements x & y only if x covers y (or) y covers x. The set of such pairs $x < y$ is called covering relation of $(s, \leq)$.
- Thus applying reflexive transitive closure on covering relation of a poset gives the poset.
- Hasse Diagram of a toset is a chain.

3.3.4 Maximum Element:

In a poset an element is called maximal if it is not related to any other element.

3.3.5 Greatest element (or) Maximum element:

In a poset an element is called greatest if every element of the set relates to that element.

3.3.6 Minimal element:

In a poset an element is called minimal, if no other element is related to it.

3.3.7 Least element (or) Minimum Element:

In a poset least element is the one which relates to every other element of the set.

- Every finite, nonempty lattice has at least one minimal and one maximal element.
- Greatest or least element if exists is unique.
- Greatest and least elements may or may not exist if they exist they are the only maximal and only minimal elements respectively.

3.3.8 Upper Bound:

If $[A : R]$ is a poset and $B \leq A$

Upper band of $B = \{x | \forall b \in B, x \leq b \text{ and } x \in A\}$

3.3.9 Least Upper Bound (lub or join or Supremum):

In a poset [A : R] lub of two elemis a, b $\in$ A is least element of upper bound of {a, b}. It is denoted as a$\vee$b.

i.e., a$\vee$b $\leq$ x $\forall$x$\in$ upper band of [a, b]

- If no least element exists in the upper bound, we say lub doesn't exist.
- In other word,

  If a$\vee$b = c then

  aRc & bRc and

  If aRd $\wedge$ bRd $\Rightarrow$ cRd

3.3.10 Greatest lower Bound (glb or meet or infimum)

In a poset [A, R] glb of a, b is greatest element of lower bound of {a, b}. It is denoted as a $\wedge$ b

- If no such element exists we say glb doesn't exist.
- Other way to define glb is

  If a $\wedge$ b = c, then

  [eRa $\wedge$ cRb] $\wedge$ [(dRa $\wedge$ dRb) $\Rightarrow$ dRc]

- For poset [D_n ; 1]

  Lub (a, b) = LCM (a, b)

  Glb (a, b) = GCD (a, b)

- For poset [P(A) ; $\leq$], where P(A) is power set of A

  lub (x, y) = x$\cup$y

  lub (x, y) = x$\cap$y

- The hasse diagram of [P(A); $\leq$] forms a hypercube on, where |A| = n

3.3.11 Join Semi Lattice:

It is a poset in which every pair of elements has lub.

3.3.12 Meet Semi Lattice:

It is a poset in which every pair of elements has glb.

3.4 Lattice

A lattice is a poset in which every pair of elements has both glb & lub.

i.e., Lattice is both join semi lattice & meet semi lattice.

A lattice L is denoted as (L, $\wedge$, $\vee$)

- It is not needed that every lattice has greatest and least element.
- Every finite lattice has least and greatest elements.

3.4.1 Properties of Lattice:

- $\left.\begin{matrix} a \wedge a = a \\ a \vee a = a \end{matrix}\right\}$ Idempotent
- $\left.\begin{matrix} a \vee b = a \vee a \\ a \wedge v = a \wedge a \end{matrix}\right\}$ Commutative
- $\left.\begin{matrix} a \vee (b \vee c) = (a \vee b) \vee c \\ a \wedge (b \wedge c) = (a \wedge b) \wedge c \end{matrix}\right\}$ Associative
- $\left.\begin{matrix} a \vee (a \wedge b) = a \\ a \wedge (a \vee b) = a \end{matrix}\right\}$ Absorption law
- $a \le b \Rightarrow a \vee c \le b \vee c$
  $a \le b \Rightarrow a \wedge c \le b \wedge c$

  If a ≤ b and c ≤ d then

  a ∨ c ≤ b ∨ d

  a ∧ c ≤ b ∧ d

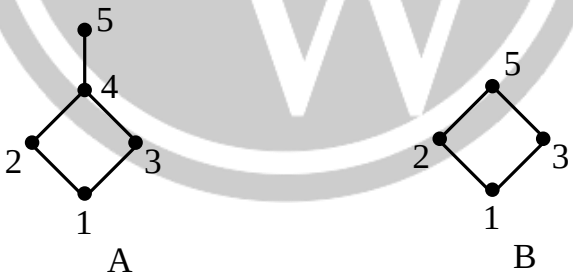
3.4.2 Sublattice:

If A is a lattice B is called

Sublattice of A iff

(i) B itself is a lattice.

(ii) lub of any two element of B is same as lub of the two elements in A.



In above figure B is subset of A and B is a lattice still B is not a sublattice of A.

Because $\left.\begin{matrix} in\ A\ glb\ (2,3) = 4 \\ in\ \mathrm{B}\ glb\ (2,3) = 5 \end{matrix}\right]$ not equal $\therefore$ not sublatice

3.4.3 Types of Lattices:

(1) Bounded Lattice:

- A lattice with greatest element and least element is called bounded lattice.
- Every finite lattice is bounded lattice.

(2) Complemented Lattice:

- Complemented lattice is a bounded in which every element has atleast one complement.
  B is said to be complement of a iff.
- glb (a, b) = greatest element and lub (a, b) = least element.

(3) Distributive Lattice:

- A lattice is said to be distributive if following distributive laws hold.
- $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.
- $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$.

Note: If L is a bounded distributive lattice then complement of an element if exists, is unique. The reverse need not to be true.

Hence if we find more than one complement for an element, we can conclude that the lattice is not distributive

(4) Boolean Algebra:

- A lattice which is both distributive and complemented is known as Boolean algebra
- It is called so because it satisfises all the properties of Boolean algebra. Thus when given lattice is a Boolean algebra we can apply all the rules that we apply in logic.
- In Boolean algebra every element has exactly one complement.
- [Dn : 1] is a distributive lattice for any n.
- It is because distributive properties hold for lcm & gcd.
- [Dn : 1] is a Boolean lattice if n is a square free number.
- [P(s); ≤] is a Boolean lattice.
- Every Boolean lattice with 2n elements, $\forall \cap \geq 0$.
  Every Boolean lattice contains 2^nelements is isomorphic to the lattice (P(s); ≤) where s is a set with n elements.
  Also this Boolean lattice is a hypercube Q_n.
- Sublattice of a complemented lattice need not to be a complemented lattice.
- Sublattice of a bounded lattice is a bounded lattice.

3.5 Function (or) Mapping (or) Transformation

A function F from set A to set B is an assignment of exactly in one element of B to each element of A. It is denoted as: $F : A \rightarrow B$

Here A is called Domain B is called codomain

Function is a special type of relation.

- Consider f(a) = b
  b is called image of a
  a is called preimage of b.

- Range: It is set of images of all the elements of A.
  Range $\neq$ Co-domain (Range need not to be equal to co-domain)
- If f_1, f_2 are two functions
  $f_1(x) + f_2(x)$ is denoted as $(f_1 + f_2)(x)$
  $f_1(x).\ f_2(x)$ is denoted as $(f_1.f_2)(x)$
  $\frac{1}{f(x)}$ is denoted as $\frac{1}{f}(x)$, ($\frac{1}{f}$ is not equal to inverse function)
- If A and B are two sets with
  $|A| = n$ $|B| = m$ then
  number of functions possible from A to B = m^n

3.6 Types of functions:

3.6.1 One - One function (Injection):

$f : A \to B$ is one to one
iff
$\forall a \forall b(f(a) = f(b) \Rightarrow a = b)$

- It $f : A \to B$ is 1–1 then $|A| \leq |B|$
- If $|A| = m$ and $|B| = n$
  no of 1 – 1 function possible from A to B = ${}_nP_m$
- If m = n = k, then number of 1:1 function = k!

3.6.2 Onto function (Surjection):

$f : A \to B$ is onto iff
$\forall b \in B \quad \exists a \in A$ such that $f(a) = b$

- If $f : A \to B$ is onto then $|A| \geq |B|$
- If $|A| = m$ $|B| = n$ then

$$\therefore \text{ number of onto functions} = \sum_{i=0}^{n} (-1)^i \, nc_i \, (n-i)^m$$

- If m = n = k, then number of 1:1 function = k!

3.6.3 One-to-One Correspondence (or) Bijection:

A function $f : A \to B$ is Bijection iff
f is both one-one and onto

- If $f : A \to B$ is bijection, then $|A| = |B|$
- If $f : A \to B$ is 1 – 1 and $|A| = |B|$
  then f is a bijection
- If $f : A \to B$ is onto and $|A| = |B|$
  then f is a bijection

- If |A| = |B| = n then no of bijections possible from A to B are n!
- If f : A → A is a function and A is a finite set then
  A is 1–1 ⇔ A is onto

3.6.4 Inverse of a function:

f : A → B is invertible if its inverse relation f^{i} is a function from B to A.

f : A → B is invertible ⇔ f is a bijection

- Inverse doesn't exist if a function is not bijection. However, we can find inverse image of subset of codomain. If f : A → B is a function and S ≤ B

Inverse image of S = {a∈A|f(a) ⊆ S}

- If f is a function from A to B and let S, T be subsets of A.

  $f(S \cup T) = f(S) \cup f(T)$

  $f(S \cap T) = f(S) \cap f(T)$

  However, $f(S \cap T) = f(S) \cap f(T)$ if f is a bijection

- If f is a function from A to B and let S, T be subsets of B

  $f^{-1}(S \cup T) = f^{-1}(s) \cup f^{-1}(T)$

  $f^{-1}(S \cap T) = f^{-1}(s) \cap f^{-1}(T)$

  $f^{-1}(\bar{s}) = \overline{f^{-1}(s)}$

3.6.5 Identity function:

A mapping I_A : A → A is called an identity function if

I_A = {(x, x)/x∈A}

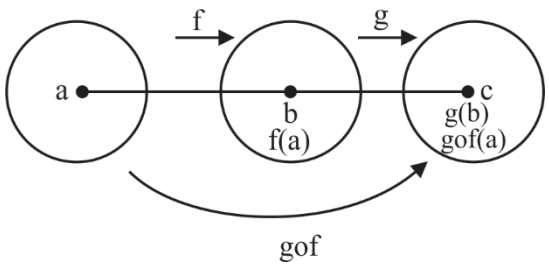
3.6.7 Constant function:

A function f : A → B is said to be a constant function if

f(x) = c ∀x ∈A

3.6.8 Composition of functions:

If f : A → B and g:B → C are two function then gof: A → C is called a composition function of f & g.

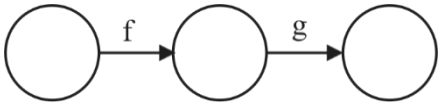


gof(x) = g(f(x))

- In gof : A → C
  A is domain of gof
  C is codomain of gof

Range of gof is image of (range of f under g)

- If $f : A \to B$ and $g : B \to C$ are tow functions
  gof is defined for every case



But fog is defined iff

range of $g \leq$ domain of f

fog maps from B to B i.e., $fog : B \to B$

$fog \neq gof$ (i.e., composition is not commutative)

(fog)oh = fo(goh) (i.e., Associative)

$(fog)^{-1} = g^{-1} of^{-1}$ (Think why)

Let $f : A \to B$ be an invertible function then

$f^{-1} : B \to A$ is a inverse of f

$f^{-1}of : A \to A$ is an identity function I_A

$fof^{-1} : B \to B$ is an identity function I_B

Note:

- f is 1–1 & g is 1–1 $\Rightarrow$ gof is 1–1
- f is onto & g is onto $\Rightarrow$ gof is onto
- f is bijection & g is bijection $\Rightarrow$ gof is bijection
- If gof is onto then g is onto
- If gof is 1–1 then f is 1–1
- If gof is a bijection then f is onto $\Leftrightarrow$ g is 1–1

3.6.9 Partial Functions:

$f: A \to B$ is called partial function if 'f' is defined only for some of $a \in A$.
The subset of A on which f is defined is called domain definition of f.

3.7 Groups:

3.7.1 Binary Operation:

The binary operator * is said to be a binary operation on a non-empty set A if the set is closed under the operation.
i.e., $(a*b) \in A \ \forall a,b \in A$

3.7.2 Binary Structure (or) Algebraic Structure:

A nonempty set A is called binary structure with respect is a binary operator *, if * is binary operation on A. it is demoted as (A, *)

3.7.3 Semi Group:

(A, *) is semigroup iff
(i) is closed operation
(ii) is an associative property

3.7.4 Monoid:

(A, *) is called monoid iff:
(i) is closed operation
(ii) is an associative property
(iii) Identity element exists

3.7.5 Group:

(A, *) is called monoid iff:
(i) is closed operation
(ii) is an associative property
(iii) Identity element exists in A
(iv) Every element of A has inverse

- Identity element if exists is unique.
- Inverse element if exist is unique.
- Finite Group:
  A group with finite number of elements
- Order of a group: It is number of elements in the group.
- In a group of 2 element
  $a^{-1} = a, \forall a \in G$
- $(\{0,1,2,3 \ldots\ldots m-1\}, \oplus_m)$ is a group
- Let S_n be set of positive integers less than n and relatively prime to n. then
  $(S_n, \otimes_n)$ is a group.
- Abelian Group:
  A group (G,*) is called abelian if * is commutative.

3.7.6 Properties of Groups

- Let (G,*) be a group
  if $ab = ac \Rightarrow b = c$ (Left cancellation property)
  $ba = ca \Rightarrow b = c$ (Right cancellation property)
  due to these property ever raw and column in Caley table has exactly one element
- If G is a group and $a,b \in G$
  then $(ab)^{-1} = b^{-1}a^{-1}$
- Group G is abelian $\Leftrightarrow (ab)^{-1} = b^{-1}a^{-1} \ \forall a, b \in G$
- $\forall a \in G, a^{\circ} = e$

3.7.7 Homomorphism

- If (G, *) and (G', $\oplus$) are two groups then a function $f : G \to G'$ is called a homomorphism
  if $f(a*b) = f(a) \oplus f(b)$
- If $f : G \to G'$ is a bijection, then we call the homomorphism isomorphism. It is denote as $G \cong G'$
- If e_1, e_2 are identity elements of G and G' respectively then $f(e_1) = e_2$
- If $a \in G$, and $f : G \to G'$ is a homomorphism $f(a^{-1}) = (f(a))^{-1}$

3.7.8 Order of an element:

The smallest positive integer n such that $a^n = e$ is called of order of an element a in the group.

- The order of an element is divisor of order of the group.
- Order (a) = order (a^{-1}) $\forall a \in G$
- $a^{-n} = b^n$ if $a^{-1} = b$ and n is any positive integer.

3.7.9 Subgroup:

A non-empty subset H of a group G is called subgroup of G if (H, *) is also a group.

- For every group G, {e} and G are called trivial subgroups of G.
- Every subgroup contains identity element of its parent group.
- If H is a subgroup of G then order(H) divides order(G) (Lagrange's theorem). The converse of Lagrange's theorem holds only for abelian groups.
- If H and k are two subgroups of same group, then
  $H \cap k$ is also, a subgroup
  $H \cup k$ need not to be a subgroup.
- If $H \subseteq G$, then H is called subgroup of G iff:
  (i) $(a*b) \in H \quad \forall a, b \in H$
  (ii) $a^{-1} \in H \qquad \forall a, b \in H$
- If $H \subseteq G$, and if H is finite $\rightarrow$ (applies only when H is finite) and nonempty, then H is called subgroup iff:
  $(a*b) \in H \quad \forall a, b \in H$
- If G is a subgroup of composite order, then G necessarily has non-trivial subgroup.

3.7.10 Cyclic Groups:

A group G is called cyclic, if $\exists a \in G$ such that every element can be written as an integral power of a such an element 'a' is called generator.

- The order of generator is order of the group.
- If 'a' is a generator then a^{-1} is also a generator.
- All subgroups of a cyclic group are cyclic.
- If G is a cyclic group of order n, then no of generator of G is given by $\phi(n)$ (Euler's phi function)
- If 'a' is a generator of G, and let m be an integer such that $1 < m \leq n$ then

$$\text{order}(a^m) = \frac{n}{\gcd(n,m)}$$

- If G is a cyclic group with generator a and let d be divisor of |G|.
  For every d there exists exactly one subgroup of order d. this subgroup is generated by $a^{n/d}$ where n = |G|

$\therefore$ No of subgroup of a cyclic group = no. of divisors of |G| each subgroup is generated by $a^{n/d}$. (d is divisor of |G|)

3.7.11 Cyclic subgroup of a group:

If (G,*) is any group and let $a \in G$.
The smallest subgroup of G containing a is <a> where
$\langle a \rangle = \{a^n / \forall n \in Z\}$
Also <a> is a cyclic subgroup.
order of the subgroup <a> = order of element 'a' in G.

- If H is any subgroup of G and if H contains 'a' then
  $\langle a \rangle \subseteq H$
- Thus if G is a group
  $\langle a \rangle$ is a cyclic subgroup of G, $\forall a \in G$.
  Such subgroups are called generating sets

Note:

- Every group of prime order is cyclic in which every element (except e) is a generator element.
- Every cyclic group is abelian.
- A group is cyclic $\Leftrightarrow$ it can't be expressed as union of two proper subgroups.
- If (G,*) and (H,$\oplus$) are two groups
  $(G \times H, \bullet)$ is a group where $\bullet$ is defined as
  $(g_1, h_1) \bullet (g_2, h_2) = (g_1 * g_2, h_1 \oplus h_2)$

This group is called direct product of G and H.

- Every group of order less than or equal to 5 is abelian.
- There is only one unique group for each of the order 1,2,3.

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4 COMBINATORICS

4.1 Introduction

Let $m \in \mathbb{N}$. For a procedure of m successive distinct and independent steps with n1 outcomes possible for the first step, n_2 outcomes possible for the second step, ... , and nm outcomes possible for the mth step, the total number of possible outcomes is

$$n_1 \cdot n_2 \cdot \cdot \cdot n_m$$

For a collection of m disjoint sets with n_1 elements in the first, n_2 elements in the second, . . ., and nm elements in the mth, the number of ways to choose one element from the collection is

$$n_1 + n_2 + \cdot \cdot \cdot + n_m$$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{n-k} x^k y^{n-k}.$$

$$\sum_{i=0}^{n} C(n,i)(-1)^i = 0$$

For n even,

$$\sum_{i=0}^{n/2} C(n,2i) = \sum_{i=0}^{n/2-1} C(n,2i+1)$$

For n odd,

$$\sum_{i=0}^{\lfloor n/2 \rfloor} C(n,2i) = \sum_{i=0}^{\lfloor n/2 \rfloor+1} C(n,2i+1)$$

$$\sum_{i=1}^{n} iC(n,i) = n2^{n-1}.$$

For positive integers n, t, the coefficient of $x_1^{n_1} x_2^{n_2} x_3^{n_3} \ldots x_t^{n_t}$ in the expansion of $(x_1 + x_2 + x_3 + \ldots + x_t)^n$ is

$$\frac{n!}{n_1!n_2!n_3!\ldots n_t!},$$

where each n_i is an integer with $0 \le n_i \le n$, for all $1 \le i \le t$, and $n_1 + n_2 + n_3 + \cdots + n_t = n$.

When we wish to select, with repetition, r of n distinct objects, we are considering all arrangements of r x's and n – 1 |'s and that their number is

$$\frac{(n+r-1)!}{r!(n-1)!}=\binom{n+r-1}{r}.$$

Consequently, the number of combinations of n objects taken r at a time, with repetition, is C(n + r – 1, r).

4.1.1 we recognize the equivalence of the following:

(a) The number of integer solutions of the equation

$x_1 + x_2 + \ldots + x_n = r,\ x_i \geq 0,\ 1 \leq i \leq n.$

(b) The number of selections, with repetition, of size r from a collection of size n.

(c) The number of ways r identical objects can be distributed among n distinct containers.

$$b_n=\binom{2n}{n}-\binom{2n}{n-1}=\frac{1}{n+1}\binom{2n}{n}, \qquad n \geq 1, b_0 = 1.$$

The numbers $b_0, b_1, b_2, \ldots$ are called the Catalan numbers, after the Belgian mathematician

Order Is Relevant	Repetitions Are Allowed	Type of Result	Formula
Yes	No	Permutation	$P(n, r) = n!/(n – r)!,\ 0 \leq r \leq n$
Yes	Yes	Arrangement	$N^r,\ n, r \geq 0$
No	No	Combination	$C(n, r) = n!/[r!(n – r)!] = \binom{n}{r},\ 0 \leq r \leq n$
No	Yes	Combination with repetition	$\binom{n+r-1}{r},\ n, r \geq 0$

If there are n objects with n_1 indistinguishable objects of a first type, n_2 indistinguishable objects of a second type, . . . and n_r indistinguishable objects of an rth type, where $n_1 + n_2 + \ldots + n_r = n$, then there are $\frac{n!}{n_1!n_2!\ldots n_r!}$ (linear) arrangements of the given n objects.

Let A and B be subsets of a finite universal set U. Then

(a) $|A \cup B| = |A| + |B| - |A \cap B|$

(b) $|A \cap B| \leq \min\{|A|, |B|\}$, the minimum of |A| and |B|

(c) $|A \backslash B| = |A| - |A \cap B| \geq |A| - |B|$

(d) $|A^c| = |U| - |A|$

(e) $|A \oplus B| = |A \cup B| - |A \cap B| = |A| + |B| - 2|A \cap B| = |A / B| + |B / A|$

(f) $|A \times B| = |A| - |B|$

Given a finite number of finite sets, $A_1, A_2, \ldots, A_n$, the number of elements in the union $A_1 \cup A_2 \cup \ldots \cup A_n$ is

$$|A_1 \cup A_2 \cup \ldots \cup A_n| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \ldots + (-1)^{n+1} |A_1 \cap A_2 \cap \ldots \cap A_n|,$$

where the first sum is over all i, the second sum is over all pairs i, j with i < j, the third sum is over all triples i, j, k with i < j < k, and so forth.

The number of derangements of $n \geq 1$ ordered symbol is

$$\boxed{D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots + (-1)^n \frac{1}{n!}\right).}$$

Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers. The function

$$f(x) = a_0 + a_1x + a_2x^2 + \ldots = \sum_{i=0}^{\infty} a_i x^i$$

is called the generating function for the given sequence.

4.1.2 Extended Binomial coefficient

$$\binom{-n}{r} = \frac{(-n)(-n-1)(-n-2)\ldots(-n-r+1)}{r!}$$

$$= \frac{(-1)^r (n)(n+1)(n+2)\ldots(n+r-1)}{r!}$$

$$= \frac{(-1)^n (n+r-1)!}{(n-1)!r!} = (-1)^r \binom{n+r-1}{r}.$$

For all m, n ∈ Z^+, a ∈ R,

(1) $(1 + x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \ldots + \binom{n}{n}x^n$

(2) $(1 + ax)^n = \binom{n}{0} + \binom{n}{1}ax + \binom{n}{2}a^2x^2 + \ldots + \binom{n}{n}a^nx^n$

(3) $(1 + x^m)^n = \binom{n}{0} + \binom{n}{1}x^m + \binom{n}{2}x^{2m} + \ldots + \binom{n}{n}x^{nm}$

(4) $(1 - x^{n+1})/(1 - x) = 1 + x + x^2 + \ldots. + x^n$

(5) $1/(1 - x) = 1 + x + x^2 + x^3 + \ldots = \sum_{i=0}^{\infty} x^i$

(6) $1/(1 - ax) = 1 + (ax) + (ax)^2 + (ax)^3 + \ldots$

$= \sum_{i=0}^{\infty} (ax)^i = \sum_{i=0}^{\infty} a^i x^i$

$= 1 + ax + a^2x^2 + a^3x^3 + \ldots$

(7) $1/(1+x)^n = \binom{-n}{0} + \binom{-n}{1}x + \binom{-n}{2}x^2 + \ldots$

$$= \sum_{i=0}^{\infty}\binom{-n}{i}x^i$$

$$= 1 + (-1)\binom{n+1-1}{1}x + (-1)^2\binom{n+2-1}{2}x^2 + \ldots$$

$$= \sum_{i=0}^{\infty}(-1)^i\binom{n+i-1}{i}x^i$$

(8) $1/(1-x)^n = \binom{-n}{0} + \binom{-n}{1}(-x) + \binom{-n}{2}(-x)^2 + \ldots$

$$= \sum_{i=0}^{\infty}\binom{-n}{i}(-x)^i$$

$$= 1 + (-1)\binom{n+1-1}{1}(-x) + (-1)^2\binom{n+2-1}{2}(-x)^2 + \ldots$$

$$= \sum_{i=0}^{\infty}\binom{n+i-1}{i}x^i$$

If $f(x) = \sum_{i=0}^{\infty}a_ix^i, g(x) = \sum_{i=0}^{\infty}b_ix^i$, and $h(x) = f(x)g(x)$, then

$h(x) = \sum_{i=0}^{\infty}c_ix^i$, where for all $k \geq 0$,

$$c_k = a_0b_x + a_1b_{x-1} + \ldots + a_{k-1}b_1 + a_kb_0 = \sum_{j=0}^{k}a_jb_{k-j}.$$

Objects Are Distinct	Containers Are Distinct	Some Container(s) May Be Empty	Number of Distributions
No	Yes	Yes	$\binom{n+m-1}{m}$
No	No	Yes	(1) p(m), for n = m (2) p(m, 1) + p(m, 2) + … + p(m, n), for n < m
No	Yes	No	$\binom{n+(m-n)-1}{(m-n)} = \binom{m-1}{m-n} = \binom{m-1}{n-1}$
No	No	No	p(m, n)

Consider the nonhomogeneous first-order relation

$$a_n + C_1a_{n-1} = kr^n,$$

where k is a constant and $n \in Z^+$. If r^n is not a solution of the associated homogeneous relation

$$a_n + C1a_{n-1} = 0,$$

then $a_n^{(p)} = Ar^n$, where A is a constant. When r^n is a solution of the associated homogeneous relation, then $a_n^{(p)} = Bnr^n$, for B a constant.

Now consider the case of the nonhomogeneous second-order relation

$$a_n + C_1a_{n-1} + C_2a_{n-2} = kr^n,$$

Where k is a constant. Here we find that

(a) $a_n^{(p)} = Ar^n$, for A a constant, if r^n is not a solution of the associated homogeneous relation;

(b) $a_n^{(p)} = Bnr^n$, where B is a constant, if $a_n^{(h)} = c_1r^n + c_2r_n^n$, where $r_1 \neq r$; and

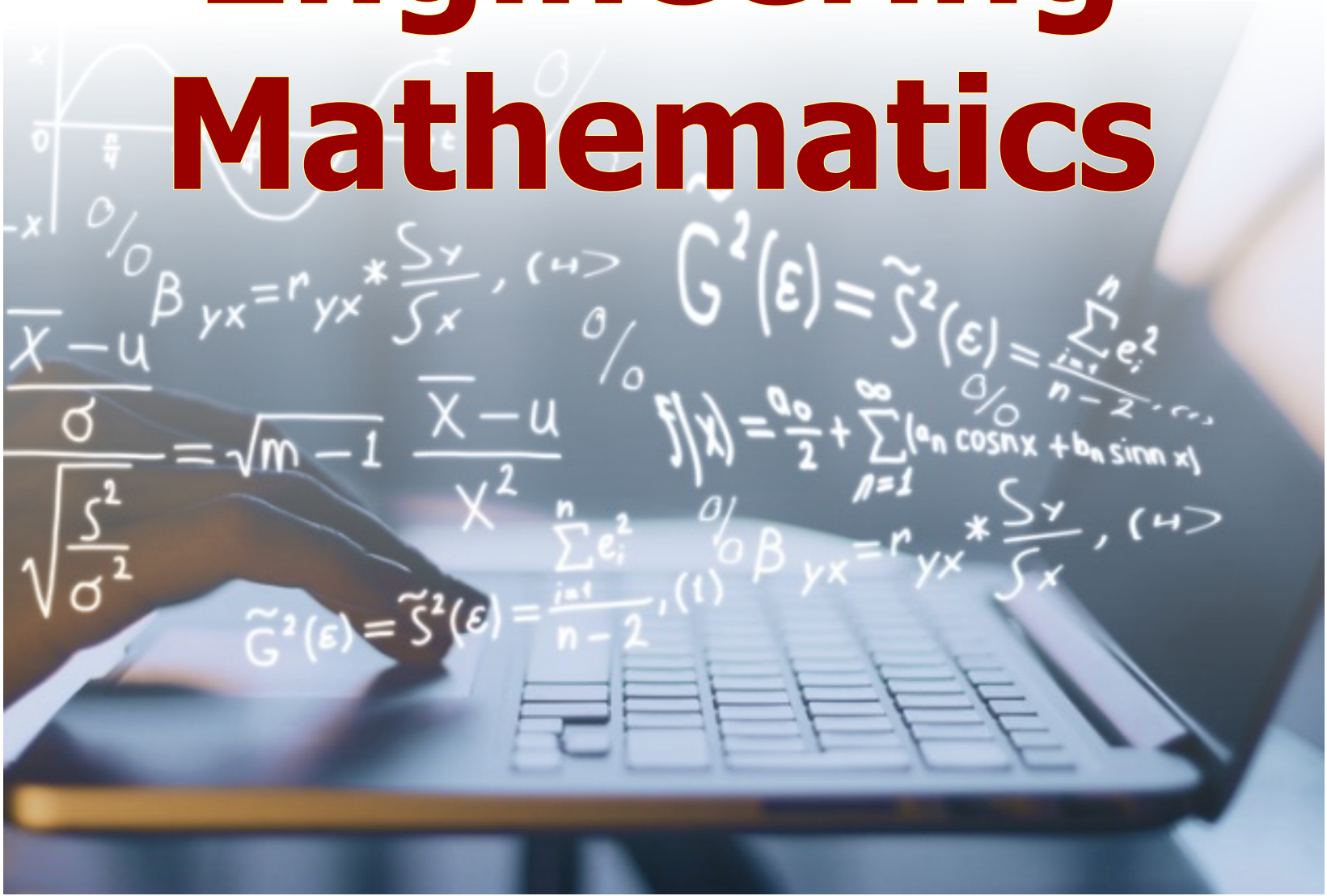
(c) $a_n^{(p)} = Cn^2r^n$, for C a constant, when $a_n^{(h)} = (c_1 + c_2n)r^n$.

Given a linear nonhomogeneous recurrence relation (with constant coefficients) of the form $C_0a_n + C_1a_{n-1} + C_2a_{n-2} + \ldots + C_ka_{n-k} = f(n)$, where $C_0 \neq 0$ and $C_k \neq 0$, let $a_n^{(h)}$ denote the homogeneous part of the solution a_n.

	$a_n^{(p)}$
c, a constant	A, a constant
n	$A_1n + A_0$
n^2	$A_2n^2 + A_1n + A_0$
n^t, $t \in Z^+$	$A_tn^t + A_{t-1}n^{t-1} + \ldots. + A_1n + A_0$
r^n, $r \in R$	Ar^n
sin θn	A sin θn + B cos θn
cos θn	A sin θn + B cos θn
$n^t r^n$	$r^n (A_tn^t + A_{t-1}n^{t-1} + \ldots + A_1n + A_0)$
r^n sin θn	Ar^n sin θn + Br^n cos θn
r^n cos θn	Ar^n sin θn + Br^n cos θn

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Engineering Mathematics



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Engineering Mathematics

INDEX

1. Basic Calculus ..... 1.1 – 1.16

2. Linear Algebra ..... 1.17 – 1.28

3. Probability and Statistics ..... 1.29 – 1.40

1 BASIC CALCULUS

1.1. Introduction

1.1.1 Limits, Continuity and Differentiability

(a) As x tends to a ($x \to a$) $\Rightarrow$ x is moving towards a

A value l is said to be limit of a function $f(x)$ at $x \to a$ if $f(x) \to l$ as $x \to a$.

It is mathematically defined as

$$\lim_{x \to a} f(x) = l = \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$$

That is, Limit exist at any point, if LHL = RHL

A function f(x) is said to be continuous at x = a, if

$$\lim_{x \to a} f(x) = l = f(a) = f(x)|_{x=a}$$

That is, for a function to be continuous at any point, RHL = LHL = Value of function at point $x = a$.

Note:
- For $\lim_{x \to a} f(x)$ to exist, the function need not be continuous at $x = a$.
- But for $f(x)$ to be continuous at x = a, $\lim_{x \to a} f(x)$ should exist.

- Continuity from Left : $\lim_{x \to a^-} f(x) = f(a)$
- Continuity from Right : If $\lim_{x \to a^+} f(x) = f(a)$

Continuity in an Open Interval

A function 'f' is said to be continuous in open interval (a, b), if it is continuous at each point of open interval.

Continuity in a Closed Interval

Let 'f' be a function defined on the closed interval (a, b) then 'f' is said to be continuous on the closed interval [a, b], if it is :

1. Continuous from the right at a and
2. Continuous from the left at b and
3. Continuous on the open interval (a, b).

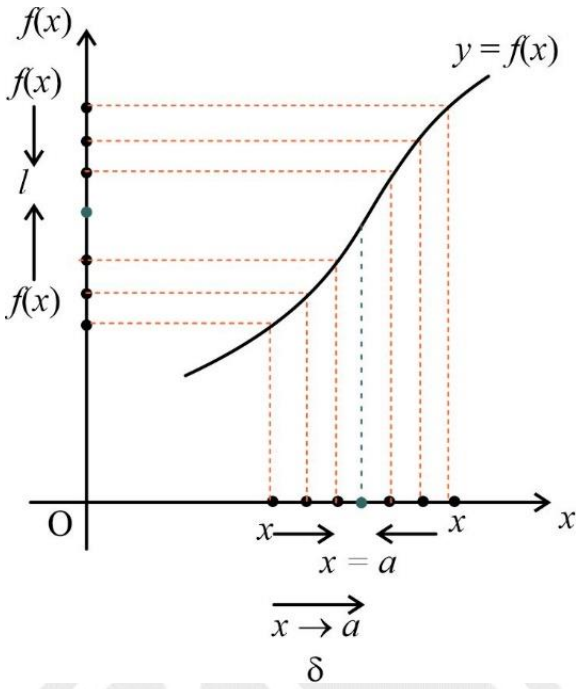


Fig. 1.1

(b) Concept of differentiability

A continuous function $f(x)$ is said to be differentiable at $x = a$, if $\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$ exists, that is, RHL and LHL exist at a point under consideration in $f'(x)$.

$f'(x)|_{x=a} = f'(a) = \lim_{x\to a}\frac{f(x)-f(a)}{x-a}$

$f'(a) = \tan\theta$, where θ is the angle made by the tangent to the curve at x=a with x – axis.

(c) Some Standard Derivatives

(i) $\frac{d}{dx}(x^n) = n.x^{n-1}$

(ii) $\frac{d}{dx}(\sin x) = \cos x$

(iii) $\frac{d}{dx}(\cos x) = -\sin x$

(iv) $\frac{d}{dx}(\tan x) = \sec^2 x$

(v) $\frac{d}{dx}(\cot x) = -\text{cosec}^2 x$

(vi) $\frac{d}{dx}(\sec x) = \sec x\,.\tan x$

(vii) $\frac{d}{dx}(\text{cosec } x) = -\text{cosec } x \cot x$

(viii) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}; -1 < x < 1$

(ix) $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, -1 < x < 1$

(x) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

(xi) $\frac{d}{dx}(cot^{-1}x) = \frac{-1}{1+x^2}$

(xii) $\frac{d}{dx}(sec^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}}$

(xiii) $\frac{d}{dx}(cosec^{-1}x) = \frac{-1}{|x|\sqrt{x^2-1}}$; $|x| > 1$

(xiv) $\frac{d}{dx}(log_a x) = \frac{1}{x\, log_e a}$

(xv) $\frac{d}{dx}(log_e x) = \frac{1}{x}$

(xvi) $\frac{d}{dx}(a^x) = a^x.log_e a$

(xvii) $\frac{d}{dx}(e^x) = e^x$

(xviii) $\frac{d}{dx}(|x|) = \frac{|x|}{x}$, $(x \neq 0)$

(xix) $\frac{d}{dx}(x^x) = x^x(1 + log_e x)$

(xx) $\frac{d}{dx}(sinh\, x) = cos\, h\, x$

(d) Product rule of differentiation

(i) $\frac{d}{dx}\big(f(x).g(x)\big) = f(x).g'(x) + f'(x).g(x)$

(ii) $d(uvw) = uvw' + uv'w + u'vw$

(e) Quotient rule of differentiation

$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x).f'(x)-f(x).g'(x)}{(g(x))^2}, (g(x) \neq 0)$

(f) Logarithmic differentiation:

Taking log might help in differentiation of a function. For example if $y = v^u$ then we can take log both side and differentiable to get $\frac{dy}{dx}$

(g) Differentiation in parametric from :

If we write x and y in term of find variable 't' that is x = f(t), y = ω(t), then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

(h) Greatest Integer function / step function / integer part function

$f(x) = [x] = n, \forall\, n \leq x < n+1 \text{where}, n \in Z$

$\lim_{x\to a}[x] = \nexists$ if a is an integer ($\therefore \nexists$ = do not exist)

L.H.L. = $\lim_{x\to a^-}[x] = a - 1$

R.H.L. = $\lim_{x\to a^+}[x] = a$

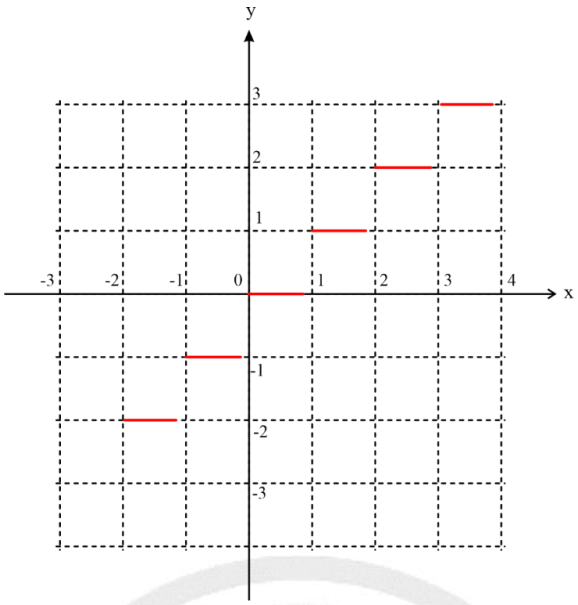


Fig.1.2. Greatest Integer

(i) Properties of Limits

(i) $\lim_{x\to a}(f(x)\pm g(x)) = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$

(ii) $\lim_{x\to a}\big(f(x).g(x)\big) = \lim_{x\to a} f(x).\lim_{x\to a} g(x)$

(iii) $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}, \left(\lim_{x\to a} g(x) \neq 0\right)$

(iv) If $\lim_{x\to a} f(x)$ exists and $\lim_{x\to a} g(x) = \nexists$, then $\lim_{x\to a} f(x).g(x)$ **may exist**

Example: Let $f(x) = \sin x$, $g(x) = \frac{1}{x}$, $\lim_{x\to 0} f(x) = 0$, $\lim_{x\to 0}\frac{1}{x} = \nexists$

But $\lim_{x\to 0} \sin x.\frac{1}{x} = 1$

(v) Indeterminate form III (0^0, 1^∞, ∞^0)

If $y = \lim_{x\to a}\left[f(x)\right]^{\phi(x)}$

Then, $\log y = \lim_{x\to a}\phi(x)\log\left[f(x)\right]$

Thus 0^0, 1^∞, ∞^0 will convert into $\infty\times 0$ from which can be solved easily.

(vi) If $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{0}{0}$ (or) $\frac{\infty}{\infty}$, then $\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)} \neq \left(\frac{0}{0}\right)$

If $\lim_{x\to a}\frac{f'(x)}{g'(x)} = \frac{0}{0}$ (or) $\frac{\infty}{\infty}$, then $\lim_{x\to a}\frac{f'(x)}{g'(x)} = \lim_{x\to a}\frac{f''(x)}{g''(x)}$ and so on

(vii) If $\lim_{x\to a}\big(f(x).g(x)\big) = 0\times\infty \Rightarrow \lim_{x\to a}\frac{f(x)}{\left(\frac{1}{g(x)}\right)} = \frac{0}{0}$ (Apply L- Hospital Rule again)

(j) Some Standard Limits

(i) $\lim_{x\to 0}\frac{\sin x}{x}=1$

(ii) $\lim_{x\to 0}\frac{\tan x}{x}=1$

(iii) $\lim_{x\to 0}\frac{1-\cos ax}{x^2}=\frac{a^2}{2}$

(iv) $\lim_{x\to \infty}\frac{\sin x}{x}=0$

(v) $\lim_{x\to \infty}\frac{\cos x}{x}=0$

(vi) $\lim_{x\to 0}(1+ax)^{b/x}=e^{ab}$

(vii) $\lim_{x\to \infty}\left(1+\frac{a}{x}\right)^{bx}=e^{ab}$

(viii) $\lim_{x\to 0}\left(\frac{a^x+b^x}{2}\right)^{1/x}=\sqrt{ab}$

(ix) $\lim_{x\to 0}\left(\frac{1^x+2^x+3^x+....+n^x}{n}\right)^{1/x}=\sqrt[n]{n!}$

(x) $\lim_{x\to 0}\frac{a^x-1}{x}=\log_e a\,;\lim_{x\to 0}\frac{e^x-1}{x}=1$

(xi) $\lim_{x\to 0}x.\sin\left(\frac{1}{x}\right)=0$

1.2 Mean Value Theorems

1.2.1 Lagrange's Mean Value Theorem (LMVT)

If $f(x)$ is continuous in [a, b] and it is differentiable in (a, b) then ∃ at least one point 'c' such that c ∈ (a, b) and

$$f'(c)=\frac{f(b)-f(a)}{b-a}$$

Here $f'(c)$ slope of tangent to $f(x)$ at $x = c$.

Tangent at x = c is parallel to the line connecting the points A and B

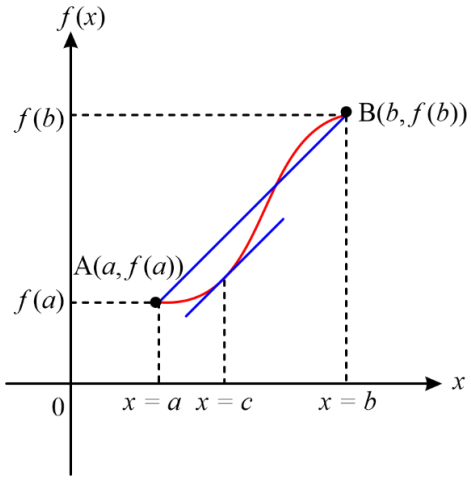


Fig.1.3. LMVT

1.2.2 Rolle's Mean Value Theorem

If $f(x)$ is continuous in $[a, b]$ and differentiable in (a, b) and $f(a) = f(b)$ then $\exists$ at least one-point c $\in (a, b)$ such that $f'(c) = 0$.

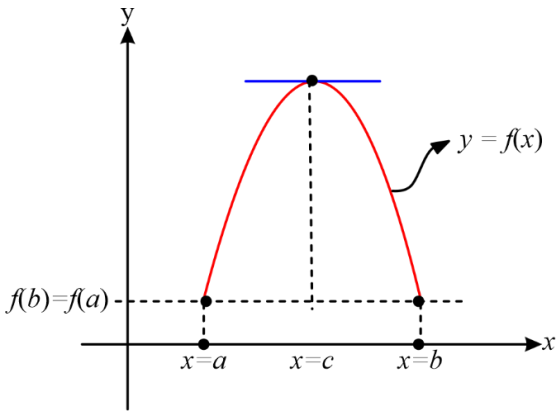


Fig. 1.4. Rolle's mean value

1.2.3 Cauchy's Mean Value Theorem

If $f(x)$ and $g(x)$ are continuous in $[a, b]$ and differentiable in (a, b) then $\exists$ at least one value of 'c' such that $c \in (a, b)$ and $\frac{g'(C)}{f'(C)} = \frac{g(b)-g(a)}{f(b)-f(a)}$

1.3 Increasing and Decreasing Functions

1.3.1 Increasing Functions

A function $f(x)$ is said to be increasing, if $f(x_1) < f(x_2)\ \forall\ x_1 < x_2$

Or

A function $f(x)$ is said to be increasing, if $f(x)$ increases as x increases.

For a function $f(x)$ to be increasing at the point x=a, $f'(a) > 0$.

Example:

e^x, $\log_e x$ $\rightarrow$ Monotonically increasing functions

sin x in $(0, \pi/2) \rightarrow$ non-monotonic functions

1.3.2 Decreasing Functions

A function $f(x)$ is said to be a decreasing function, if $f(x_1) > f(x_2) \forall x_1 < x_2$

A function $f(x)$ is said to be decreasing function, if $f(x)$ decreases as x increases.

Example: e^{-x} $\rightarrow$Monotonically decreasing function, sin x in $\left(\frac{\pi}{2}, \pi\right)$

1.4. Concept of Maxima and Minima

Let $f(x)$ be a differentiable function, then to find the maximum (or) minimum of $f(x)$.

(1) Find $f'(x)$ and equate to zero.

(2) Solve the resulting equation for x. Let its roots be $a_1, a_2, \ldots$ then $f(x)$ is stationary at $x = a_1, a_2, \ldots\ldots$ Thus $x = a_1, a_2, \ldots \ldots$ are the only points at which $f(x)$ can be maximum or a minimum.

(3) Find $f''(x)$ and substitute in it by terms $x = a_1, a_2, \ldots\ldots$ wherever $f''(x)$ is negative, we have a maximum and wherever $f''(x)$ is positive, we have a minimum.

(4) If $f''(a_1) = 0$, find $f'''(x)$ put $x = a_1$ in it. If $f'''(a_1) \neq 0$, there is neither a maximum nor a minimum at $x = a_1$. If $f'''(a_1) = 0$, find $f^{iv}(x)$ and put $x = a_1$ in it. If $f^{iv}(a_1)$ is negative, we have maximum at $x = a_1$, if it is positive there is a minimum at $x = a_1$. If $f^{iv}(a_1)$ is zero, we must find $f^{v}(x)$, and so on. Repeat the above process for each root of the equation $f'(x) = 0$.

Example: $x = 0$ is a critical point of $f(x) = x^3$

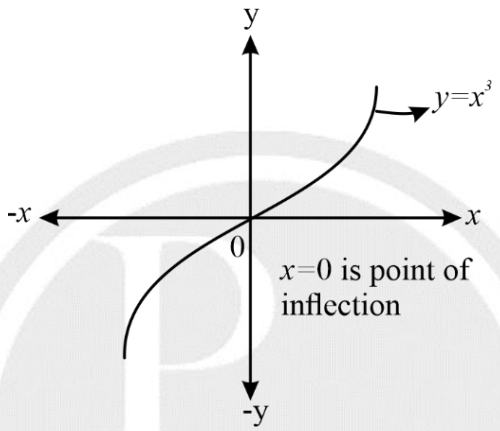


Fig. 1.5. Graph of x^3

$$f(x) = x^3$$

$$\Rightarrow \quad f'(x) = 3x^2 = 0 \quad \Rightarrow \quad x = 0$$

$$f''(x) = 6x \Rightarrow f''(0) = 6(0) = 0$$

- **Global maxima and minima :**
  We first find local maxima and minima and then calculate the value of 'f' at boundary points of interval given e.g. (a, b) we find f(a) and f(b) and compare it with the values of local maxima and minima. The absolute maxima and minima can be decided then.

1.5. Taylor Series

If $f(x)$ is continuously differentiable $(f'(x), f''(x), f'''(x), \ldots\ldots \text{exists})$ then the Taylor series expansion of $f(x)$ about the point $x = a$ is given by

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 . + \ldots \infty$$

If $a = 0$, then $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots\ldots \infty$ (Remember that Mc-Lauren Series is same as Taylor Series if a = 0)

The coefficient of $(x - a)^n$ in the Taylor series expansion of $f(x)$ is $\frac{f^n(a)}{n!}$.

The general expansion of Taylor series is given by $f(x+h) = f(x) + h.\frac{f'(x)}{1!} + h^2.\frac{f''(x)}{2!} + h^3.\frac{f'''(x)}{3!} + \ldots\ldots \infty$

- Finding the expansion of e^x about $x = 0$

$f(x) = e^x \Rightarrow f(0) = e^0 = 1$

$f'(x) = e^x \Rightarrow f'(0) = e^0 = 1; f"(0) = f'''(0) = f''''(0) = \ldots = 1$

$f(x) = e^x = 1 + (x-0)\frac{1}{1!} + (x-0)^2.\frac{1}{2!} + (x-0)^3.\frac{1}{3!} + \ldots..$

$\Rightarrow \quad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots..$

1.6 Integral Calculus

If $F(x)$ is anti-derivative of $f(x)$. That is, continuous and differentiable in (a, b), then we write $\int_{x=a}^{x=b} f(x)\,dx = F(b) - F(a)$. Here $f(x)$ is integrand

If $f(x) > 0\ \forall a \le x \le b, then\ \int_a^b f(x)\,dx$ represents the shaded area in the given figure.

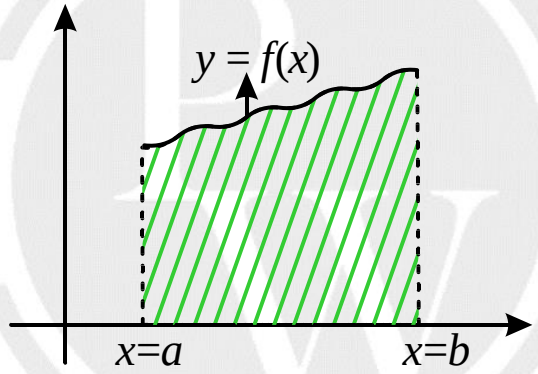


Fig.1. 6. Integration of continuous function

1.6.1 Mean Value Theorem of Integration

If $f(x)$ is continuous in $[a, b]$ and differentiable in (a, b) then '$\exists$' atleast one-point $c \in (a, b)$ such that

$f(c) = \frac{\int_a^b f(x)dx}{(b-a)}$

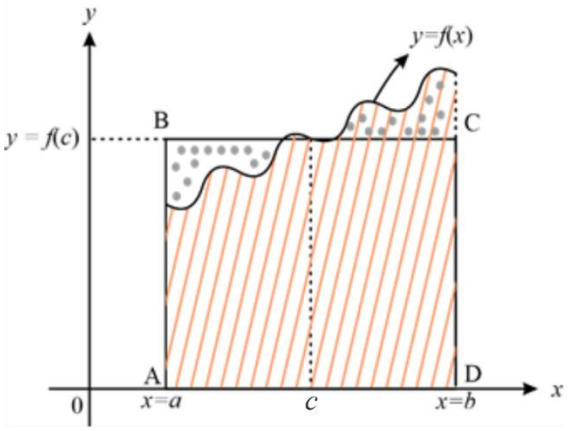


Fig. 1.7. Mean value of integration

1.7. Newton-Leibnitz Rule

If $f(x)$ is continuously differentiable and $\phi(x)$, $\Psi(x)$ are two functions for which the 1st derivative exists, then

$$\frac{d}{dx}\left(\int_{\phi(x)}^{\psi(x)} f(x)dx\right) = f\big(\psi(x)\big).\psi'(x) - f\big(\phi(x)\big).\phi'(x)$$

Example: $\frac{d}{dx}\left(\int_{x}^{x^2} \sin x\, dx\right) = \sin(x^2).2x - \sin x.1 = 2x\sin(x^2) - \sin x$

1.8. Some Standard Integrals

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C,\ (n \neq -1)$

2. $\int \frac{1}{x} dx = \log_e |x| + C$

3. $\int \sin x\, dx = -\cos x + C$

4. $\int \cos x\, dx = \sin x + C$

5. $\int \frac{f'(x)}{f(x)} dx = \log_e |f(x)| + C$

6. $\int \tan x\, dx = -\int -\frac{\sin x}{\cos x} dx = -\log_e |\cos x| + C$

$\Rightarrow \int \tan x\, dx = \log_e |\sec x| + C$

7. $\int \cot x\, dx = \int \frac{\cos x}{\sin x} dx = \log_e |\sin x| + C = -\log_e |\operatorname{cosec} x| + C$

8. $\int \sec x\, dx = \int \frac{\sec x(\sec x + \tan x)}{(\sec x + \tan x)} dx = \log_e |\sec x + \tan x| + C$

9. $\int \operatorname{cosec} x dx = \log_e |\operatorname{cosec} x - \cot x| + C$

10. $\int a^x dx = \frac{a^x}{\log_e a} + C$

11. $\int \frac{1}{x.\log_e a} dx = \log_a x + C$

12. $\int x^x (1 + \log_e x)\, dx = x^x + C$

13. $\int f(x).f'(x) dx = \frac{1}{2}\big(f(x)\big)^2 + C$

14. $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2.\sqrt{f(x)} + C$

15. If $f(x)$, $g(x)$ are two functions. that are differentiable, then

$\int f(x)\, g(x) dx = f(x).\int g(x)\, dx - \int [f'(x)\, g(x)] dx + C$

Before integrating the product, the functions $f(x)$ and $g(x)$ are to be arranged according to the ILATE Principle.

Here, ILATE stands for INVERSE LOGARITHMIC ALGEBRAIC TRIGONOMETRIC EXPONENTIAL.

1.9 Properties of Definite Integrals

1. If $f(x)$ is differentiable in interval (a, b), then $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

2. If $\exists$ a point $c \in (a, b)$ such that $f(x)$ is not differentiable, then

$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

3. If $f(x)$ is continuously differentiable function,

$\int_{-a}^{a} f(x)\,dx = 2 \times \int_0^a f(x)\,dx$; if $f(-x) = f(x)$, ("$f(x)$ is even function")

$= 0$; if $f(-x) = -f(x)$, ("$f(x)$ is odd function")

4. $\int_0^{2a} f(x)\,dx = 2 \times \int_0^a f(x)\,dx$, if $f(2a - x) = f(x)$

5. $\int_a^b f(x)\,dx = \int_a^b f(a + b - x)\,dx$

6. $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} = \left(\frac{b-a}{2}\right)$

Example:

(i) $\int_0^{\pi/2} \frac{\sin x}{\sin x+\cos x} = \frac{\pi}{4}$

(ii) $\int_0^{\pi/2} \frac{1}{1+\sqrt{\tan x}}\,dx = \int_0^{\pi/2} \frac{1}{1+\left(\frac{\sqrt{\sin x}}{\sqrt{\cos x}}\right)}\,dx = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx = \frac{\pi}{4}$

(iii) $\int_2^3 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{5-x}} = \left(\frac{3-2}{2}\right) = \frac{1}{2}$

(iv) $\int_0^{\pi/2} \frac{\sqrt{\tan x}}{\sqrt{\tan x}+\sqrt{\cot x}}\,dx = \frac{\pi}{4}$

7. $\int_0^{\pi/2} \sin^m x\,dx = \int_0^{\pi/2} \cos^m x\,dx = \frac{(m-1)\times(m-3)\times(m-5)}{m\times(m-2)\times(m-4)} \times \ldots \left(\frac{1}{2}\right)$ (or) $\frac{2}{3} \times K$

Where $K = \pi/2$ if m is even

$= 1$ if m is odd.

8. $\int_0^{\pi} \frac{dx}{a^2\cos^2 x+b^2\sin^2 x} = \frac{\pi}{ab}$

9. $\int_0^{\pi/2} \frac{dx}{a^2\cos^2 x+b^2\sin^2 x} = \frac{\pi}{2ab}$

1.10 Length of a Curve

(a) The length of the arc of the curve $y = f(x)$ between the points where x = a and x = b is $s = \int_a^b \sqrt{\left[1+\left(\frac{dy}{dx}\right)^2\right]}\,dx$

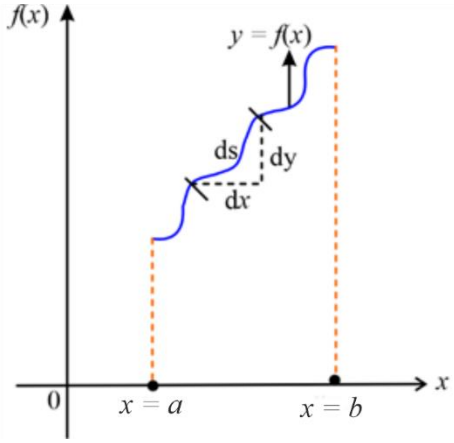


Fig.1.8. Length of the curve

(b) The length of the arc of the curve $x = f(y)$ between the points where y = a and y = b, is $s = \int_a^b \sqrt{\left[1 + \left(\frac{dx}{dy}\right)^2\right]} dy$

(c) The length of the arc of the curve $x = f(t), y = f(t)$ between the points where t = a and t = b, is $s = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$

(d) The length of the arc of the curve $r = f(\theta)$, between the points where $\theta = \alpha$ and $\theta = \beta$, is $s = \int_\alpha^\beta \sqrt{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]} d\theta$

1.11 Surface Area of Solid generated by revolving a curve about a fixed axis

Elemental Surface Area

$dA = 2\pi y \times ds = 2\pi y ds$

$\Rightarrow$ Total surface area = A = $\int_{x=a}^{x=b} 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

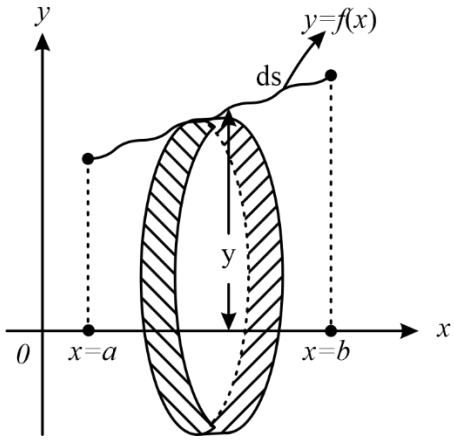


Fig.1.9. Surface area

1.12 Volume of the solid

A. The volume of the solid obtained by revolving the curve y = $f(x)$ between the lines $x = a$ and $x = b$ is given by

$\Rightarrow \quad V \approx \int_{x=a}^{x=b} \pi y^2 dx$

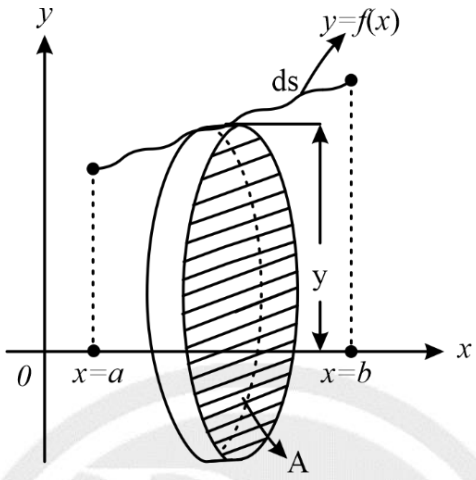


Fig. 1.10. Volume of the solid

B. **Revolution about the y-axis.** Interchanging x and y in the above formula, we see that the volume of the solid generated by the revolution, about y-axis, of the area, bounded by the curve $x = f(y)$, the y-axis and the abscissa y = a, y = b is $\int_a^b \pi x^2 dy$.

1.13 Gamma Function

The integral $\int_0^\infty e^{-x}.x^{n-1}\,dx, (n > 0)$ is called Gamma function of n. It is denoted by $\Gamma n = \int_0^\infty e^{-x}x^{n-1}dx$.

Note : $\int_0^{\pi/2} \sin^m x \cos^n x dx = \dfrac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left[\frac{m+n+2}{2}\right]}$

Where Γ(x) is called the gamma function.

1.13.1 Properties of Gamma Function

(i) $\Gamma n = (n-1)!$

(ii) $\Gamma(n+1) = (n)!$

(iii) $\Gamma(n+1) = n\Gamma n$

(iv) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

1.14 Beta Function

The function β (m, n) = $\int_0^1 x^{m-1}.(1-x)^{n-1}\,dx$ (m, $n > 0$) is called β function of m and n.

1.14.1 Properties of β function

(i) $\beta(m,n) = \frac{\Gamma m.\Gamma n}{\Gamma(m+n)}$

(ii) $\beta(m,n) = \beta(n,m)$

(iii) $\beta(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx$

$\beta(n,m) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}dx$

(iv) $sin^p\theta.cos^q\,\theta\,dx = \frac{1}{2}\beta\left(\frac{p+1}{2},\frac{q+1}{2}\right),(p,q>-1)$

1.15 Area between the curves

If the function f(x) > g(x) for all values of x between x=a and x=b then

$A = \int_a^b f(x)dx - \int_a^b g(x)dx \quad \Rightarrow \quad A = \int_a^b \big(f(x)-g(x)\big)\,dx$

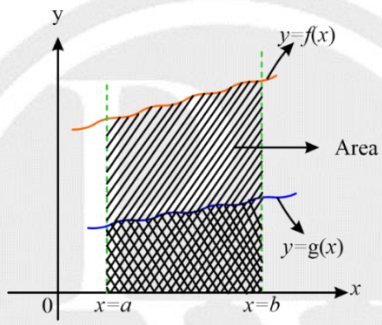


Fig. 1.11. Area under curve

Note : Area bounded by curve $r = f(\theta)$ between $\theta = \alpha$ and β is $\frac{1}{2}\int_\alpha^\beta r^2 d\theta$

1.16 Multi Variable Calculus

(a) Continuity of a function

A function *f*(x, y) is said to be continuous at (*a*, *b*), if $\lim_{\substack{x\to a\\ y\to b}} f(x,y) = f(a,b)$

(b) Differentiation of a two-variable function

If *f*(*x*, *y*) is a continuous function, then the derivative of *f*(*x*, *y*) with respect to *x* treating *y* as constant is given by

$$p = \frac{\partial f}{\partial x} = \lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h}$$

The derivative of *f*(*x*, *y*) with respect to *y* treating *x* as constant is given by

$$q = \frac{\partial f}{\partial y} = \lim_{k\to 0}\frac{f(x,y+k)-f(x,y)}{k}$$

(c) Homogenous Function

A function *f* (x, y) is said to be homogenous function of degree 'n' if $f(kx,ky) = k^n.f(x,y)$.

Example: $f(x,y) = x^3 - 3x^2y + 3xy^2 + y^3$

$\Rightarrow f(kx,ky) = (kx)^3 - 3(kx)^2(ky) + 3(kx).(ky)^2 + (ky)^3$

$= k^3(x^3 - 3x^2y + 3xy^2 + y^3)$

$= k^3.f(x,y) \Rightarrow f(x,y)$ is a homogenous function of degree '3'.

(d) Euler's Theorem

If $f(x, y)$ is a homogeneous function of degree 'n' then

(i) $x.\frac{\partial f}{\partial x} + y.\frac{\partial f}{\partial y} = nf$

(ii) $x^2.\frac{\partial^2 f}{\partial x^2} + 2xy\frac{\partial^2 f}{\partial x \partial y} + y^2\frac{\partial^2 f}{\partial y^2} = n(n-1)f$

If $f(x, y) = g(x, y) + h(x, y) + \phi(x, y)$ where $g(x, y)$, $h(x, y)$ and $\phi(x, y)$ are homogenous functions of degrees m, n and p respectively, then

$x.\frac{\partial f}{\partial x} + y.\frac{\partial f}{\partial y} = m.g(x,y) + n.h(x,y) + p.\phi(x,y)$

$x^2.\frac{\partial^2 f}{\partial x^2} + 2xy.\frac{\partial^2 f}{\partial x \partial y} + y^2.\frac{\partial^2 f}{\partial y^2} = m(m-1).g(x,y) + n(n-1).h(x,y) + p(p-1).\phi(x,y)$

(e) Total derivative:

(i) If u = f(x, y) and if x = ω(t), y = v(t) then $\frac{du}{dt} = \frac{\partial u}{\partial x}.\frac{dx}{dt} + \frac{\partial u}{\partial y}.\frac{dy}{dt}$

(ii) If u be a function of x and y, where y is a function of x, then $\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}.\frac{dy}{dx}$

(iii) If u = f(x, y) and $x = f_1(t_1, t_2)$ and $y = f_2(t_1, t_2)$, then

$\frac{\partial u}{\partial t_1} = \frac{\partial u}{\partial x}.\frac{\partial x}{\partial t_1} + \frac{\partial u}{\partial y}.\frac{\partial y}{\partial t_1}$ and $\frac{\partial u}{\partial t_2} = \frac{\partial u}{\partial x}.\frac{\partial x}{\partial t_2} + \frac{\partial u}{\partial y}.\frac{\partial y}{\partial t_2}$

(iv) If x and y are connected by an equation of the form f(x, y) = 0, then $\frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y}$

(f) Concept of Maxima and Minima in Two Variables

If $f(x, y)$ is a two-variable differentiable function, then to find the maxima (or) minima.

Step-1: Calculate $p = \frac{\partial f}{\partial x}$ and $q = \frac{\partial f}{\partial y}$ and equate $p = 0$, $q = 0$

Let (x_0, y_0) be a stationary point.

Step-2: Calculate r, s, t where $r = \frac{\partial^2 f}{\partial x^2}\Big|_{(x_0,y_0)}$; $s = \frac{\partial^2 f}{\partial x.\partial y}\Big|_{(x_0,y_0)}$; $t = \frac{\partial^2 f}{\partial y^2}\Big|_{(x_0,y_0)}$

Case (i): If $rt - s^2 > 0$ and $r > 0$, then the function $f(x, y)$ has minimum at (x_0, y_0) and the minimum value is $f(x_0, y_0)$.

Case (ii): If $rt - s^2 > 0$ and $r < 0$, then the function $f(x, y)$ has maximum at (x_0, y_0) and the maximum value is

$f(x_0, y_0)$.

Case (iii): If $rt - s^2 < 0$; then we cannot comment on the existence of maxima and minima.

Such stationary points where $rt - s^2 = 0$ are called **saddle points**.

(g) Concept of Constraint Maxima and Minima (Method of Lagrange's unidentified multipliers).

If $f(x, y, z)$ is a continuous and differentiable function, such that the variables x, y and z are related/constrained by the equation $\phi(x, y, z) = C$ then to calculate the extreme value of $f(x, y, z)$ using Lagrange's Method of unidentified multipliers.

Step-1: Form the function $F(x, y, z) = f(x, y, z) + \lambda\{\phi(x, y, z) - C\}$, where λ is a multiplier.

Step-2: Calculate $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$ and $\frac{\partial F}{\partial z}$ and equate them to zero

Step-3: Equate the values of λ from the above 3 equations and obtain the relation between the variables x, y and z.

Step-4: Substitute the relation between x, y and z in $\phi(x, y, z) = C$ and get the values of x, y, z. Let they be (x_0, y_0, z_0).

Step-5: Calculate $f(x_0, y_0, z_0)$

The value $f(x_0, y_0, z_0)$ is the extreme value of $f(x, y, z)$.

(h) Multiple Integrals

If $f(x, y)$ is continuous and differentiable at every point within a region 'R' bounded by some curves is given by

$$I = \iint_R f(x,y)\,dxdy$$

If there is a double integral,

$$I = \int_{x=a}^{x=b}\int_{y=\phi(x)}^{y=\psi(x)} f(x,y)dydx \quad [\text{Let } \psi(x) > \phi(x)]$$

Then I = area of the region bounded by the curves, $y = \phi(x)$; $y = \Psi(x)$, $x = a$ and $x = b$ if $f(x, y) = 1$

Value of x – co-ordinate of centroid of the region bounded by $y = \phi(x)$; $y = \psi(x)$; $x = a$, $x = b$ if $f(x, y) = x$

(i) Change of Orders of Integration

$$I = \int_{x=a}^{x=b}\int_{y=\phi(x)}^{y=\psi(x)} f(x,y)dydx \;\rightarrow\; I = \int_{y=c}^{y=d}\int_{x=g(y)}^{x=h(y)} f(x,y)dxdy$$

In change of order of Integration, the order of the Integrating variables changes and the limits as well.

Note : When limits are constants, the order of integration does not matter,

$$\int_{y=c}^{y=d}\int_{x=a}^{x=b} f(x,y)dxdy = \int_{x=a}^{x=b}\int_{y=c}^{y=d} f(x,y)dydx$$

1.17 Jacobian of the Transformation

(i) The Jacobian of the transformation, $x = f_1(u,v), y = f_2(u,v)$ is defined as,

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$$

(ii) The Jacobian of the transformation,

$x = f_1(u,v,w), y = f_2(u,v,w), z = f_3(u,v,w)$ is defined as

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$$

1.18 Change of Variables Formula

(i) $\iint\limits_R f(x,y)dx\,dy = \iint\limits_R f(f_1(u,v), f_2(u,v))\,|J|\,du\,dv$

(ii) $\iiint\limits_R f(x,y,z)dxdydz = \iiint\limits_R f(f_1(u,v,w), f_2(u,v,w), f_3(u,v,w))\,|J|\,du\,dv\,dw$

1.19 Change of Variables

(i) Cartesian to polar co-ordinates :

$x = r\cos\theta$

$y = r\sin\theta$

$J = r$

$dx\,dy = rdrd\theta$

(ii) Cartesian to cylindrical polar co-ordinate :

$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$J = r$

$dxdydz = rdr\,d\theta\,dz$

(iii) Cartesian to spherical polar co-ordinates :

$x = \rho\sin\phi\cos\theta$

$y = \rho\sin\phi\sin\theta$

$z = \rho\cos\phi$

$J = \rho^2\sin\phi$

$dx\,dy\,dz = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$

❑❑❑

2 LINEAR ALGEBRA

2.1 Matrix

An array of elements in horizontal lines (Rows) and Vertical Lines (Columns) is called a Matrix.

Example: $A = \begin{bmatrix} i & n & d & i & a \\ j & a & p & a & n \end{bmatrix}$

2.1.1 Size of Matrix

If a matrix has 'm' rows and 'n' columns, then we say that the size of the matrix is m × n (read as m by n)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13}\ldots\ldots\ldots a_{1n} \\ a_{21} & a_{22} & a_{23}\ldots\ldots\ldots a_{2n} \\ \cdot & \cdot & \cdot\cdot \\ \cdot & \cdot & \cdot\cdot \\ a_{m1} & a_{m2} & a_{m3}\ldots\ldots\ldots a_{mn} \end{bmatrix}; \; A = [a_{ij}]_{m\times n} \text{ such that } 1 \le i \le m, 1 \le j \le n \text{ and } a_{ij} = f(i,j)$$

2.1.2 Addition of Matrices

(i) Two matrices $A = [a_{ij}]_{m\times n}$ & $B = [b_{ij}]_{p\times q}$ can be added only if m = p & n = q.

(ii) Matrix Addition is commutative (A + B = B + A)

(iii) Matrix Addition is Associative. A + (B + C) = (A + B) + C

(iv) Existence of additive identity : If O be m × n matrix each of whose elements are zero. Then, A + O = A = O + A for every m × n matrix A.

(v) Existence of additive inverse : Let $A=[a_{ij}]_{m\times n}$ then the negative of matrix A is defined as matrix $[-a_{ij}]_{m\times n}$ and is denoted by –A.

⇒ Matrix –A is additive inverse of A. Because (–A) + A = O = A + (–A). Here O is null matrix of order m × n.

(vi) Cancellation laws holds good in case of addition of matrices, which is X = –A.

⇒ A + X = B + X ⇒ A = B

⇒ X + A = X + B ⇒ A = B

(vii) The equation A + X = 0 has a unique solution in the set of all m × n matrices.

2.1.3 Multiplication of Matrices

The multiplication of two matrices $A = [a_{ij}]_{m\times n}$ and $B = [b_{ij}]_{p\times q}$ $(\Rightarrow AB_{m\times q})$ is feasible only if n = P.

$A_{m\times n} \cdot B_{p\times q} = C$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times3} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}_{3\times2} \quad A_{3\times3} \times B_{3\times2}$$

$$\Rightarrow \begin{bmatrix} a_{11}.b_{11} + a_{12}\cdot b_{21} + a_{13}.b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}.b_{11} + a_{22}.b_{21} + a_{23}.b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \\ a_{31}.b_{11} + a_{32}\cdot b_{21} + a_{33}\cdot b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} \end{bmatrix}_{3\times2}$$

2.1.4 Properties of Multiplication of Matrices

(i) Matrix Multiplication Need not be commutative.

(ii) Matrix Multiplication is Associative (A(BC)) = ((AB)C)

(iii) Matrix Multiplication is distributive A(B + C) = (AB + AC)

(iv) The product of two Matrices $A_{m\times n}, B_{n\times q}$ (i.e. $AB_{m\times q}$) can be a zero matrix even if $A \neq O \& B \neq O$.

Example: $A = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}; B = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} \Rightarrow AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

- For the multiplication of two matrices $A_{m\times n}$ & $B_{n\times q}$

  (i) The No. of Multiplications required = m n q

  (ii) The number of Additions required = m (n –1) q

2.2 Types of Matrices

(1) Upper triangular Matrix: A matrix $A = [a_{ij}]; 1 \leq i, j \leq n$ is said to be an upper triangular matrix if

$$a_{ij} = 0 \; \forall \; i > j$$

(2) Lower Triangular Matrix: A matrix $A = [a_{ij}]_{n\times n}; 1 \leq i, j \leq n$ is said to be a lower Triangular Matrix

if $a_{ij} = 0 \; \forall \; i < j$

(3) Diagonal Matrix: A matrix $A = [a_{ij}], \forall \; 1 \leq i, j \leq n$ is said to be a diagonal matrix if $a_{ij} = O \; \forall \; i \neq j$

Example: $A = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$. The diagonal Matrix is also denoted as $A = diag\,[d_1, d_2, d_3]$

(4) Scalar Matrix: A Matrix 'A' = $[a_{ij}]$; $1 \leq i, j \leq n$ is said to be a scalar Matrix if $a_{ij} = \begin{cases} K; i = j \\ 0; 1 \neq j \end{cases}$

If K = 1, then A → Identity Matrix,

If K = 0, then A → Null Matrix.

(5) Idempotent Matrix:

A Matrix '$A_{n\times n}$' is said to be an idempotent matrix if $A^2 = A$.

Example: $A = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix}$

$\Rightarrow A \cdot A = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix}\begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} = A$

(6) Nilpotent Matrix: A matrix A is said to be nilpotent of class x or index if $A^x = 0$ and $A^{x-1} \neq 0$ i.e. x is the smallest index which makes $A^x = 0$.

Example: The matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is nilpotent class 3, since $A \neq 0$ and $A^2 \neq 0$, but $A^3 = 0$.

(7) Orthogonal Matrix: A matrix A is said to be orthogonal if $A.A^T = I$

Example: $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

(8) Involutory Matrix: A matrix A is said to be involutory if $A^2 = I$

Example: $\begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$

2.3 Transpose of a Matrix

For a given matrix $= [a_{ij}]; 1 \leq i \leq m, 1 \leq j \leq n$, we can say that 'B' where $B = [b_{ij}], i \leq i \leq n \;\; i \leq j \leq m$ is the transpose of the Matrix 'A' if $a_{ij} = b_{ji}$

2.3.1 Properties of Transpose of a Matrix

(i) $(A^T)^T = A$

(ii) $(AB)^T = B^T \cdot A^T$

(iii) $(KA)^T = KA^T$ where 'K' is a scalar.

2.4 Conjugate of a matrix

The matrix obtained by replacing each element of matrix by its complex conjugate.

2.4.1 Properties of conjugate matrix

(a) $(\bar{\bar{A}}) = A$ (b) $(\overline{A+B}) = \bar{A} + \bar{B}$

(c) $(\overline{KA}) = \bar{K}\bar{A}$ (d) $(\overline{AB}) = \bar{A}\bar{B}$

(e) $\bar{A} = A$ if A is real matrix

$\bar{A} = -A$ if A is purely imaginary matrix

2.5 Transposed Conjugate of a Matrix

The transpose of conjugate of a matrix is called transposed conjugate. It is represented by A^θ.

(a) $(A^\theta)^\theta = A$

(b) $(A+B)^\theta = A^\theta + B^\theta$

(c) $(KA)^\theta = \bar{K}A^\theta$ (K : Complex number)

(d) $(AB)^\theta = B^\theta A^\theta$

2.6 Trace of a Matrix

Trace is simply sum of all diagonal elements of a matrix.

2.6.1 Properties of Trace of a matrix

Let A and B be two square matrices of order η and λ is scalar then

1. $Tr(\lambda A) = \lambda Tr(A)$
2. $Tr(A+B) = Tr(A) + Tr(B)$
3. $Tr(AB) = Tr(BA)$ [If both AB and BA are defined]

2.7 Type of Real Matrix

(a) Symmetric matrix : $(A)^T = A$

(b) Skew symmetric matrix : $(A^T) = -A$

(c) Orthogonal matrix : $(A^T = A^{-1}, AA^T = I)$

Note : (a) If A and B are symmetric, then (A + B) and (A – B) are also symmetric.

(b) For any matrix AA^T is always symmetric.

(c) For any matrix, $\left(\frac{A+A^T}{2}\right)$ is symmetric and $\left(\frac{A-A^T}{2}\right)$ is skew symmetric.

(d) For orthogonal matrices, $|A| = \pm 1$

(e) We can write any matrix A as a sum of symmetric and skew symmetric matrix $A = \frac{A+A^T}{2} + \frac{A-A^T}{2}$

2.8 Type of complex matrix

(a) Hermitian matrix : $(A^\theta = A)$

(b) Skew-Hermitian matrix: $A^\theta = -A$

(c) Unitary matrix : $(A^\theta = A^{-1}, AA^\theta = I)$

Note : (a) $\frac{A+A^\theta}{2}$ is Hermitian and $\frac{A-A^\theta}{2}$ is skew Hermitian matrix.

(b) We can write any matrix as a sum of Hermitian and skew Hermitian matrix $A = \frac{A+A^\theta}{2} + \frac{A-A^\theta}{2}$

2.9 Determinant

The summation of the product of elements of a row(or) column of a matrix with their corresponding Co-factors.

$$A \cdot adj(A) = |A| \cdot I$$

Determinant can be calculated only if matrix is a square matrix.

Suppose, we need to calculate a 3 × 3 determinant,

$$\Delta = \sum_{j=1}^{3} a_{1j} cof(a_{1j}) = \sum_{j=1}^{3} a_{2j} cof(a_{2j}) = \sum_{j=1}^{3} a_{3j} cof(a_{3j})$$

We can calculate determinant along any row or column of the matrix.

2.9.1 Properties of Determinants

(i) If 'A' is a Square Matrix of size ' $n \times n$ ' and 'k' is a Scalar then

$|K \cdot A_{n\times n}| = K^n \cdot |A_{n\times n}|$

(ii) $|adj(A)| = |A|^{(n-1)}$

(iii) $\left|adj\left(adj(A)\right)\right| = (|A|)^{(n-1)^2}$

(iv) $|AB| = |A| \cdot |B|$

(v) $|(AB)^T| = |B^T| \cdot |A^T|$

(vi) If two rows (or) two columns of a determinant are interchanged, then the determinant changes its sign.

(vii) The determinant of an upper triangular Matrix/a lower triangular Matrix/a diagonal Matrix is the product of the principal diagonal elements of the Matrix.

(viii) The determinant of Every Skew-Symmetric Matrix of odd order $(A_{n\times n})\left('n' is\ odd\right)$ is zero.

(ix) The determinant of an orthogonal Matrix $A_{n\times n}$ is ± 1

(x) The determinant of an Idempotent Matrix is either 0 (or) 1.

(xi) The determinant of an Involuntary Matrix is ± 1

(xii) The determinant of a Nilpotent Matrix is always zero.

(xiii) If the product of two Non-zero Matrices $A_{n\times n} \neq 0; B_{n\times n} \neq 0$ is a zero Matrix $((AB)_{n\times n} = 0)$, then both $|A| = 0$ & $|B| = 0$.

(xiv) If two rows (or) two columns of a Matrix are either equal or Proportional, then the determinant of the Matrix is equal to zero.

(xv) The number of terms in the general expansion of an 'n × n' determinant is $\underline{n!}$

(xvi) Value of the determinant is invariant under row and column interchange i.e., $\left|A^T\right| = |A|$

(xvii) If any row or column is completely zero, then |A| = 0.

(xviii) If any single row or column of the matrix is multiplied by k then the determinant the of new matrix = K|A|

(xix) In a determinant the sum of the product of the element of any row or column with its cofactor gives a determinant of the matrix.

(xx) In determinant the sum of the product of the element of any row or column with a cofactor of another row or column will give zero.

(xxi) |AB| = |A| × |B|

(xxii) Elementary operations don't effect the determinant that is $A \xrightarrow{R_i = R_i + KR_j} B$ then |A| = |B|

$A \xrightarrow{C_i = C_i + KC_j} B$ then |A| = |B|

2.10 Minors, Cofactor and Adjoint of a Matrix

Minor of an element is equal to the determinant of the remaining elements of the matrix, after excluding the row and column containing the particular element. The cofactor of an element can be calculated from the minor of the element. The cofactor of an element is equal to the product of the minor of the element, and -1 to the power of position values of row and column of the element.

$$\text{Cofactor of an Element} = (-1)^{i+j} \times \text{Minor of an Element}$$

Here i and j are the positional values of the row and column of the element.

Example :

If $$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Minor of element a_{21}: $M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$

Co-factor of an element, $a_{ij} = (-1)^{i+j} M_{ij}$

- To design co-factor matrix, we replace each element by its co-factor.
- Adjoint of a matrix = transpose of cofactor matrix
- $A^{-1} = \dfrac{Adj(A)}{|A|}$

2.11 Inverse of a matrix

Inverse of a matrix only exists for square matrices.

$$\left(A^{-1}\right) = \frac{Adj(A)}{|A|} \text{ and } |A| \neq 0$$

Properties:

(a) $AA^{-1} = A^{-1}A = I$

(b) $(AB)^{-1} = B^{-1}A^{-1}$

(c) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

(d) $(A^T)^{-1} = (A^{-1})^T$

(e) The inverse of 2 × 2 matrix should be remembered,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{(ad-bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

(i) Interchange the diagonal elements and put negative sign on the rest.

(ii) Divide by determinant.

2.12 Rank of a Matrix

- The rank of the matrix refers to the number of linearly independent rows or columns in the matrix. ρ(A) is used to denote the rank of matrix A.
- A matrix is said to be of rank zero when all of its elements become zero.
- The rank of the matrix is the dimension of the vector space obtained by its columns.
- The rank of a matrix cannot exceed more than the number of its rows or columns. The rank of the null matrix is zero.
- The nullity of a matrix is defined as the number of vectors present in the null space of a given matrix. In other words, it can be defined as the dimension of the null space of matrix A called the nullity of A. Rank + Nullity is the number of all columns in matrix A.

A real Number 'r' is said to be the rank of a matrix '$A_{m\times n}$' if

(1) There is at least one square sub-matrix of A of order r whose determinant is not equal to zero.

(2) If the matrix A contains any square sub-matrix of order $(r + 1)$ and above, then the determinant of such a matrix should be zero.

It is mathematically denoted by $\rho(A) = r$

2.12.1 Properties of Rank of a Matrix

(i) $\rho(A_{m\times n}) \leq (m, n)$

(ii) $\rho(AB) \leq min\{\rho(A), \rho(B)\}$

(iii) Rank of transpose of matrix is equal to rank of matrix

(iv) Elementary operations do-not affect the rank the matrix

(v) $\rho(A + B) \leq \{\rho(A) + \rho(B)\}$

2.12.2 Row Echelon Form

A Matrix $A_{m\times n}$ is said to be in row-echelon form if

(i) Number of zeroes before the 1st Non-zero element in any row is less than the number of such zeroes in its succeeding row.

(ii) Zero rows (if any) should lie at the bottom of the Matrix.

$\rho(A_{m\times n})$ = Number of non-zero rows in the Row-Echelon form of A.

2.13 System of Equations

The given system of equations

$$a_{11}x_1 + a_{12}x_{12} + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

can be written in Matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\boxed{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}} \boxed{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} = \boxed{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}$$

Ax = B

Coefficient Matrix — Variable Matrix — Constants Matrix

The system Ax = B is said to be a homogeneous system if B = 0.

The system of Ax = B is said to be a non-homogeneous system if $B \neq 0$.

2.13.1 Consistency of a non-homogeneous system of Equations

For the above system of non – homogeneous equations, Ax = B; Augmented Matrix = [A/B] = $\begin{bmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{bmatrix}$

(i) If $\rho(A) = \rho(A/B) =$ Number of unknowns, then the system Ax = B has a unique solution.

(ii) If $\rho(A) = \rho(A/B) <$ Number of unknowns, then the system has infinitely many solutions.

(iii) If $\rho(A) \neq \rho(A/B)$, then the system has no solution.

Number of linearly independent solutions for a system of 'n' equations given by Ax = B is $n - \rho(A)$

2.13.2 Consistency of Homogeneous System of Equations

$a_{11}x + a_{12}y + a_{13}z = 0$

$a_{21}x + a_{22}y + a_{23}z = 0$

$a_{31}x + a_{32}y + a_{33}z = 0$

$$\text{Ax} = 0 \Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} [A/B] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ a_{31} & a_{32} & a_{33} & 0 \end{bmatrix}_{3\times4}$$

If $\rho(A) = \rho(A/B) = n$ $(i.e\ |A| \neq 0)$; the system has a unique solution.

(Trivial solution; x = 0, y = 0, z = 0)

If $\rho(A) = \rho(A/B) < n(|A| = 0)$; the system has infinitely many solutions (Non-trivial solution exists for the system).

2.14 Linear Combination of Vectors

If $x_1, x_2, x_3, \ldots\ldots, x_n$ are 'n' rows vectors, then the combination $k_1x_1 + k_2x_2 + k_3x_3 + \ldots\ldots + k_nx_n$ is called a linear combination of $x_1, x_2, \ldots\ldots, x_n (k_1, k_2, k_3, \ldots\ldots k_n$ are scalars)

(1) The linear combination $k_1x_1 + k_2x_2 + k_3x_3 + \ldots\ldots + k_nx_n$ is said to be linearly dependent if $k_1x_1 + k_2x_2 + k_3x_3 + \ldots\ldots + k_nx_n = 0$ when $k_1, k_2, k_3, \ldots\ldots, k_n$ (NOT All zeroes).

If $x_1 = [a_1 \quad b_1 \quad c_1]; x_2[a_2 \quad b_2 \quad c_2]; x_3 = [a_3 \quad b_3 \quad c_3]$, then the vectors x_1, x_2, x_3 are said to be linearly dependent if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$

(2) The combination $k_1x_1 + k_2x_2 + \ldots\ldots + k_nx_n$ is said to be linearly independent if $k_1x_1 + k_2x_2 + \ldots\ldots\ldots + k_nx_n = 0$ when $k_1 = k_2 = k_3 = \ldots\ldots k_n = 0$

2.14.1 Eigen Values and Eigen Vectors

For any square Matrix $A_{n\times n}$, the equation $|A - \lambda I| = 0$ where 'λ' is a scalar is called the characteristic equation.

The roots of the characteristic equation of a Matrix are called Eigen Values.

2.14.2 Properties of Eigen Values

(i) If $\lambda_1, \lambda_2, \lambda_3, \ldots\ldots, \lambda_n$ are 'n' Eigen Values of $A_{n\times n}$, then

(a) Sum of Eigen Values of 'A' = $\lambda_1 + \lambda_2 + \lambda_3 + \ldots + \lambda_n = \sum_{i=1}^{n} \lambda_i = trace(A)$ = Sum of Principal diagonal elements

(b) Product of all the Eigen Values of 'A' = $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 \cdot \ldots\ldots \lambda_n = \prod_{i=1}^{n} \lambda_i = |A|$

(c) Eigen Values of A^m are $\lambda_1^m, \lambda_2^m, \lambda_3^m, \ldots\ldots \lambda_n^m$

(d) Eigen Values of adj(A) are $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \frac{|A|}{\lambda_3}, \ldots\ldots, \frac{|A|}{\lambda_n}$

(e) Eigen Values of A & A^T are the same.

(f) Eigen Values of $k_1A + k_2I$ (Where k_1 and k_2 are scalar) are

$$k_1\lambda_1 + k_2, k_1\lambda_2 + k_2, k_1\lambda_3 + k_2, k_1\lambda_4 + k_2, \ldots\ldots\ldots k_1\lambda_n + k_2$$

(ii) '0' is always an Eigen Value of an odd-order Skew-Symmetric Matrix.

(iii) Eigen Values of a Real Symmetric Matrix are always real.

(iv) Eigen Values of the Skew-Symmetric Matrix are either zero (or) purely Imaginary.

(v) The Eigen values of an Orthogonal Matrix are of unit modulus.

(vi) If the sum of all the elements in a row (or Column) is constant (= k) for all the rows (or columns) in the matrix respectively, then 'k' is an Eigen Value of the Matrix.

Example: If $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ and if $a_1 + b_1 + c_1 = a_2 + b_2 + c_2 = a_3 + b_3 + c_3 = k,$

then 'k' is an Eigen Value of 'A'.

(vii) The Eigen Values of an upper triangular Matrix, a lower triangular Matrix, a diagonal Matrix are the Principal diagonal elements of the Matrix.

2.15 Eigen Vector

A non-zero column vector $X_{n\times1}$ is said to be an Eigen Vector of $A_{n\times n}$ corresponding to the Eigen Value 'λ', if $AX = \lambda X (X \neq 0)$.

2.15.1 Properties of Eigen Vectors

(i) Eigen Vectors of A & A^T are not the same.

(ii) Eigen Vectors of A & A^M are same.

(iii) The Eigen Vectors of a Real Symmetric Matrix are always orthogonal.

(iv) The number of linearly independent Eigen Vectors of '$A_{n\times n}$' is equal to the number of distinct Eigen Values of '$A_{n\times n}$'.

2.15.2 Cayley Hamilton Theorem

Every Matrix satisfies its characteristic equation.

This means that, if $c_0\lambda^n + c_1\lambda^{-1} + ... + c_{n-1}\lambda + c_n = 0$ is the characteristic equation of a square matrix *A* of order *n*, then

$c_0A^n + c_1A^{n-1} + ... + c_{n-1}A + c_nI = 0$ …(i)

Note: When λ is replaced by A in the characteristic equation, the constant term c_n should be replaced by c_nI to get the result of the Cayley-Hamilton theorem, where *I* is the unit matrix of order *n*.

Also, 0 in the R.H.S. of (i) is a null matrix of order *n*.

2.16. Subspace (Basis of Dimensions)

2.16.1 Vector

An ordered n-tuple of numbers is called an n-vector.

2.16.2 Linearly Independent and Dependent Vector

Let X_1 and X_2 be the non-zero vectors:

- $\{x_1, x_2, …., x_k\}$ are linearly independent if $r_1x_1 + r_2x_2 + … + r_k x_k = 0$ only for $r_1 = r_2 = … + r_k = 0$.
- The vectors $x_1, r_2, …., x_k$ = are linearly dependent if they are not linearly independent; that is, if there exist scalars $r_1, r_2, … , r_k$ which are not all zero such that

  $r_1x_1 + r_2x_2 + …. + r_k x_k = 0$

Note: Let $X_1, X_2…….X_n$ be 'n' vector of matrix *A*.

- If rank (*A*) = number of vectors then vector $X_1, X_2…..X_n$ are linearly independent.
- If rank (*A*) ≠ number of vectors then vector $X_1, X_2 ….. X_n$ are linearly dependent.

2.16.3 Vector Space R^n

If n positive integer, then an ordered n-tuple is a sequence of n real numbers ($\alpha_1, \alpha_2, \ldots \alpha_n$). the set of all ordered *n*-tuples is called n-space and is denoted by $\mathbb{R}^n$.

2.16.4 Subspaces of an N-vector space V_n

A non-empty set S, of vectors of $V_n(F)$,is called a subspace of $V_n(F)$, if

- ξ_1, ξ_2 are any two members of S, then $\xi_1 + \xi_2$ is also a member of S; and
- ξ is a member of S, and k is a scalar then $k\xi$ is also a member of S.

  Briefly, we may say that a set S of vectors $V_n(F)$ is a subspace of $V_n(F)$ it closed w.r.t. the compositions of "addition" and "multiplication with scalars".

  Every subspace of V_n contains the zero vector; being the product of any vector with the scalar zero.

2.16.5 Construction of Subspaces

- **A subspace Spanned by a Set of Vectors:** A subspace that arises as a set of all linear combinations of any given set of vectors is said to be spanned by the given set of vectors.
- **Basis of a subspace:** A set of vectors is said to be a basis of a subspace, if
  - The subspace is spanned by the set, and
  - The set is linearly independent.

Note: If we have N vectors and they are independent then they span N-dimension space. But if they are dependent then they span only a subspace of N-dimension space.

2.16.6 Orthogonality of Vectors

- Two vectors are orthogonal if each is non-zero and $X_1^T X_2 = 0$
- If n vectors $X_1, X_2 \ldots X_n$ each of n dimensions is orthogonal then they are surely linearly independent and form the basis for n-dimension space.
- The set of the vector is orthonormal if they are orthogonal and have unit magnitude.

2.17 Similar Matrices

- Two matrix A and B are similar if there exist a non singular matrix P such that $B = P^{-1}AP$
- Similar matrix has same eigen valves
- If A is similar to B then B is also similar to A
- If A is similar to B and B is similar to C then A is similar to C.

2.18 Diagonalization of a matrix

Finding a matrix D which is a diagonal matrix and which is similar to A is called diagonalization i.e., we wish to find a non-singular matrix M such that $A = M^{-1}DM$ where D is a diagonal matrix.

2.18.1 Condition for a Matrix to be Diagonalizable

1. A necessary and sufficient condition for a matrix $A_{n \times n}$ to be diagonalizable is that the matrix must have n linearly independent eigen vectors.

2. A sufficient (but not necessary) condition for a matrix $A_{n \times n}$ to be diagonalizable is that the matrix must have n linearly independent eigen values.

   This is because if a matrix has n linearly independent eigen values then it surely has n linearly independent eigen vectors (although the converse of this is not true).

❑❑❑

3 PROBABILITY AND STATISTICS

3.1 Random Experiment

The experiment in which the outcome is uncertain is called a Random Experiment (RE).

Example: Flipping a coin, rolling a pair of dice, Picking a ball from a bag.

3.1.1 Sample Space

The set contains all the possible outcomes of a random experiment. It is denoted by 'S'.

If RE is flipping a coin, S = {Head, Tail}

If RE is rolling a dice, S = {1,2,3,4,5,6}

3.2 Event

Any subset of sample space 'S' is called an Event.

Example: If RE is flipping a coin, then the occurring of a Head is an Event.

If RE is rolling a dice, then getting an odd number is an Event.

3.2.1 Probability of an Event

If 'A' is any event with in the sample space 'S' of a Random experiment, then the probability of event 'A' is given by

$$P(A) = \frac{\text{No. of outcomes favouring event 'A' to happen}}{\text{Total number of elements in 'S'}} = \frac{n(A)}{n(S)}$$

Probability of getting an Even Number when a dice is rolled.

$$P(\text{Even Number}) = \frac{3}{6} = 0.5$$

S = {1,2,3,4,5,6},

A = {2,4,6}

Note: Probability can also be expressed as odds if favour and odds against an event:

- **Odds is favour of an event:**
  Odds in **favour** of an event = Number of successes : Number of failures = m: (n – m).
- **Odds against an event:**
  Odds against an event = Number of failures : Number of successes = (n – m) : m.

3.2.2 Axioms Probability

(i) If 'A' is any event with in the sample space 'S' of a RE, then $0 \le P(A) \le 1$

$$\frac{0}{n(S)} \le \boxed{\frac{n(A)}{n(S)}} \le \frac{n(S)}{n(S)}$$

$$\boxed{0 \le P(A) \le 1}$$

(ii) P(S) = 1

When a RE is conducted the experiment yields a possible outcome.

3.2.3 Types of Events

(i) Mutually Exclusive Events:

If A, B are two events within a sample space 'S', then A & B are said to be mutually exclusive if $A \cap B = \phi$.

Example: If 'A' is the event of getting a prime number when a dice is rolled and 'B' is the event of getting a composite number when a dice is rolled then

$S = \{1,2,3,4,5,6\}$, $A = \{2,3,5\}$, $B = \{4,6\} \Rightarrow A \cap B = \phi \Rightarrow P(A \cap B) = 0$

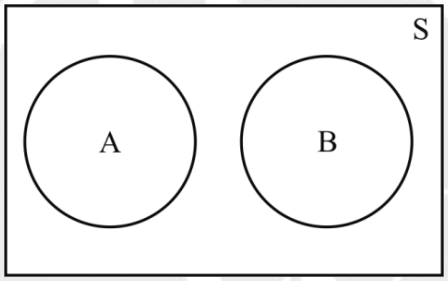


Fig. 5.1. Mutually exclusive event

(ii) Mutually Exhaustive Events:

If 'A', and 'B' are two events within a sample space 'S', then 'A' & 'B' are said to be mutually exhaustive if $A \cup B = S$

Example: If 'A' is the event of getting an odd number when a dice is rolled and 'B' is the event of getting an Even Number, then

$A \cup B = S$

$S = \{1,2,3,4,5,6\}$

$A = \{1,3,5\}$, $B = \{2,4,6\}$

$\Rightarrow$ $A \cup B = S$

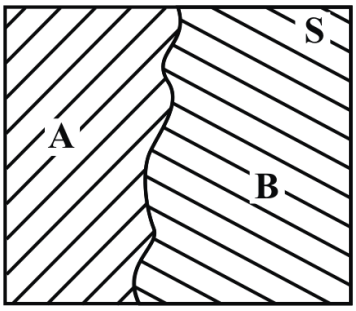


Fig. 5.2. Mutually exhaustive event

(iii) Independent Events:

Two events 'A' & 'B' within the sample space 'S' (or) within two different sample spaces 'S_1' & 'S_2' are said to be independent if $P(A \cap B) = P(A) \cdot P(B)$.

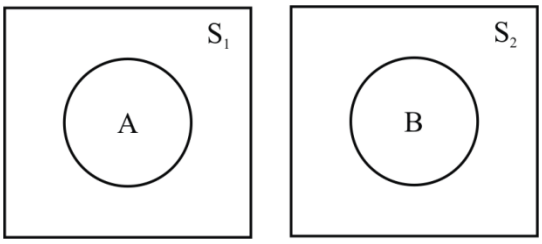


Fig. 5.3. Independent Event

(iv) Impossible Event (ϕ):

The event with zero probability is called an Impossible Event $P(\phi) = 0$.

3.3 Addition Theorem of Probability

If A, and B are two events with a sample space 'S' of a Random Experiment, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{n(A \cup B)}{n(S)} = \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - \frac{n(A \cap B)}{n(S)}$$

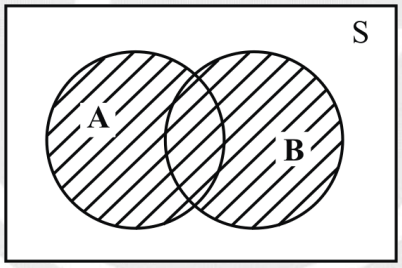


Fig. 5.4. Addition theorem

$\Rightarrow$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

When A, and B are mutually exclusive events, $A \cap B = \phi$.

$\Rightarrow$ $P(A \cap B) = 0$

$P(A \cup B) = P(A) + P(B)$

- If $E_1, E_2, E_3, \ldots\ldots E_n$ are mutually exclusive events ($E_i \cap E_j = \phi$), then $P(E_1 \cup E_2 \cup E_3 \cup \ldots\ldots \cup E_n) = \sum_{i=1}^{n} P(E_i)$

  $= P(E_1) + P(E_2) + P(E_3) + \ldots\ldots p(E_n)$

3.3.1 De Morgan's Law

- $(A \cup B)^C = A^C \cap B^C$
- $(A \cap B)^C = A^C \cup B^C$

3.3.2 Union and Intersection properties

For any two events A and B:

(a) $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

(b) $P(A^c\cap B^c)=1-P(A\cup B)$

For any three events A, B and C:

(a) $P(A\cup B\cup C)=P(A)+P(B)+P(C)-P(A\cap B)-P(B\cap C)-P(C\cap A)+P(A\cap B\cap C)$

(b) $P(A^c\cap B^c\cap C^c)=1-P(A\cup B\cup C)$

3.3.3 Conditional Probability:

The probability of the happening of event 'A' when it is known that event 'B' has already occurred is given by P(A/B)

$$P(A/B)=\frac{P(A\cap B)}{P(B)}=\frac{n(A\cap B)}{n(B)}$$

3.3.4 Joint Probability:

- $f(x, y)$ is the joint probability of two RV'S x, y.
- If the two RV are Independent then
  $f(x, y) = f(x) \cdot f(y)$
- $P(a\le x\le b, c\le y\le d)=\int_a^b\int_c^d f(x,y)dydx$
- $f(x)=\int_{-\infty}^{\infty} f(x,y)dy$
- $f(y)=\int_{-\infty}^{\infty} f(x,y)dx$

3.3.5 Multiplication Theorem of Probability:

If A, and B are two events within a sample space 'S', then P(A/B) · P(B) = P(B/A) · P(A)

P(A/B) = $\frac{P(A\cap B)}{P(B)} \Rightarrow P(A\cap B)=P(A/B)\cdot P(B) \rightarrow (1)$

P(B/A) = $\frac{P(B\cap A)}{P(A)} \Rightarrow P(B\cap A)=P(B/A)\cdot P(A) \rightarrow (2)$

From (1) & (2)

$P(A/B)\cdot P(B)=P(B/A)\cdot P(A)$

3.3.6 Total Theorem of Probability:

If $E_1, E_2, E_3, \ldots\ldots E_n$ are 'n' mutually exclusive ($E_i\cap E_j=\phi; \forall i\neq j$) and collectively exhaustive event ($E_1\cup E_2\cup E_3\cup\ldots\ldots\cup E_n = S$) and 'A' is any event with in the sample space 'S', then

$P(A)=P(E_1)\cdot P(A/E_1)+P(E_2)\cdot P(A/E_2)+\ldots\ldots+P(E_n)\cdot P(A/E_n)$

$$\boxed{P(A)=\sum_{i=1}^{n}P(E_i)\bullet P(A/E_i)}$$

3.3.7 Baye's Theorem

If $E_1, E_2, E_3, \ldots\ldots E_n$ are mutually exclusive ($E_i \cap E_j = \phi \forall i \neq j$) and collectively exhaustive event ($E_1 \cup E_2 \cup E_3 \cup \ldots\ldots \cup E_n = S$) and 'A' is any event with in the sample space 'S', then

$$P(E_i/A) = \frac{P(E_i) \cdot P(A/E_i)}{\sum_{i=1}^{n} P(E_i) \cdot P(A/E_i)}$$

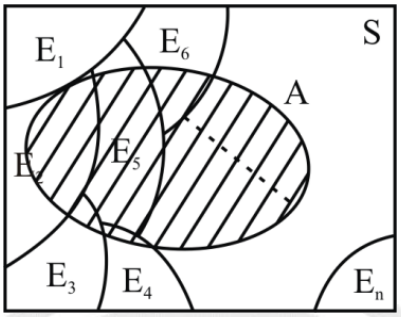


Fig. 5.5. Baye's theorem

3.3.8 Use of permutation and combination

What is combination?

A combination of 'n' objects taken 'r' at a time (r-combination of 'n' objects is an unordered selection of 'r' of the objects).

Number of ways of combining of 'r' object out of 'n' objects without repetition

$$^{\eta}C_r = \frac{n!}{(n-r)!r!}$$

What is permutation?

A combination of 'n' objects taken 'r' at a time (r-combination of 'n' objects is an ordered selection of 'r' of the objects).

Number of ways of selection of r object out of n objects without repetition

$$^{\eta}P_r = \frac{n!}{(n-r)!}$$

Result:

(i) $^{n}C_r = {}^{n}C_{n-r}$

(ii) $^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \ldots. + {}^{n}C_n = 2^n$

(iii) $^{n}C_0 + {}^{n}C_2 + {}^{n}C_4 + \ldots. += 2^{n-1}$

(iv) $^{n}C_1 + {}^{n}C_3 + {}^{n}C_5 + \ldots. += 2^{n-1}$

(v) $0.{}^{n}C_0 + 1.{}^{n}C_1 + 2.{}^{n}C_2 + \ldots. + n.{}^{n}C_n = n.2^{n-1}$

Permutations with Repetition

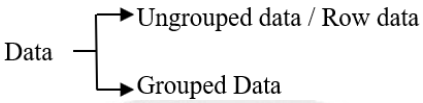
The number of permutations of n objects, where *p* objects are of one kind, q objects are of another kind and the rest, if any, are of a different kind is $\frac{{}^{n}P_{r}}{p!q!}$.

Combination with Repetition

Number of combinations of 'n' distinct things taking 'r' at a time when each thing may be repeated any number of times is given by ${}^{n-1+r}C_{r}$.

3.4 STATISTICS

Statistics → Collection and Analysis of Data



3.4.1 Analysis of Ungrouped Data

If $x_1, x_2, x_3, \ldots\ldots, x_n$ are 'n' observations, then

(1) The range of the data = R = max{$x_1, x_2, \ldots\ldots, x_n$} – min{$x_1, x_2, x_3, \ldots\ldots, x_n$}

(2) Arithmetic mean **:** Mean of the data is equal to sum of observaions divided by the total number of observations.

$$\bar{x}(or)\mu = \frac{x_1 + x_2 + \ldots\ldots + x_n}{n} = \boxed{\frac{\sum_{i=1}^{n} x_i}{n} = \bar{x} = \mu}$$

- The mean of 1st 'n' natural numbers = $\frac{\left(\frac{n(n+1)}{2}\right)}{n} = \frac{n+1}{2}$
- The mean of 1st 'n' odd numbers = $\frac{n^2}{n} = n$
- The mean of 1st 'n' even numbers = n +1

3.4.2 Median

The middle most observation of the data $(x_1, x_2, x_3, \ldots\ldots x_n)$ When the data is arranged in either ascending or descending order.

If $x_1, x_2, x_3, x_4, \ldots\ldots x_n$ are 'n' observations that are arranged in ascending/descending order then

(i) Median of the Data = $\left(\frac{n+1}{2}\right)^{th}$ observation, if 'n' is odd.

(ii) Median of the Data = Mean of $\left(\frac{n}{2}\right)^{th}$ & $\left(\frac{n}{2}+1\right)^{th}$ observations, if 'n' is even.

3.4.3 Mode

The observation with highest frequency is called mode.

Any Data with two Modes is called → Bimodel Data

If $x_1, x_2, x_3, \ldots\ldots, x_n$ are 'n' data points, $\bar{x} = \mu = \frac{x_1 + x_2 + \ldots\ldots + x_n}{n}$

Mean Deviation of the observation $(x_i) = d_i = x_i - \bar{x}$

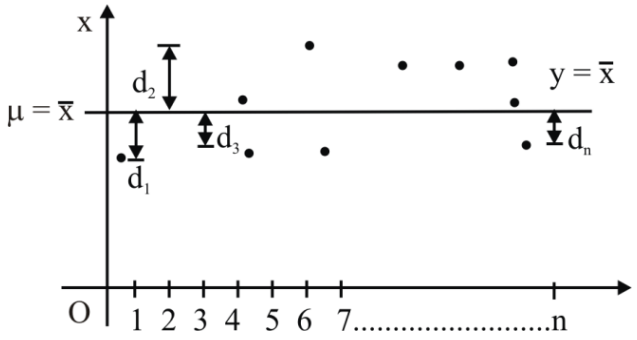


Fig. 5.6. Discrete data

Sum of derivations of all the observations = $\Sigma d_i = (x_1 - \bar{x}) + (x_2 - \bar{x}) + \ldots\ldots 0 + (x_n - \bar{x})$

$$= \Sigma d_i = (x_1 + x_2 + \ldots\ldots + x_n) - n\bar{x}$$

$$\boxed{\Sigma d_i = 0}$$

The sum of mean deviations of all the observations is equal to zero.

3.4.4 Absolute Mean Deviation

If $x_1, x_2, x_3, \ldots\ldots, x_n$ are 'n' data points with Mean = $\bar{x}$, then the absolute mean deviation of x_i about $\bar{x}$ is given by $|d_i| = |x - \bar{x}|$

The sum of absolute mean derivations of given data is not zero.

$$(\Sigma |d_i| \neq 0) \Rightarrow (|x_1 - \bar{x}| + |x_2 - \bar{x}| + \ldots\ldots\ldots + |x_n - \bar{x}| \neq 0)$$

3.4.5 Standard Deviation

If $x_1, x_2, x_3, \ldots\ldots, x_n$ ('n' is very large), then the standard deviation of the data is given by

Population Standard Deviation $\sigma = \sqrt{\frac{1}{n}\Sigma(x_i - \bar{x})^2}$, n $\rightarrow$ size of population

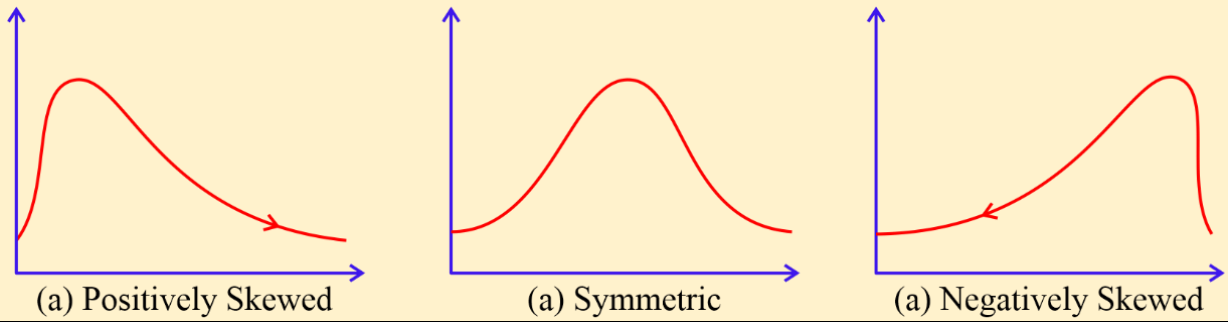
Sample Standard derivation: $\sigma = \sqrt{\frac{1}{(n-1)}\Sigma(x_i - \bar{x})^2}$, n$\rightarrow$ size of sample

Generally (n > 29 $\rightarrow$ population) (n < 29 $\rightarrow$sample)

Note: **Measures of skewness (The degree of asymmetry)**

A lack of symmetry is skewness.

- For symmetric distribution mean (M) = Median (M_d) = Mode (M_e)
- For negatively skewed distribution mean (M) < Median (M_d) < Mode (M_e)
- For positively skewed distribution Mean (M) > Median (M_d) > Mode M_e).



(a) Positively Skewed (a) Symmetric (a) Negatively Skewed

3.5 Random Variables

The variable that connects the outcome of a Random Experiment to a real number.

Example: 'x' is the value of the number that a dice shows when it is rolled.

Discrete RV → The RV whose value is obtained by counting, defined by PMF

Random Variable

Continuous RV → The RV whose value is obtained by Measuring, defined by PDF

- If a data consists of 'f_1' data points with value $'x_1'$, $'f_2'$ data points with value $'x_2'$……$'f_n'$ data point with value $'x_n'$, then
  (i) Expectation of 'x' $= E(x) = \sum_{i=1}^{n} x_i P(x = x_i)$
  (ii) Variance of 'x' = $\sigma^2 = E(x^2) - (E(x))^2$ and σ is the standard deviation.

3.5.1 Probability Mass Function (PMF)

The PMF p(x) of a discrete random variable X taking values $x_1, x_2, ...x_n$ is defined such that,

(i) $p(x_i) \geq 0$

(ii) $\sum_{i=1}^{n} p(x_i) = 1$

(iii) $p(x_i) = p(X = x_i)$

3.5.2 Probability Density Function (PDF)

The pdf f(x) of a continuous random variable X is defined such that,

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x)dx = 1$

(iii) $P(a < X < b) = \int_{a}^{b} f(x)dx$

3.5.3 Expected Value

1. Expected value of a random variable X, E [X], is defined as, $E[X]\begin{cases} \sum xp(x); & \text{X is discrete rv} \\ \int_{-\infty}^{\infty} xf(x)dx; & \text{X is continuous rv} \end{cases}$

2. Expected value of X^2 is,

$$E[X^2] = \begin{cases} \sum x^2 p(x); & \text{X is discrete rv} \\ \int_{-\infty}^{\infty} x^2 f(x)dx; & \text{X is continuous rv} \end{cases}$$

Note: $E[X^n]$ is called nth moment.

3.5.4 Mean of Random Variable 'X'

Mean $= \mu = E[X]$

3.5.5 Variance of a Random Variable 'X'

Var $(X) = E[(X - \mu)^2]$

Or, Var $(X) = E[X^2] - \mu^2$

3.5.6 Properties of Expectation

(i) E[c] = c, c is a constant.

(ii) E[ax] = aE[X]

(iii) E(aX + b) = aE(X) + b

(iv) If X and Y are random variable E[X ± Y] = E(x) ± E(Y).

(v) If X and Y are random variables E(X. Y) = E(X). E(Y / X).

(vi) If X and Y independent random variables E(X. Y) = E(X). E (Y).

3.5.7 Properties of Variance

(i) Var[C] = 0, C is constant.

(ii) Var (aX) = a^2V(X) where X is random variable and 'a' constant.

Var(–X) = $(-1)^2$ Var(X) = Var(X) Variance is always positive.

(iii) Var(ax + b) = a^2 Var(X) + 0

(iv) If X and Y are independent random variables.

Var(X + Y) = Var(X) + Var(Y)

Var(X – Y) = Var(X) + Var(Y)

(v) Var(ax + by) = a^2 v(x) + b^2 v(y) + 2ab Cov (x, y)

(vi) Cov (x, y) = E(x, y) – E(x) E(y)

(vii) For independent random variables Cov(x, y) = 0

3.5.8 Continuous RV

The value of the Random Variable is obtained by Measuring.

3.6 Probability Distribution Function (PDF)

A continuous & differentiable function P(x) is said to be a probability distribution/density function of a continuous random variable 'x' if $P(a \leq x \leq b) = \int_a^b P(x)dx$

3.6.1 Mean (or) Expectation

If P(x) is a probability distribution/density function of a continuous Random Variable 'x' then the Mean of 'x' = E(x) $= \int_{-\infty}^{\infty} x \cdot P(x)dx$

3.6.2 Median

The value of 'x' for which the total probability is exactly divided into two equal halves is called Median.

3.6.3 Mode

The valueof 'x' at which P(x) is maximum is called mode.

3.6.4 Variance

$= \sigma^2 = E(x^2) - (E(x))^2$

$$\Rightarrow \sigma^2 = \int_{-\infty}^{\infty} x^2 \cdot P(x)dx - \left(\int_{-\infty}^{\infty} x \cdot P(x)dx\right)^2$$

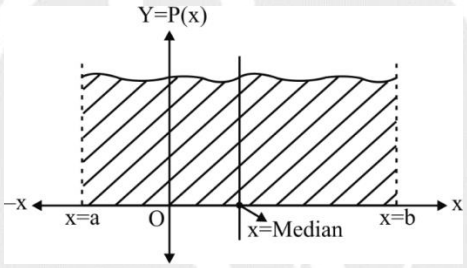


Fig. 5.7. Continuous random variables

3.7 Continous RV distributions

(1) Gaussian/Normal Distributon:

If 'x' is a continuous Random variable with mean 'μ' and standard deviation 'σ', then the probability distribution/density function of normally distributed variable 'x' is given by

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)}$$

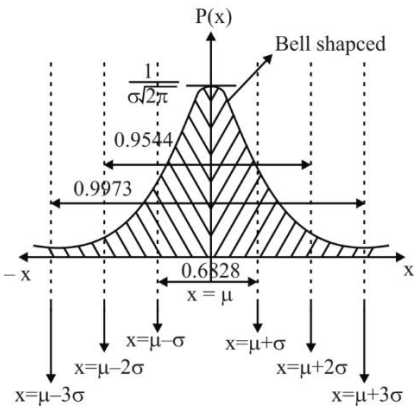


Fig. 5.8. Normal distribution

Mean = Median = Mode = μ

$$P(\mu - \sigma \le x \le \mu + \sigma) = 0.6828$$

$$P(\mu - 2\sigma \le x \le \mu + 2\sigma) = 0.9544$$

$$P(\mu - 3\sigma \le x \le \mu + 3\sigma) = 0.9973$$

$$P(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} . e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

(2) Standard Normal Distribution:

Assuming $z = \frac{x-\mu}{\sigma}; \mu = 0; \sigma = 1, P(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{\frac{-z^2}{2}}$

$$P(-1 \le z \le 1) = 0.6828$$

$$P(-2 \le z \le 2) = 0.9544$$

$$P(-3 \le z \le 3) = 0.9973$$

Note:

1. The normal distribution curve is bell shaped curve
2. The points of infelection of the normal distribution curve are at $x = \mu + \sigma \; and \; x = \mu - \sigma$.
3. The cumulative function graph is of 'S' Shape.
4. For a given normal distribution, Mean = median = Mode

(3) Uniform Distribution:

If 'x' is a uniformly distrbuted random variable such that $a \le x \le b$ then the Pdf is given by

$$P(x) = \frac{1}{(b-a)}$$

Mean = $\int_a^b x \cdot P(x)dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{(b-a)} \int_a^b x \cdot dx$

$$\boxed{\left(\frac{b+a}{2}\right) = \text{Mean}}$$

$\Rightarrow \; Variance = \sigma^2 = \frac{(b-a)^2}{12}$

Std. deviation = $\sigma = \frac{(b-a)}{\sqrt{12}}$

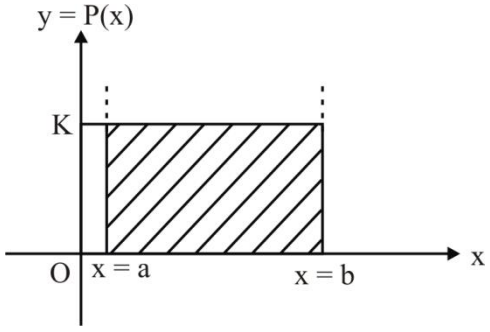


Fig. 5.9. Uniform Distribution

3.7.1 Properties of Mean and Variance:

$E(ax + by) = a \cdot E(x) + b \cdot E(y)$

$$V(ax + by) = a^2 \cdot V(x) + b^2 \cdot V(y) - 2abCOV(x, y)$$

where $COV(x, y) = E(xy) - E(x) \cdot E(y)$

If x,y are indpendent random variables, then $E(xy) = E(x) \cdot E(y) \Rightarrow COV(x,y) = 0$

(1) Exponential Dirtibution:

If 'x' is a continous random variable with mean as $\frac{1}{\lambda}$ then the exponential distribution of 'x' is given by the function

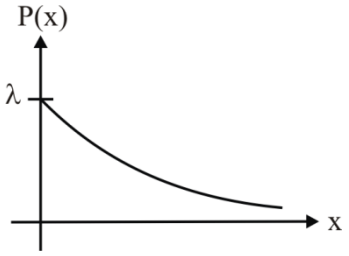


Fig. 5.10. Exponential distribution

$$f(x) = \begin{cases} \lambda \cdot e^{-\lambda x} & ; x \geq 0 \\ 0 & : otherwise \end{cases}$$

Mean = $\frac{1}{\lambda}$

$\sigma^2 = \frac{1}{\lambda^2}$

Mean = Standard Deviation = $\frac{1}{\lambda}$

3.8 Discrete Random Variable Distributions

If a Random experiment has **only two Possible outcomes**, (one is Success & other is failure) and the Probability of Success doesn't depend on time, then the probability of occuring of exactly 'r-successes' in 'n-trials' is given by

$$P(X = r) = {}^{n}C_r \cdot P^r \cdot q^{n-r}$$

Where, P $\rightarrow$ Probability of Success,

q $\rightarrow$ Probability of Failure

p + q = 1

Mean = np,Variance = npq = σ^2, standard deviation = $\sigma = \sqrt{npq}$

3.8.1 Poisson Distribution

If a random experiment has only two possible outcomes, and the average number of successes in a given time 't' is λ, then the probability that exactly 'r' successes occur within the same time 't' given by

$P(x = r)\frac{e^{-\lambda} \cdot \lambda^r}{r!}$

Mean = λ.

Mean = Variance = λ

$\Rightarrow \sigma = \sqrt{\lambda}$

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